

Linear mixed effects models 3



A cartoon character of a bar chart with a smiling face, arms, and legs, appearing to be jumping or dancing.

COLLABORATIVE PLAYLIST

psych252

<https://tinyurl.com/psych252spotify>

PLAY

...

02/23/2026

Plan for today

- Quick recap
- Linear Mixed Model
 - Let's simulate some lmer()s
 - Some more examples
 - Different random effect structures
 - lmer() standard operating procedures

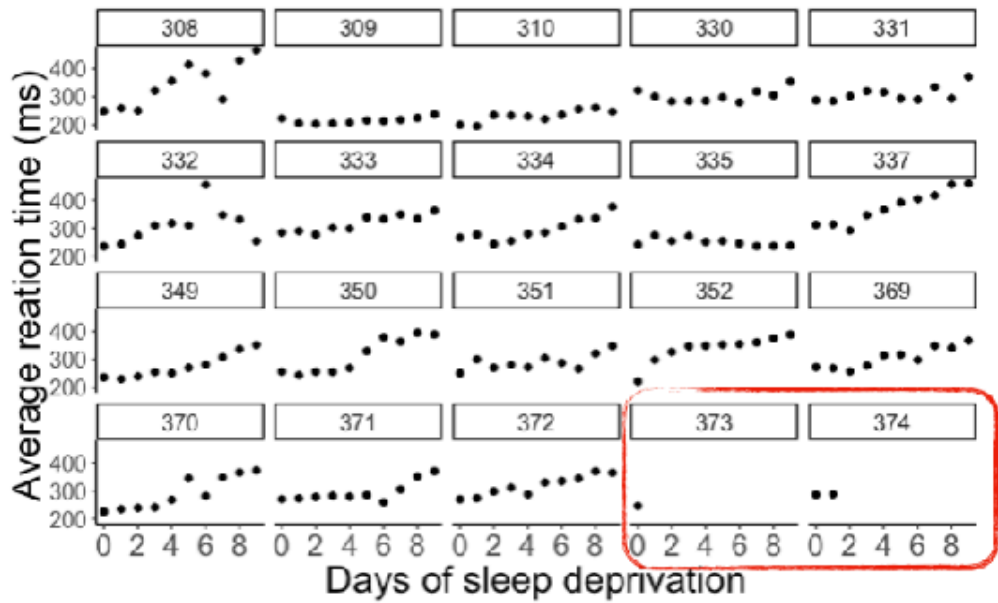
Quick recap

Quick recap: worked example

Data set

How does sleep deprivation affect reaction time?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72



20 participants 2 with incomplete information

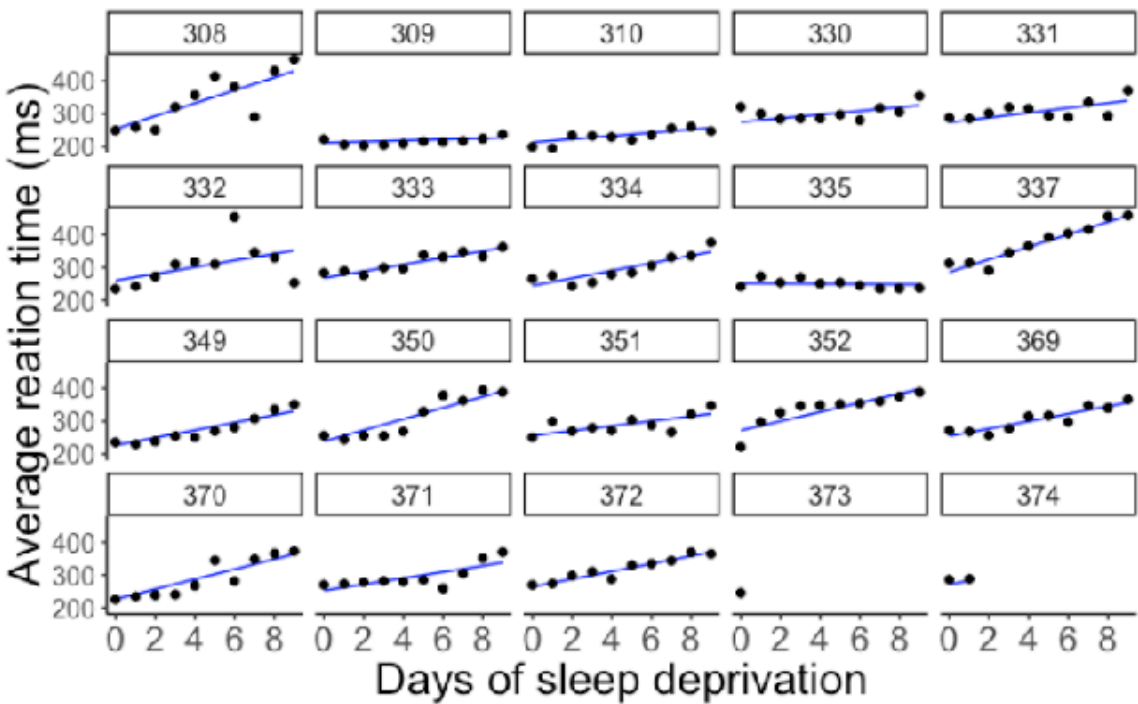
26

Partial pooling: Fit mixed effects model

intercepts and slopes differ
between participants

random intercept random slope

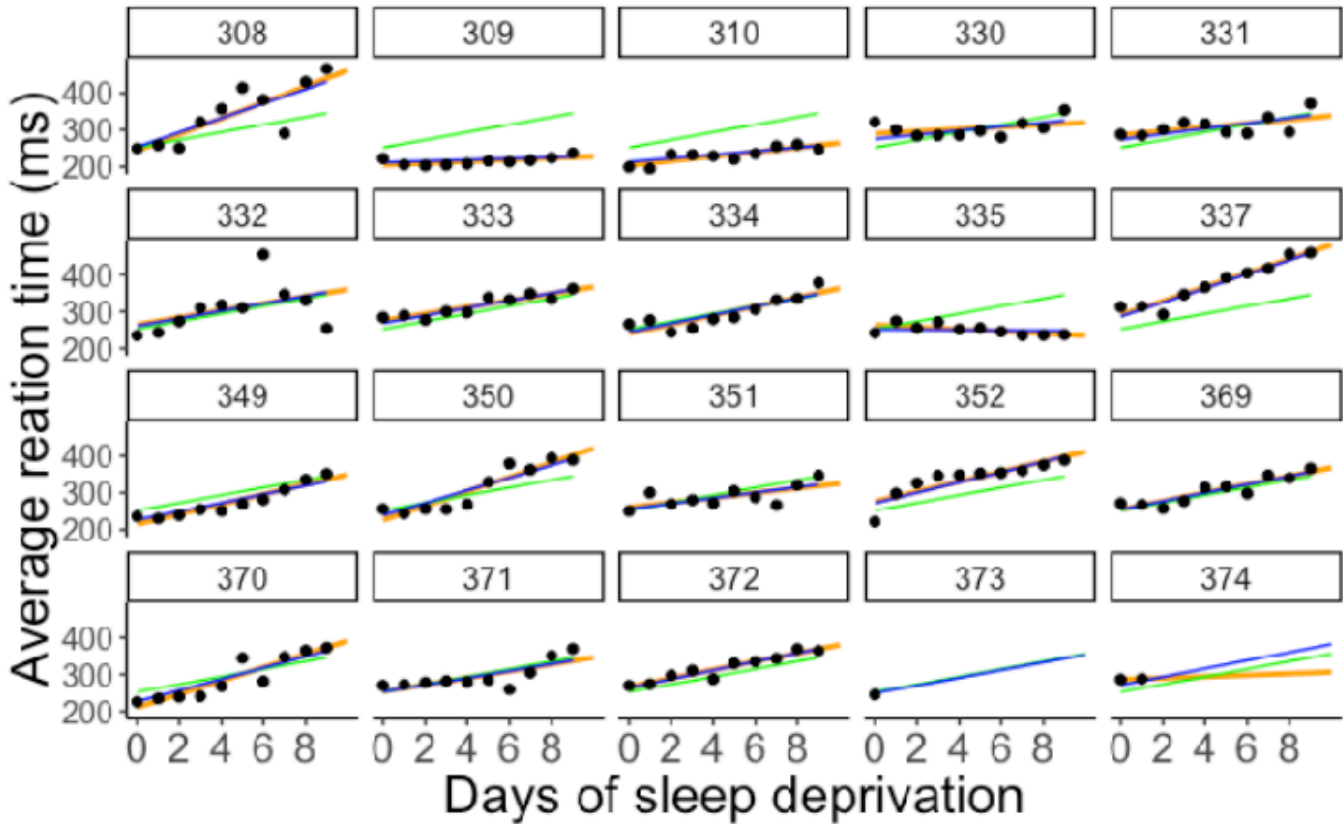
```
lmer(formula = reaction ~ 1 + days + (1 + days | subject),  
      data = df.sleep)
```



33

Comparison

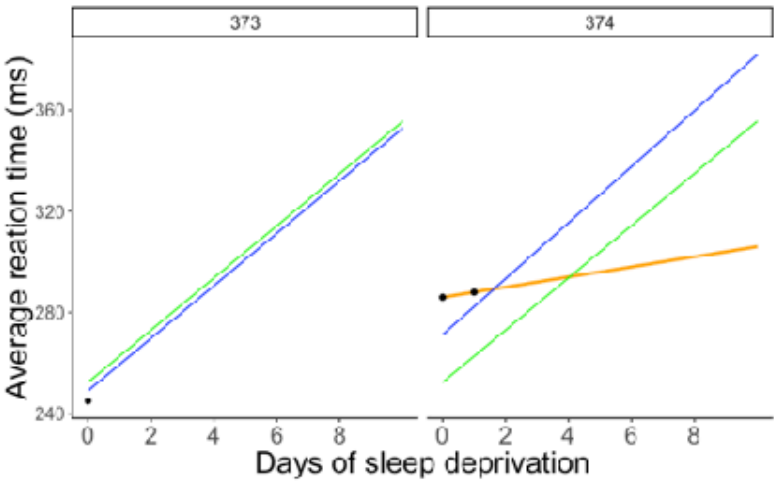
complete pooling
no pooling
partial pooling



37

Quick recap: shrinkage

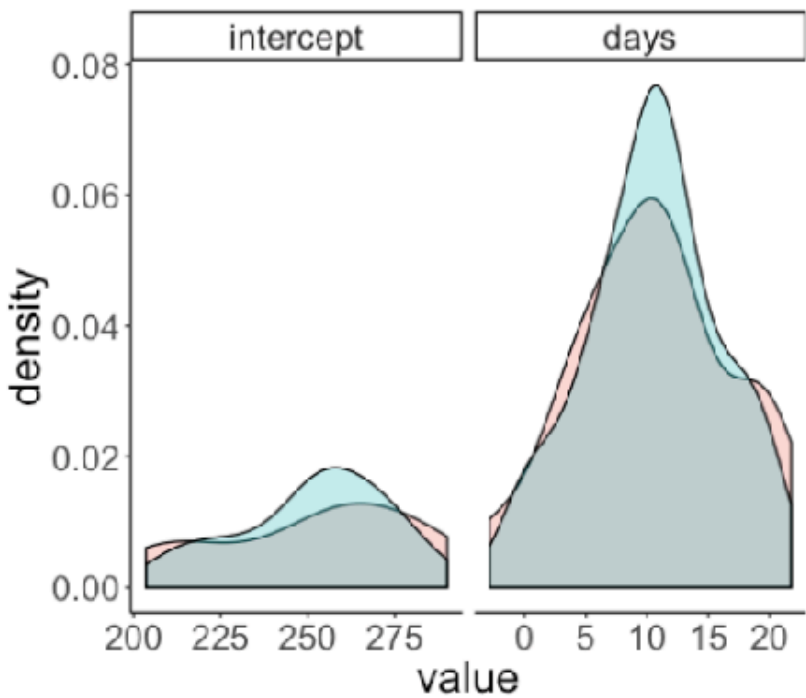
Comparison



- complete pooling:**
 - doesn't account for any individual variation
- no pooling:**
 - doesn't yield predictions when we only have observation
 - doesn't consider the general effect of sleep deprivation when making predictions
- partial pooling:**
 - draws on all the information in the data
 - extrapolates based on information about the individual participants, as well as information based on the whole sample

38

Shrinkage



separate regression for each participant

subject	intercept	days
1	284.207	11.74970
2	285.040	2.35270
3	285.040	6.14880
4	284.207	3.44871
5	285.040	9.44871
6	284.207	9.44870
7	285.040	6.44870
8	285.040	10.44870
9	285.040	2.44870
10	285.040	2.44870
11	285.040	10.44870
12	285.040	10.44870
13	285.040	10.44870
14	285.040	10.44870
15	285.040	10.44870
16	285.040	10.44870
17	285.040	10.44870
18	285.040	10.44870
19	285.040	10.44870
20	285.040	10.44870

lmer()

(1 + days | subject)
random intercepts and slopes (+ correlation)

method

- no pooling
- partial pooling

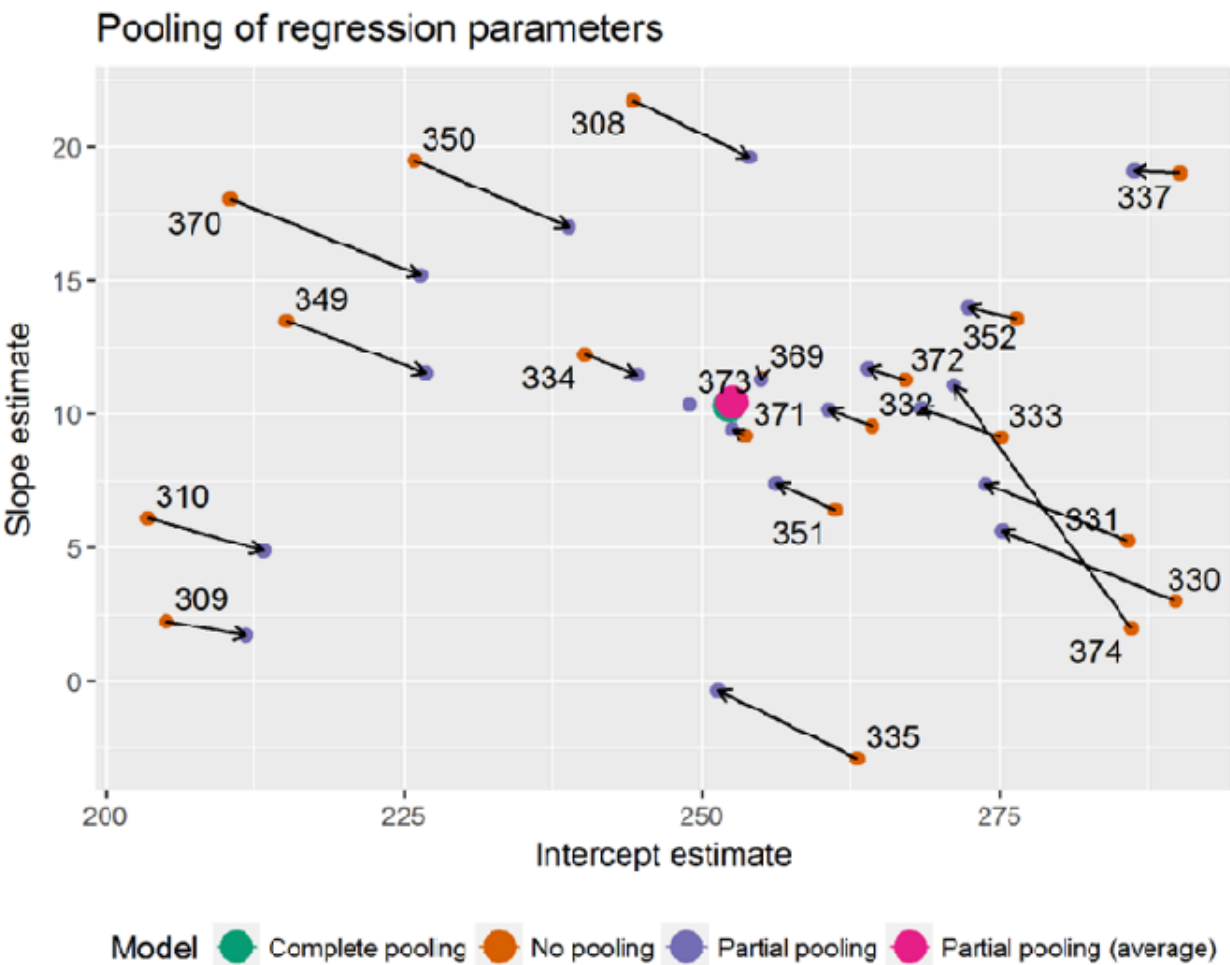
standard deviation

method	intercept	days
no pooling	28.95	6.56
partial pooling	21.59	5.48

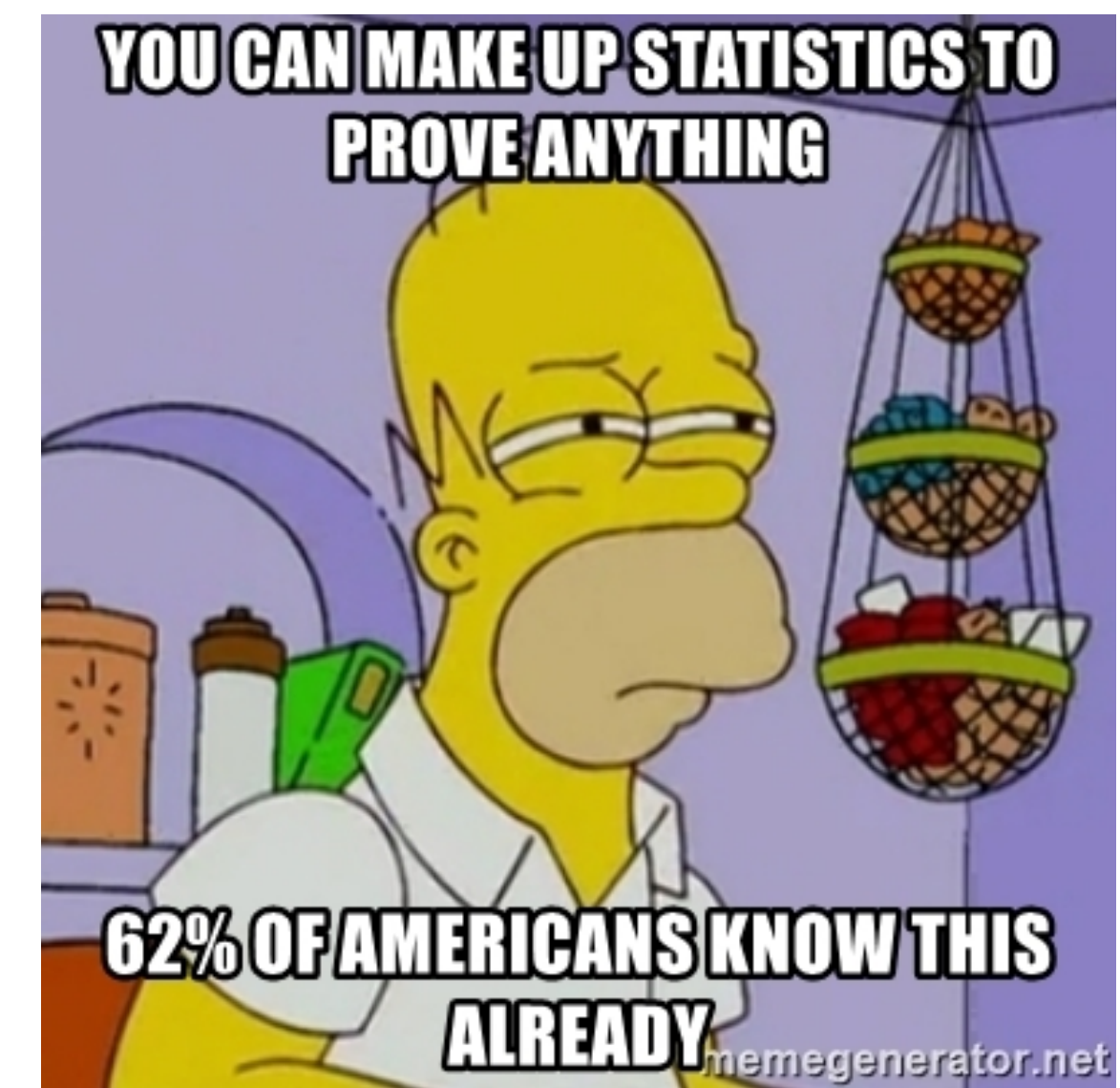
variance "shrinks"

39

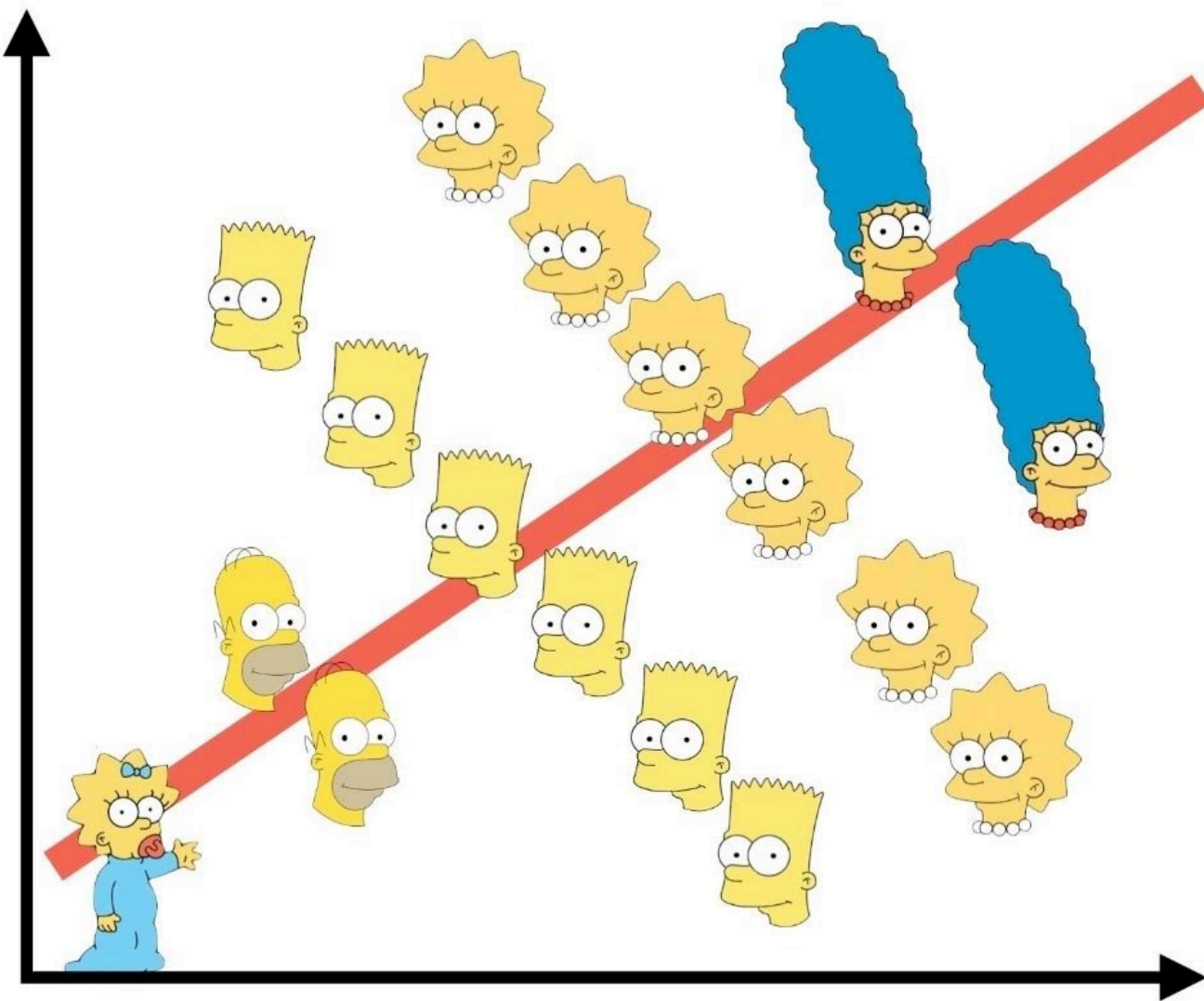
Shrinkage



40

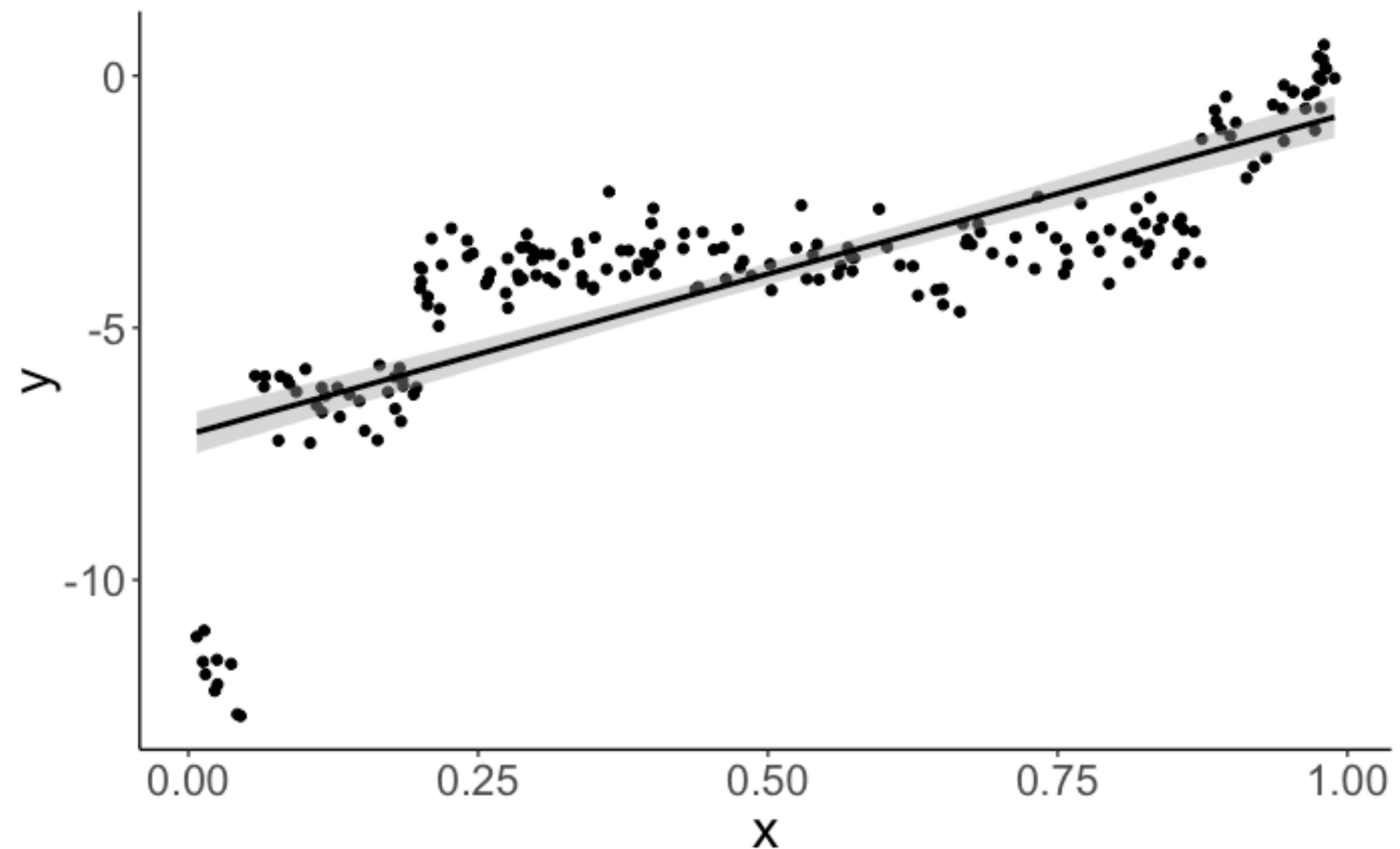


Simpsons Paradox



Simpson's paradox

```
1 lm(formula = y ~ 1 + x,  
2    data = df.simpson) %>%  
3    summary()
```



```
Call:
lm(formula = y ~ x, data = df.simpson)

Residuals:
    Min       1Q   Median       3Q      Max
-5.8731 -0.6362  0.2272  1.0051  2.6410

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -7.1151     0.2107  -33.76  <2e-16 ***
x              6.3671     0.3631   17.54  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.55 on 198 degrees of freedom
Multiple R-squared:  0.6083,    Adjusted R-squared:  0.6064 
F-statistic: 307.5 on 1 and 198 DF,  p-value: < 2.2e-16
```

**positive relationship
between x and y**

Simpson's paradox

```
1 lmer(formula = y ~ 1 + x + (1 | participant),  
2     data = df.simpson) %>%  
3 summary()
```

```
Linear mixed model fit by REML ['lmerMod']  
Formula: y ~ 1 + x + (1 | participant)  
Data: df.simpson
```

REML criterion at convergence: 345.1

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.43394	-0.59687	0.04493	0.62694	2.68828

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	21.4898	4.6357
	Residual	0.1661	0.4075

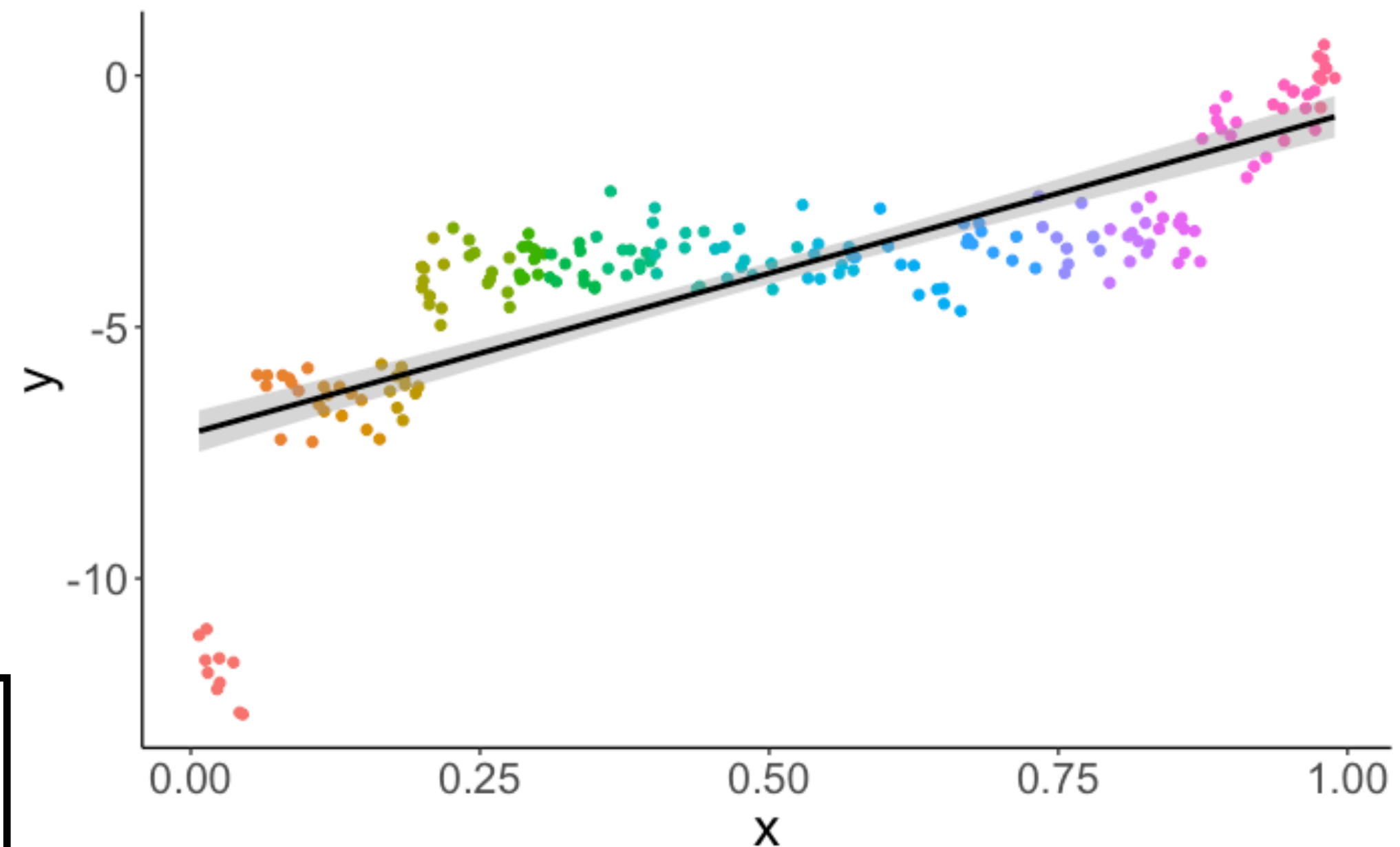
Number of obs: 200, groups: participant, 20

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-0.1577	1.3230	-0.119
x	-7.6678	1.6572	-4.627

Correlation of Fixed Effects:

(Intr)	
x	-0.621



**negative (!)
relationship between
x and y**

Simpson's paradox

```
1 lmer(formula = y ~ 1 + x + (1 | participant),  
2       data = df.simpson) %>%  
3 summary()
```

```
Linear mixed model fit by REML ['lmerMod']  
Formula: y ~ 1 + x + (1 | participant)  
Data: df.simpson
```

REML criterion at convergence: 345.1

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.43394	-0.59687	0.04493	0.62694	2.68828

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	21.4898	4.6357
	Residual	0.1661	0.4075

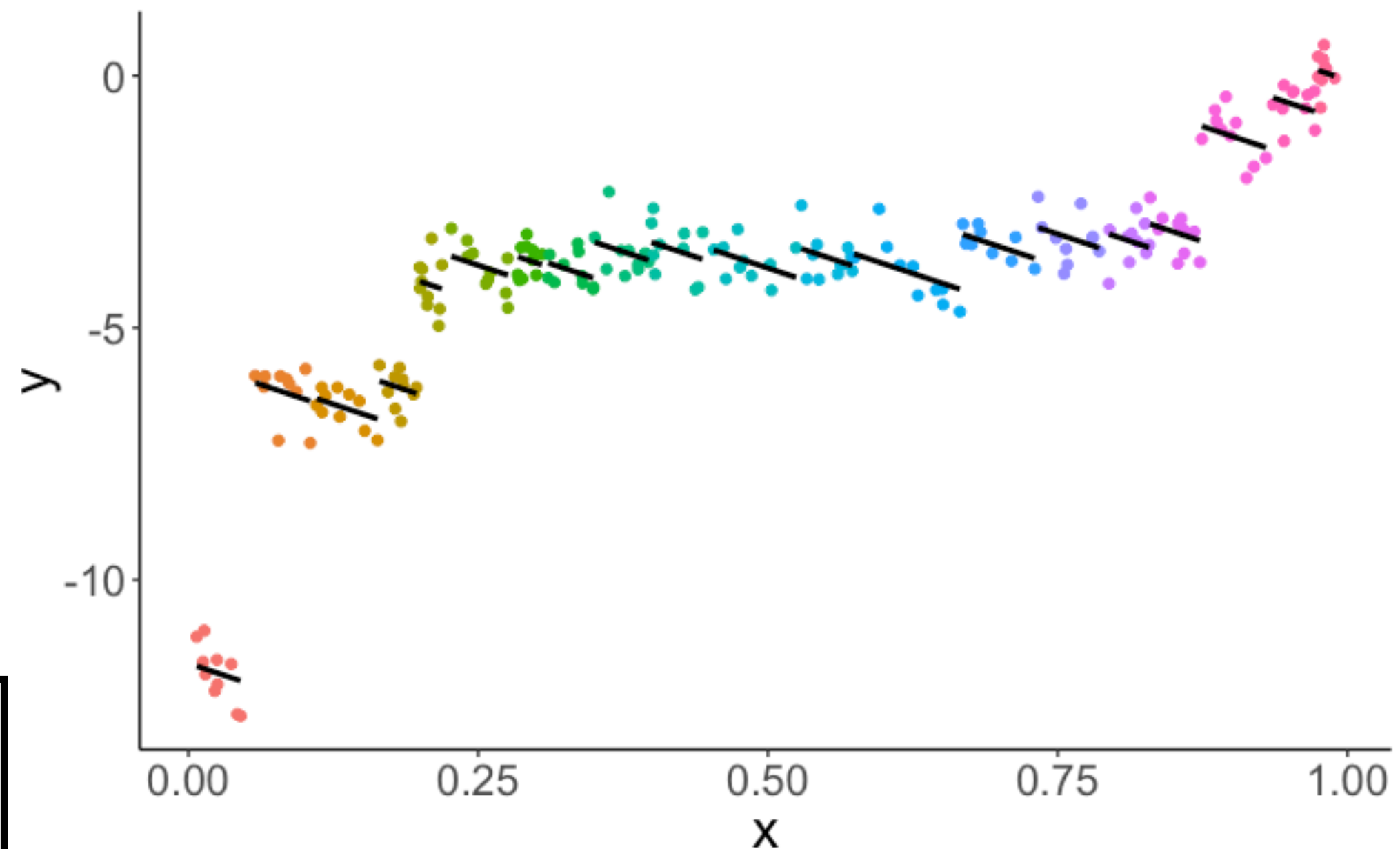
Number of obs: 200, groups: participant, 20

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	-0.1577	1.3230	-0.119
x	-7.6678	1.6572	-4.627

Correlation of Fixed Effects:

(Intr)	
x	-0.621



**negative (!)
relationship between
x and y**

**(once we take into
account individual
differences)**

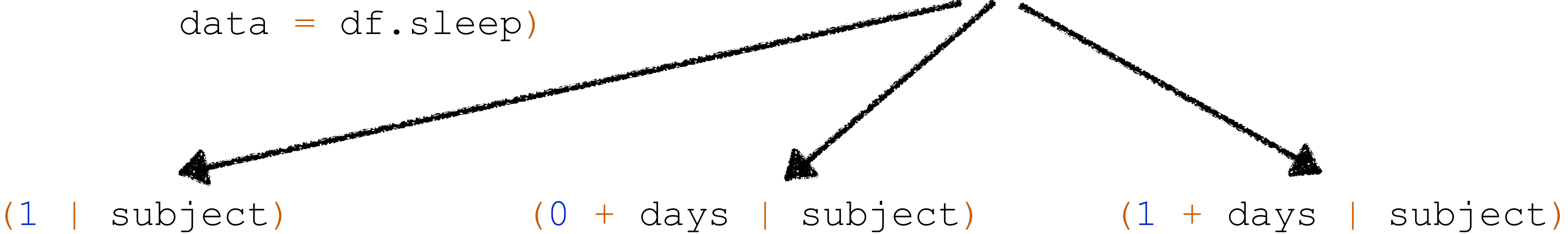
Understanding `lmer()` syntax

lmer () syntax summary

formula	description
<code>dv ~ 1 + x1 + (1 g)</code>	Random intercept for each level of `g`
<code>dv ~ 1 + x1 + (0 + x1 g)</code>	Random slope for each level of `g`
<code>dv ~ 1 + x1 + (1 + x1 g)</code>	Correlated random slope and intercept for each level of `g`
<code>dv ~ 1 + x1 + (1 + x1 g)</code>	Uncorrelated random slope and intercept for each level of `g`
<code>dv ~ 1 + x1 + (1 part) + (1 item)</code>	Random intercept for each level of `participant` and for each level of `item` (crossed)
<code>dv ~ 1 + x1 + (1 school) + (1 school:class)</code>	Random intercept for each level of `school` and for each level of `class` in `school` (nested)

Coefficients

```
lmer(formula = reaction ~ 1 + days + ...,  
      data = df.sleep)
```



random intercepts

random slopes

random intercepts and
slopes (+ correlation)

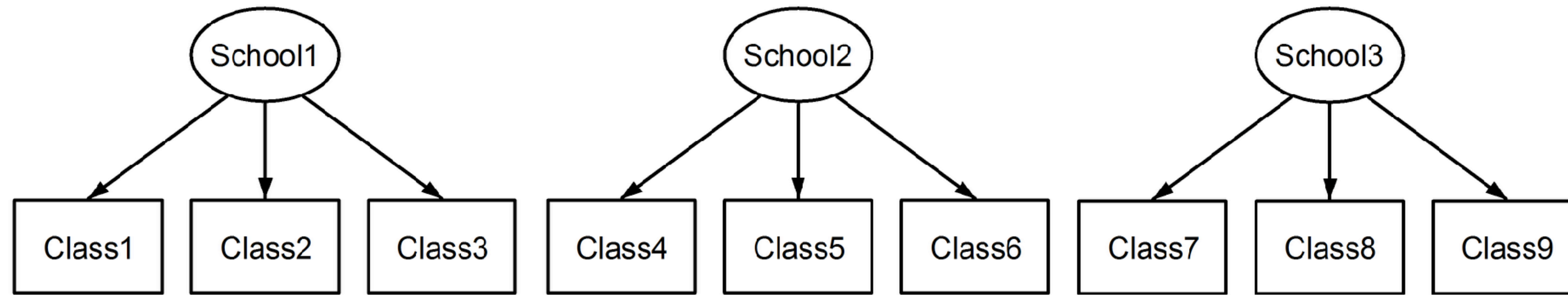
\$subject	(Intercept)	days
308	292.2749	10.43191
309	174.0559	10.43191
310	188.7454	10.43191
330	256.0247	10.43191
331	261.8141	10.43191
332	259.8262	10.43191
333	268.0765	10.43191
334	248.6471	10.43191
335	206.5096	10.43191
337	323.5643	10.43191
349	230.5114	10.43191
350	265.6957	10.43191
351	243.7988	10.43191
352	287.8850	10.43191
369	258.6454	10.43191
370	245.2931	10.43191
371	248.3508	10.43191
372	269.6861	10.43191
373	248.2086	10.43191
374	273.9400	10.43191

\$subject	(Intercept)	days
308	252.2965	19.9526801
309	252.2965	-4.3719650
310	252.2965	-0.9574726
330	252.2965	8.9909957
331	252.2965	10.5394285
332	252.2965	11.3994289
333	252.2965	12.6074020
334	252.2965	10.3413879
335	252.2965	-0.5722073
337	252.2965	24.2246485
349	252.2965	7.7702676
350	252.2965	15.0661415
351	252.2965	7.9675415
352	252.2965	17.0002999
369	252.2965	11.6982767
370	252.2965	11.3939807
371	252.2965	9.4535879
372	252.2965	13.4569059
373	252.2965	10.4142695
374	252.2965	11.9097917

\$subject	(Intercept)	days
308	253.9479	19.6264139
309	211.7328	1.7319567
310	213.1579	4.9061843
330	275.1425	5.6435987
331	273.7286	7.3862680
332	260.6504	10.1632535
333	268.3684	10.2245979
334	244.5523	11.4837825
335	251.3700	-0.3355554
337	286.2321	19.1090061
349	226.7662	11.5531963
350	238.7807	17.0156766
351	256.2344	7.4119501
352	272.3512	13.9920698
369	254.9484	11.2985741
370	226.3701	15.2027922
371	252.5051	9.4335432
372	263.8916	11.7253342
373	248.9752	10.3915245
374	271.1451	11.0782697

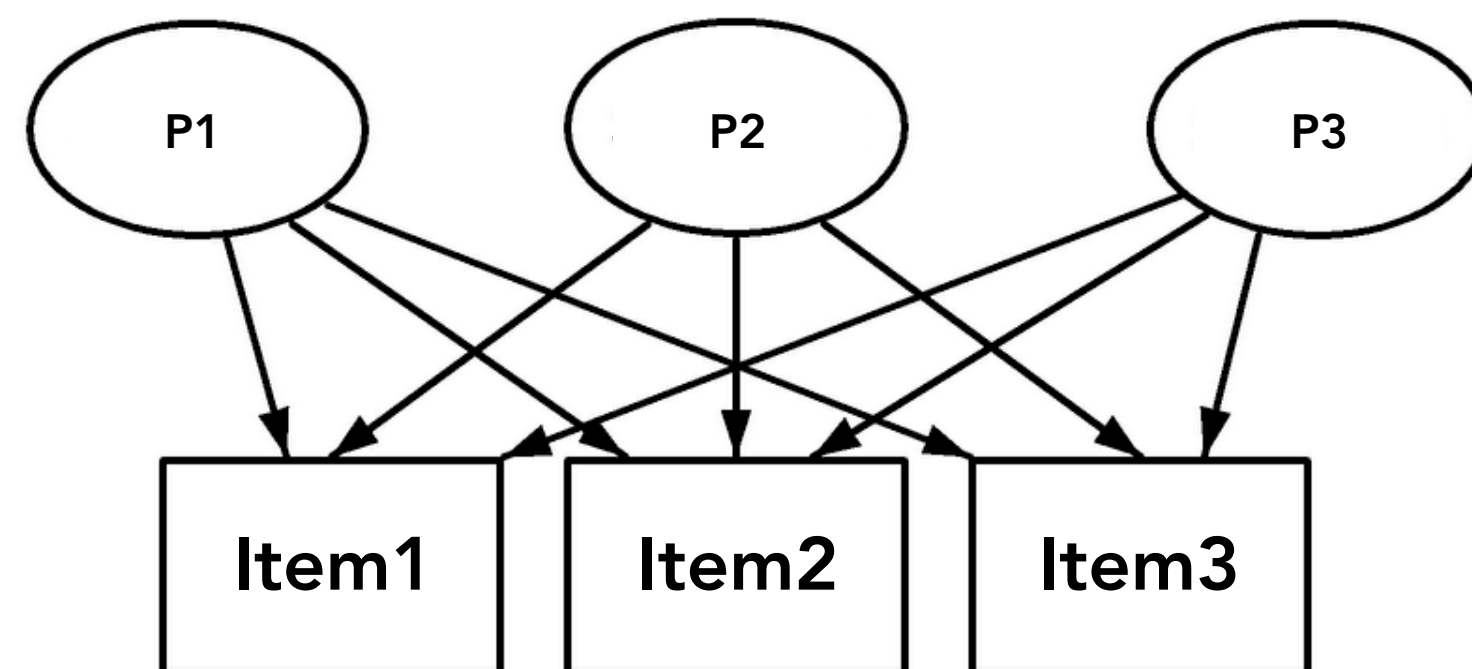
Multi-level models

nested $(1 | \text{School}) + (1 | \text{School:Class})$



each class only appears within one school

crossed $(1 | \text{participant}) + (1 | \text{item})$

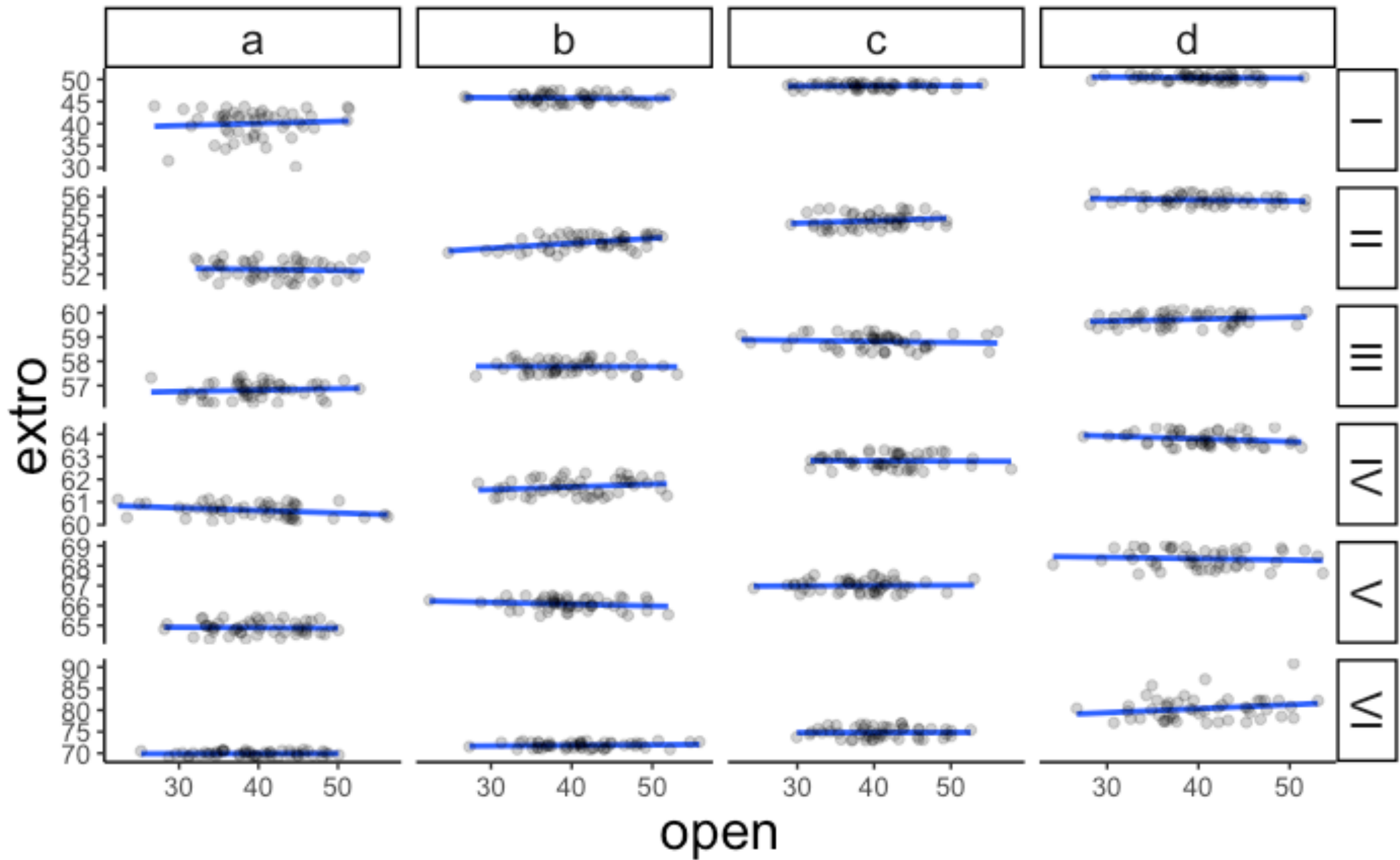


**each participant
rates each item**

Multi-level models

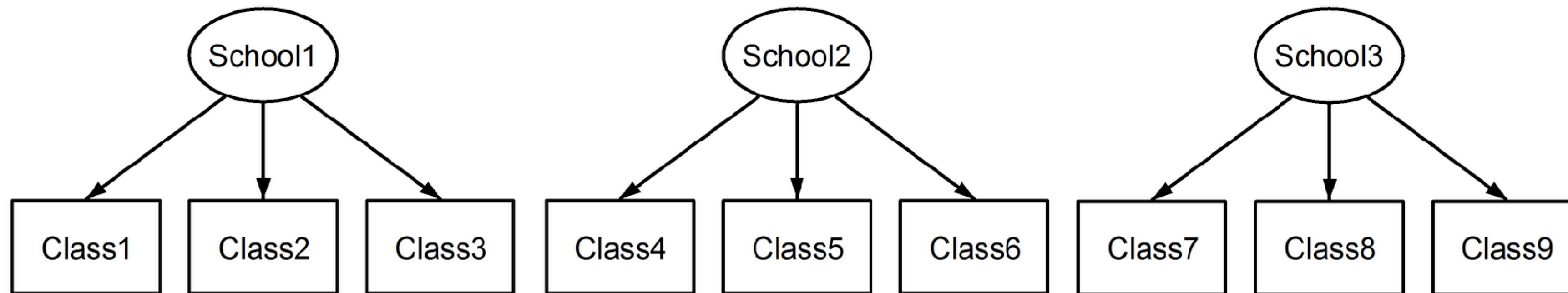
	id	extro	open	agree	social	class	school
1	1	63.69356	43.43306	38.02668	75.05811	d	IV
2	2	69.48244	46.86979	31.48957	98.12560	a	VI
3	3	79.74006	32.27013	40.20866	116.33897	d	VI
4	4	62.96674	44.40790	30.50866	90.46888	c	IV
5	5	64.24582	36.86337	37.43949	98.51873	d	IV
6	6	50.97107	46.25627	38.83196	75.21992	d	I
7	7	60.14740	37.04243	38.55959	95.91299	d	III
8	8	64.17886	42.16530	34.88235	91.45257	d	IV
9	9	56.67670	32.84933	31.68027	115.25167	a	III
10	10	47.23914	44.25764	24.99970	122.70848	b	I

relationship between
openness and extraversion



Multi-level models

nested (1 | School) + (1 | School:Class)



```
1 # fit nested model
2 fit.nested = lmer(extro ~ open + agree + social + (1 | school) + (1 | school:class),
3                   data = df.school)
4 # print model summary
5 fit.nested %>% summary()
6
7 # model coefficients
8 fit.nested %>% coef()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: extro ~ open + agree + social + (1 | school/class)
Data: df.school

REML criterion at convergence: 3554.6

Scaled residuals:
    Min       1Q   Median       3Q      Max
-9.9949 -0.3348  0.0057  0.3394 10.6476

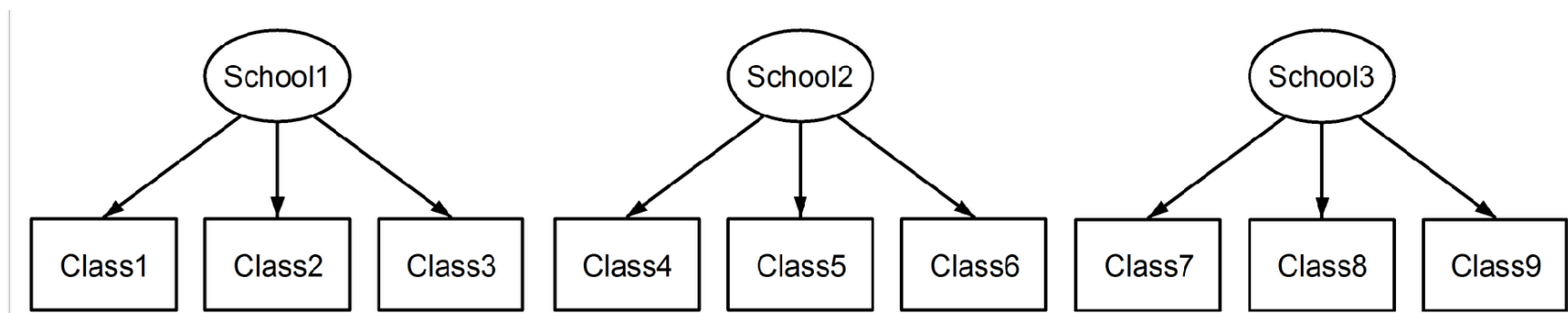
Random effects:
Groups      Name      Variance Std.Dev.
school:class (Intercept)  8.2046  2.8644
school      (Intercept) 93.8433  9.6873
Residual                    0.9684  0.9841
Number of obs: 1200, groups:  school:class, 24; school, 6
```

Multi-level models

random intercepts

nested

$(1 | \text{School}) + (1 | \text{School}:\text{Class})$



random intercepts
of class within
each school

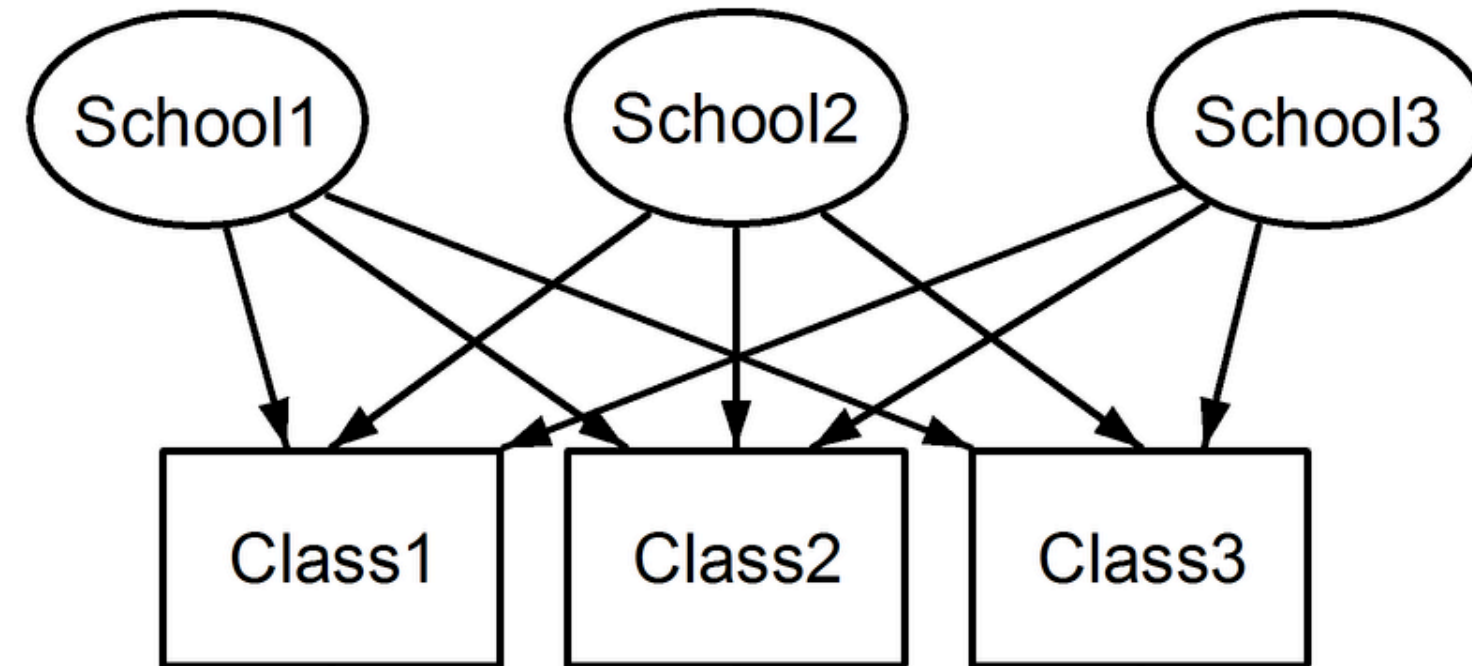
random intercepts of
schools

\$`class:school`				
	(Intercept)	open	agree	social
a:I	53.77106	0.006106514	-0.007665927	0.0005404069
a:II	58.23842	0.006106514	-0.007665927	0.0005404069
a:III	58.71819	0.006106514	-0.007665927	0.0005404069
a:IV	58.68813	0.006106514	-0.007665927	0.0005404069
a:V	58.68268	0.006106514	-0.007665927	0.0005404069
a:VI	56.23088	0.006106514	-0.007665927	0.0005404069
b:I	59.54852	0.006106514	-0.007665927	0.0005404069
b:II	59.62643	0.006106514	-0.007665927	0.0005404069
b:III	59.70219	0.006106514	-0.007665927	0.0005404069
b:IV	59.73276	0.006106514	-0.007665927	0.0005404069
b:V	59.87036	0.006106514	-0.007665927	0.0005404069
b:VI	58.14865	0.006106514	-0.007665927	0.0005404069
c:I	62.28460	0.006106514	-0.007665927	0.0005404069
c:II	60.74743	0.006106514	-0.007665927	0.0005404069
c:III	60.70970	0.006106514	-0.007665927	0.0005404069
c:IV	60.86062	0.006106514	-0.007665927	0.0005404069
c:V	60.80225	0.006106514	-0.007665927	0.0005404069
c:VI	61.10164	0.006106514	-0.007665927	0.0005404069
d:I	64.14113	0.006106514	-0.007665927	0.0005404069
d:II	61.81189	0.006106514	-0.007665927	0.0005404069
d:III	61.65165	0.006106514	-0.007665927	0.0005404069
d:IV	61.83703	0.006106514	-0.007665927	0.0005404069
d:V	62.13593	0.006106514	-0.007665927	0.0005404069
d:VI	66.66561	0.006106514	-0.007665927	0.0005404069
\$school				
	(Intercept)	open	agree	social
I	46.44407	0.006106514	-0.007665927	0.0005404069
II	54.20862	0.006106514	-0.007665927	0.0005404069
III	58.29847	0.006106514	-0.007665927	0.0005404069
IV	62.15074	0.006106514	-0.007665927	0.0005404069
V	66.41348	0.006106514	-0.007665927	0.0005404069
VI	73.91156	0.006106514	-0.007665927	0.0005404069

model coefficients
fit.nested %>% coef()

Multi-level models

crossed (1 | School) + (1 | Class)



**each class appears
in each of the
schools**

```
1 # fit crossed model
2 fit.crossed = lmer(extro ~ open + agree + social + (1 | school) + (1 | class), data = df.school)
3
4 # print model summary
5 fit.crossed %>% summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: extro ~ open + agree + social + (1 | school) + (1 | class)
Data: df.school

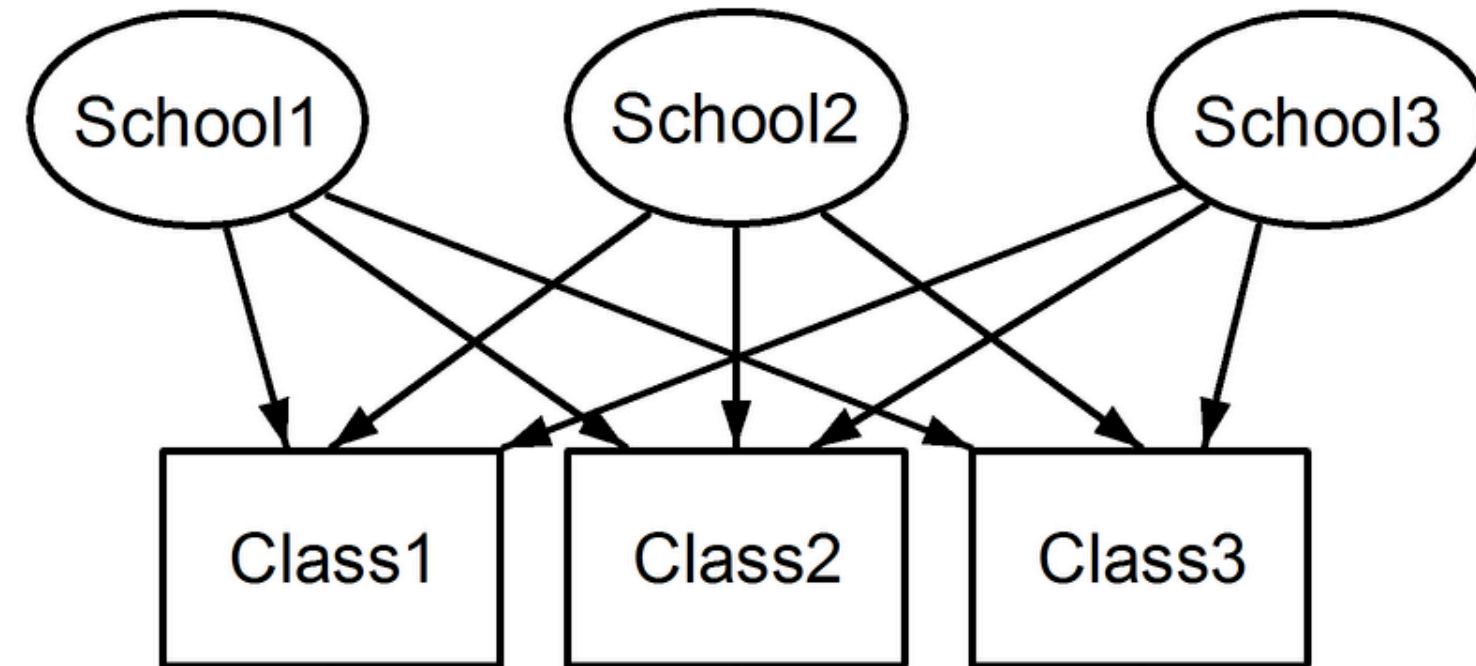
REML criterion at convergence: 4723.9

Scaled residuals:
    Min       1Q   Median       3Q      Max
-7.8677 -0.5421  0.0101  0.5218  8.2282

Random effects:
   Groups     Name      Variance Std.Dev.
school  (Intercept)  95.914     9.794
class   (Intercept)   5.787     2.406
Residual                    2.787     1.669
Number of obs: 1200, groups: school, 6; class, 4
```

Multi-level models

crossed (1 | School) + (1 | Class)



each class is in
each of the schools

random intercepts

random intercepts
of school

random intercepts of
class

```
$school
  (Intercept)      open      agree      social
I      46.10663 0.01083374 -0.005420032 -0.001761963
II     54.02956 0.01083374 -0.005420032 -0.001761963
III    58.22277 0.01083374 -0.005420032 -0.001761963
IV     62.15508 0.01083374 -0.005420032 -0.001761963
V      66.51062 0.01083374 -0.005420032 -0.001761963
VI     74.16838 0.01083374 -0.005420032 -0.001761963

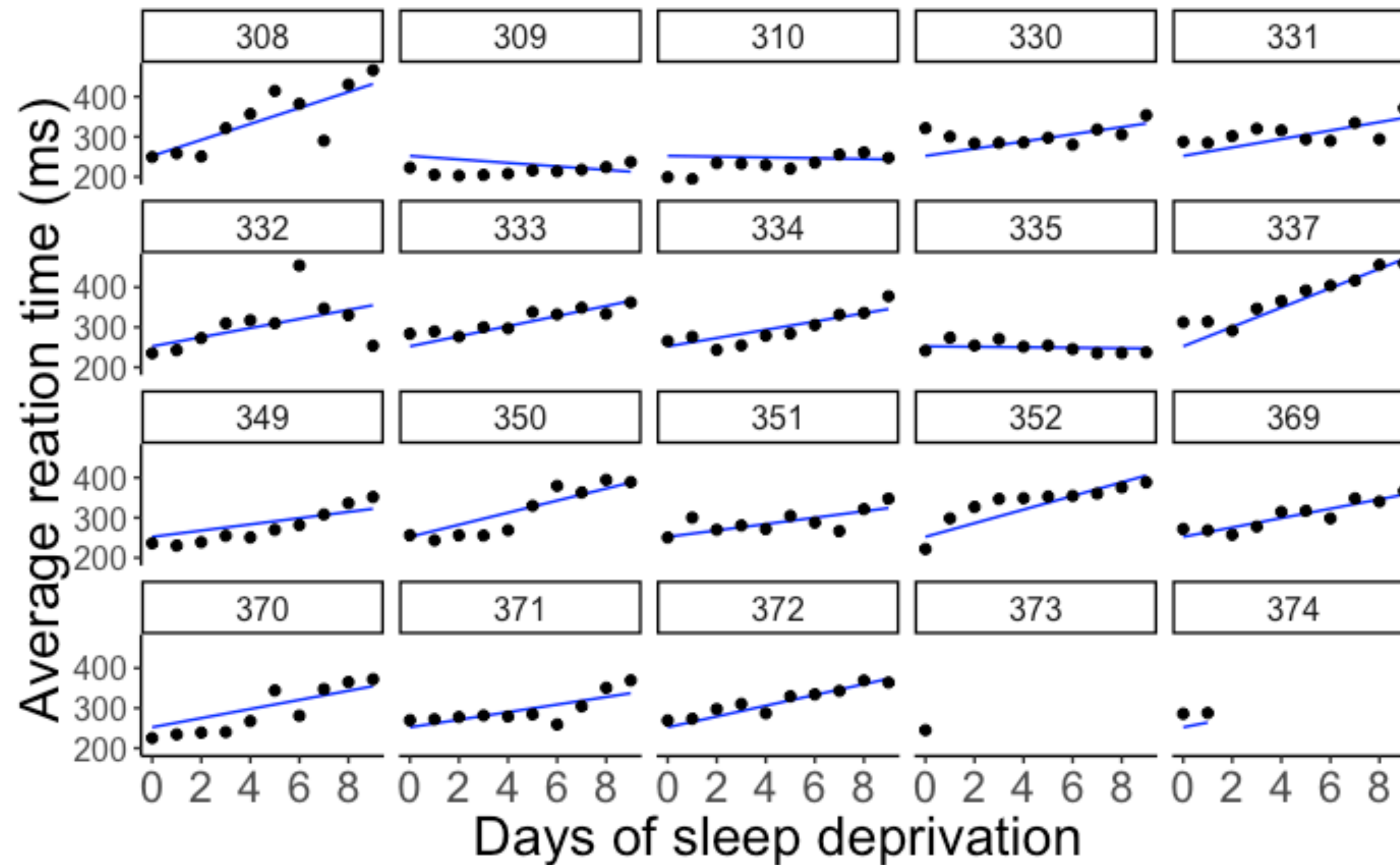
$class
  (Intercept)      open      agree      social
a      57.35175 0.01083374 -0.005420032 -0.001761963
b      59.39261 0.01083374 -0.005420032 -0.001761963
c      61.04758 0.01083374 -0.005420032 -0.001761963
d      63.00342 0.01083374 -0.005420032 -0.001761963
```

```
# model coefficients
fit.crossed %>% coef()
```


Reporting results

Reporting results

Visualization

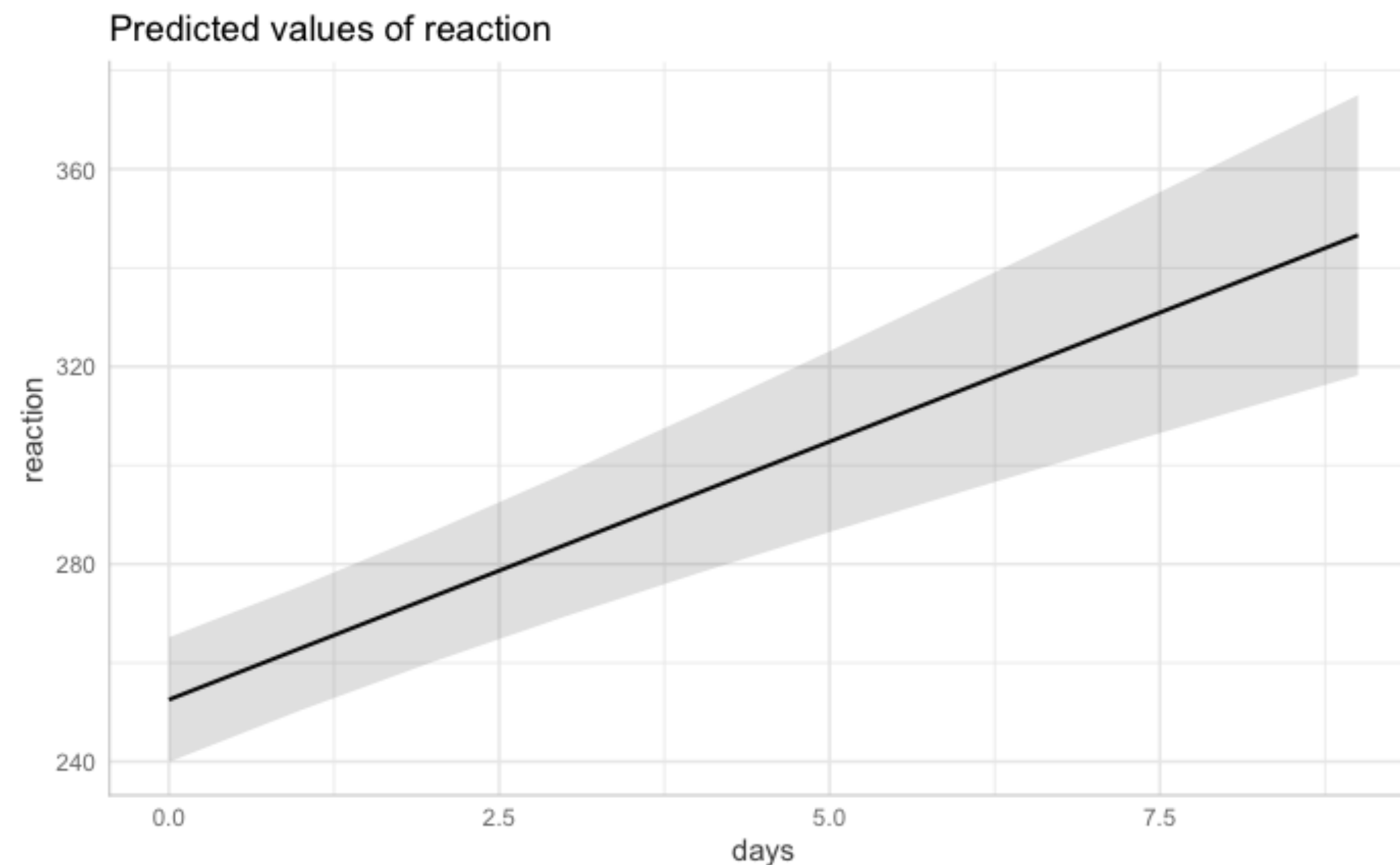


show the data together with the model predictions

Visualization



```
1 library("ggeffects")
2
3 ggpredict(model = fit.random_intercept_slope,
4           terms = "days",
5           type = "fe") %>%
6   plot()
```



- the relationship between the variables of interest (marginalizing over other variables)

show the (marginalized) model prediction

Reporting results

7.1. In Writing

Our reports include a description of the following parts (also see [Meteyard & Davies, 2019](#); [Barr et al., 2013](#)):

- Model specification, including:
 - Dependent variable, and all fixed and random effects (intercepts, slopes, correlations), both in words and possibly also by providing the model equation/ R-pseudo code (so-called Wilkinson notation)
 - Transformation of variables, e.g., standardizing or centering variables
 - Contrast coding (typically sum-to-zero coding)
- Inference:
 - Description of how p -values were obtained (in case of a frequentist approach) or what other (Bayesian) decision rule was used for inference.
 - Description of what post-hoc or follow-up tests were performed
 - Any convergence issues that may arise while running the model (in particular if they require adjustments in the model specification) and how they were dealt with should be described, as well as the subsequent adjustments that were made.
- Model output, at minimum the following:
 - Model results: (un)standardized regression coefficients, standard errors and/or confidence / credible intervals, test statistics, degrees of freedom, p -values

Meteyard, L. & Davies, R. A. I. (2020). Best practice guidance for linear mixed-effects models in psychological science. *Journal of Memory and Language*.

Reporting results

Including Random Effects
in the Table

Table 2
Results From Final Models Examining How Familiarity of the Social Context Influences Momentary Affect Valence and Affect Arousal, and the Moderating Role of Age Therein

Parameters	Parameter estimates (SE)	
	Affect valence	Affect arousal
Fixed effects		
Intercept, γ_{00}	77.5231* (.836)	62.8809* (.953)
Familiarity ecology, γ_{01}	.2571* (.077)	-.1005 (.092)
Familiarity situation, γ_{10}	.0805* (.007)	-.0199* (.008)
Familiarity situation², γ_{20}	.0004* (.0001)	-.0007* (.0001)
Age, γ_{02}	-.5752* (.234)	.2413 (.265)
Age ² , γ_{03}	.0064* (.002)	-.0014 (.003)
Age × familiarity ecology, γ_{04}	.0027 (.004)	.0033 (.005)
Age × familiarity situation, γ_{11}	.0003 (.0004)	.0005 (.0004)
Health, γ_{05}	.5364* (.2021)	.3569 (.230)
Familiarity ecology × familiarity situation, γ_{12}	—	-.0018* (.001)
Valence trait, γ_{06}	—	.6008* (.094)
Valence momentary, γ_{13}	—	.2797* (.006)
Familiarity ecology × valence momentary, γ_{14}	—	-.0044* (.0005)
Familiarity situation × valence trait, γ_{15}	—	.0012 (.001)
Familiarity situation × valence momentary, γ_{16}	—	-.0002 (.000)
Familiarity situation ² × valence momentary, γ_{17}	—	-.0000* (.0000)
Age × Familiarity ecology × valence momentary, γ_{18}	—	-.0003* (.000)
Age × Familiarity situation × valence momentary, γ_{19}	—	-.00002* (.000)
Random effects		
Variance intercept, σ_{u0}^2	103.50* (12.30)	134.02* (16.01)
Variance familiarity situation, σ_{u1}^2	.007* (.001)	.008* (.001)
Covariance intercept, familiarity situation, $\sigma_{u0, u1}$	-.096 (.074)	.29* (.097)
Autocorrelation, ar(1)	.21* (.004)	.29* (.004)
Residual variance, σ_e^2	190.16* (1.117)	272.08* (1.671)
-2LL	514,339	534,375
Note. SE = standard error. Data are reports on 64,213 social interactions nested within 150 persons. Bold lines indicate tests for the main hypotheses of interest. * $p < .05$.		

Let's simulate some `lmer()` s

Let's simulate an `lmer()`

```
1 # make example reproducible
2 set.seed(1)
3
4 # parameters
5 sample_size = 100
6 b0 = 1
7 b1 = 2
8 sd_residual = 1
9 sd_participant_intercept = 0.5
10
11 # generate the data
12 df.mixed = tibble(participant = rep(1:sample_size, 2),
13                   condition = rep(0:1, each = sample_size)) %>%
14   group_by(participant) %>%
15   mutate(participant_intercepts = rnorm(n = 1, sd = sd_participant_intercept)) %>%
16   ungroup() %>%
17   mutate(value = b0 + b1 * condition + participant_intercept + rnorm(n(), sd = sd_residual)) %>%
18   arrange(participant, condition)
```

participant	condition	participant_intercept	value
1	0	-0.31	0.07
1	1	-0.31	3.10
2	0	0.09	1.13
2	1	0.09	4.78
3	0	-0.42	-0.33
3	1	-0.42	4.17
4	0	0.80	1.96
4	1	0.80	3.47
5	0	0.16	0.51
5	1	0.16	0.88

$$\text{value}_{ij} = b_0 + b_1 \cdot \text{condition}_{ij} + U_i + e_{ij}$$

$$e_{ij} \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_{\text{error}})$$

$$U_i \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_U)$$

simulating data from a model and trying to recover the parameters is a great way to check one's understanding of what the model does

Let's simulate an `lmer()`

```
1 # make example reproducible
2 set.seed(1)
3
4 # parameters
5 sample_size = 100
6 b0 = 1
7 b1 = 2
8 sd_residual = 1
9 sd_participant_intercept = 0.5
10
11 # generate the data
12 df.mixed = tibble(participant = rep(1:sample_size, 2),
13                   condition = rep(0:1, each = sample_size)) %>%
14   group_by(participant) %>%
15   mutate(participant_intercept = rnorm(n = 1, sd = sd_participant_intercept)) %>%
16   ungroup() %>%
17   mutate(value = b0 + b1 * condition + participant_intercept + rnorm(n(), sd = sd_residual)) %>%
18   arrange(participant, condition)
```

```
1 # fit model
2 fit.test = lmer(formula = value ~ 1 + condition + (1 | participant),
3                 data = df.mixed)
4
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
Data: df.mixed

REML criterion at convergence: 606

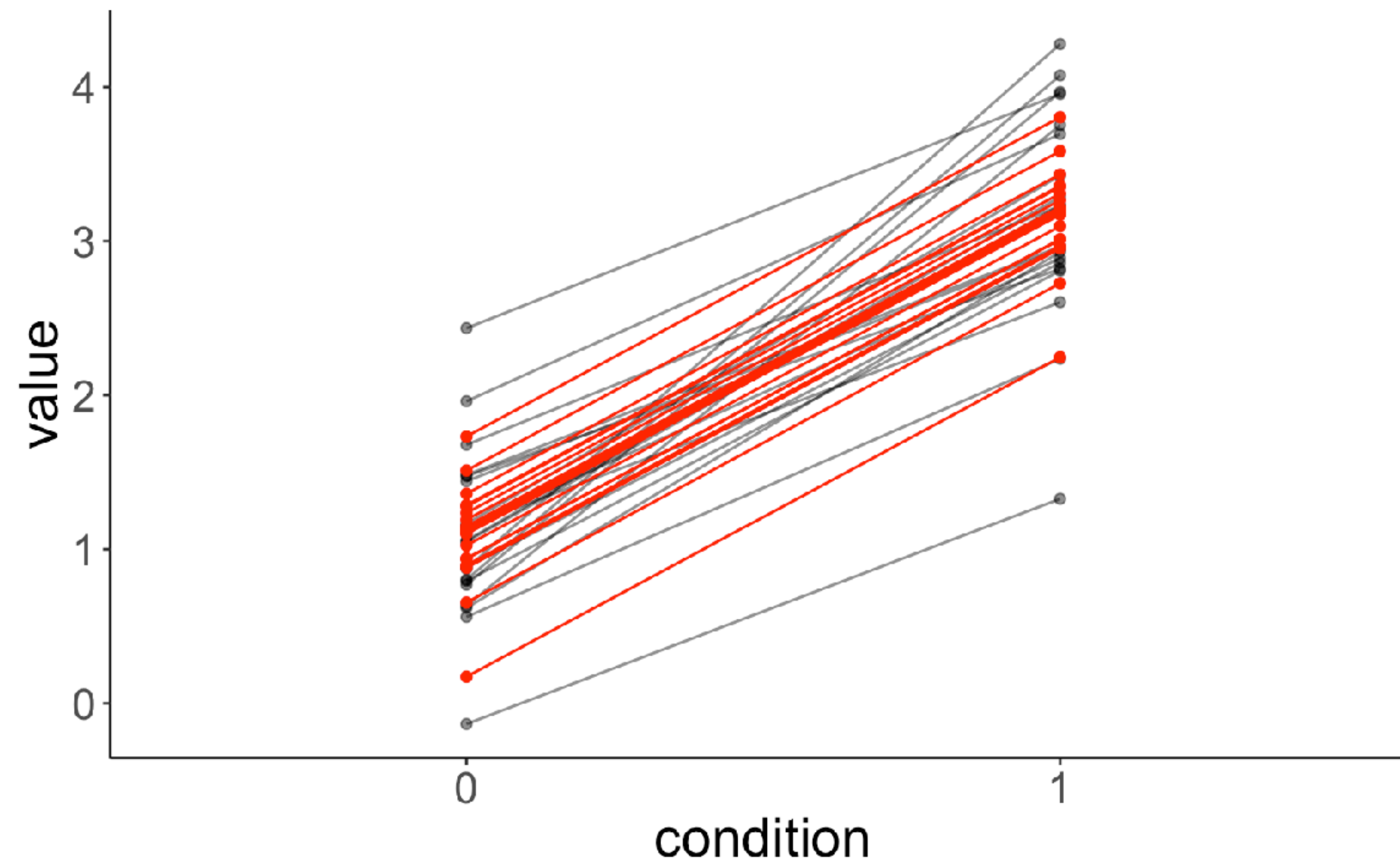
Scaled residuals:
    Min       1Q   Median       3Q      Max
-2.53710 -0.62295 -0.04364  0.67035  2.19899

Random effects:
 Groups      Name      Variance Std.Dev.
participant (Intercept) 0.1607   0.4009
Residual                1.0427   1.0211
Number of obs: 200, groups: participant, 100

Fixed effects:
              Estimate Std. Error t value
(Intercept)   1.0166    0.1097    9.267
condition      2.0675    0.1444   14.317

Correlation of Fixed Effects:
              (Intr)
condition -0.658
```


No outlier



```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
Data: df.test

REML criterion at convergence: 74.9

Scaled residuals:
    Min       1Q   Median       3Q      Max
-1.9268 -0.5412 -0.1103  0.4868  1.7747

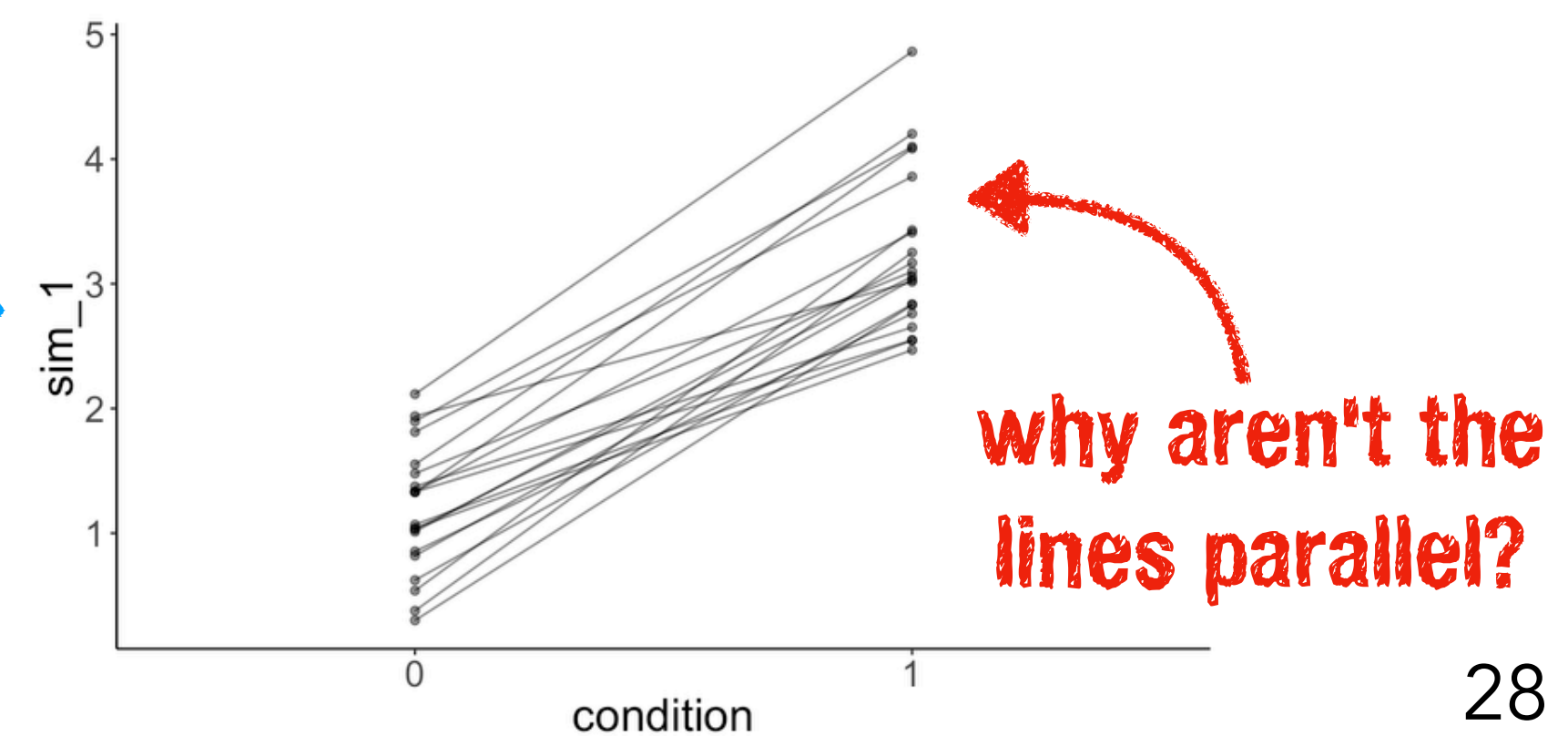
Random effects:
 Groups      Name      Variance Std.Dev.
participant (Intercept) 0.1702   0.4125
Residual                0.2270   0.4764
Number of obs: 40, groups: participant, 20

Fixed effects:
              Estimate Std. Error t value
(Intercept)   1.0920     0.1409    7.75
condition1    2.0726     0.1507   13.76

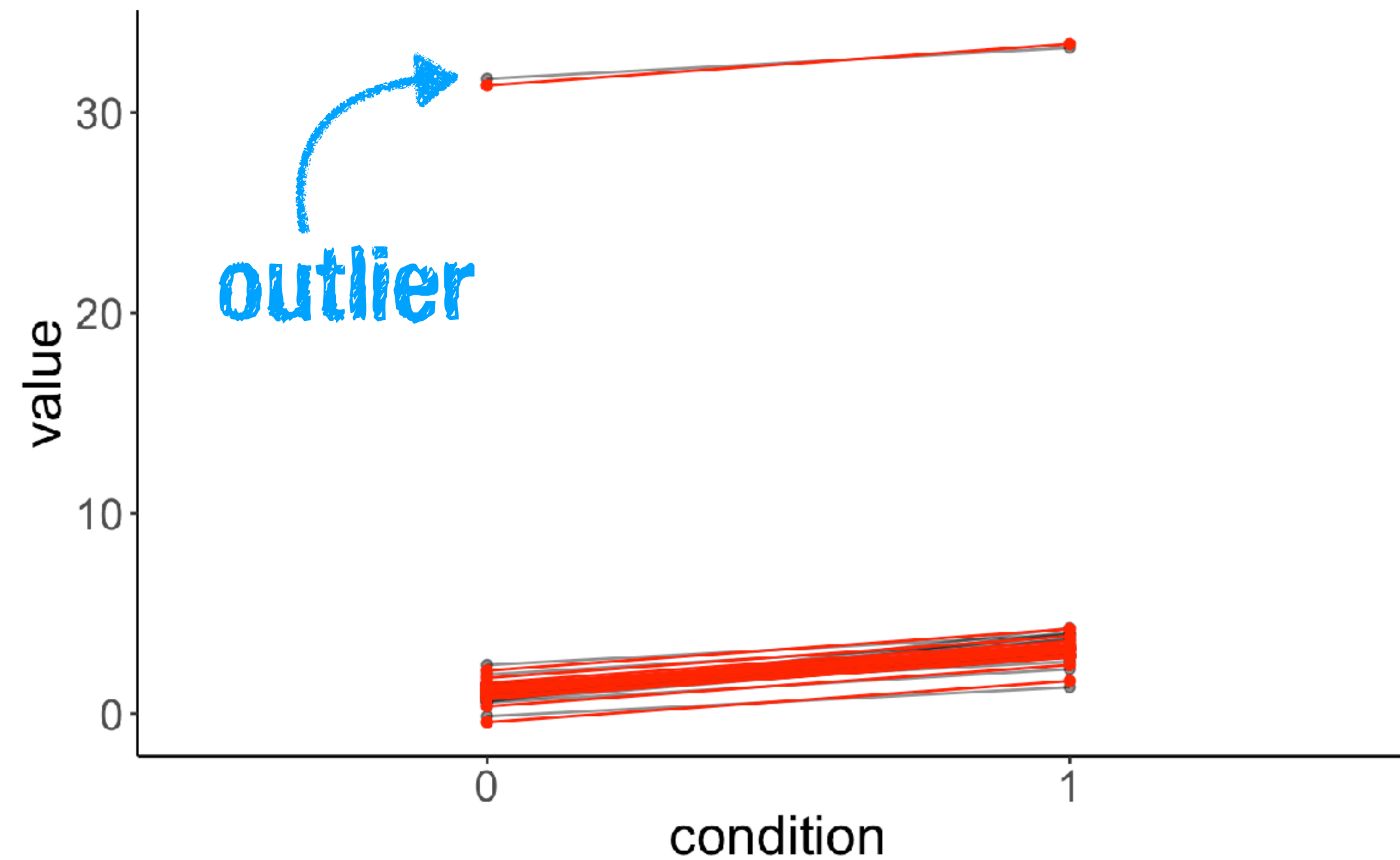
Correlation of Fixed Effects:
              (Intr)
condition1 -0.535
```

```
1 # fit model
2 fit.test = lmer(formula = value ~ 1 + condition + (1 | participant),
3                 data = df.test)
4
5 # simulate data
6 fit.test %>%
7   simulate()
```

simulated data



With outlier



```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
Data: df.test

REML criterion at convergence: 171.7

Scaled residuals:
    Min       1Q   Median       3Q      Max
-1.4038 -0.4678 -0.0094  0.5800  1.3930

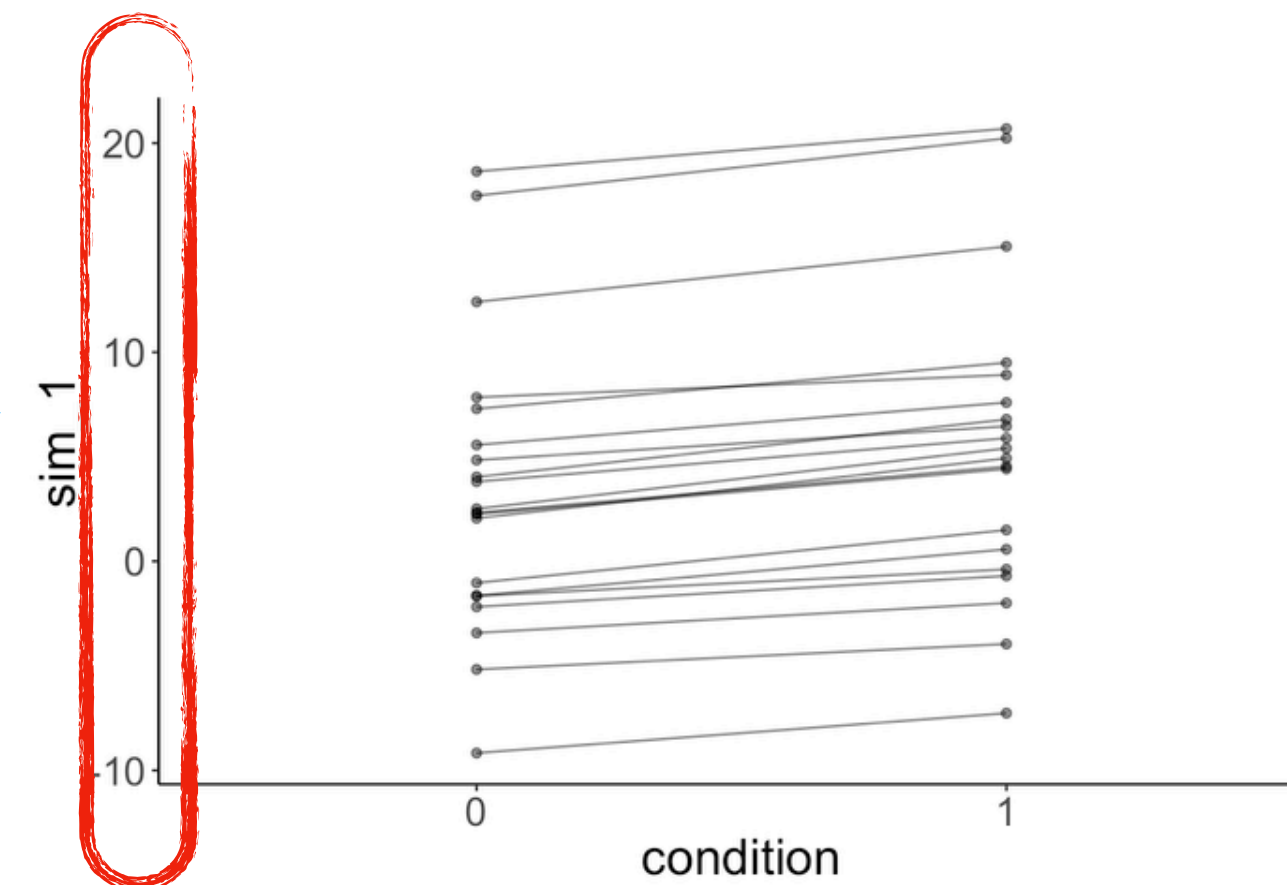
Random effects:
Groups      Name      Variance Std.Dev.
participant (Intercept) 46.198   6.7969
Residual    0.227    0.4764
Number of obs: 40, groups: participant, 20

Fixed effects:
              Estimate Std. Error t value
(Intercept)   2.5920     1.5236   1.701
condition1    2.0726     0.1507  13.758

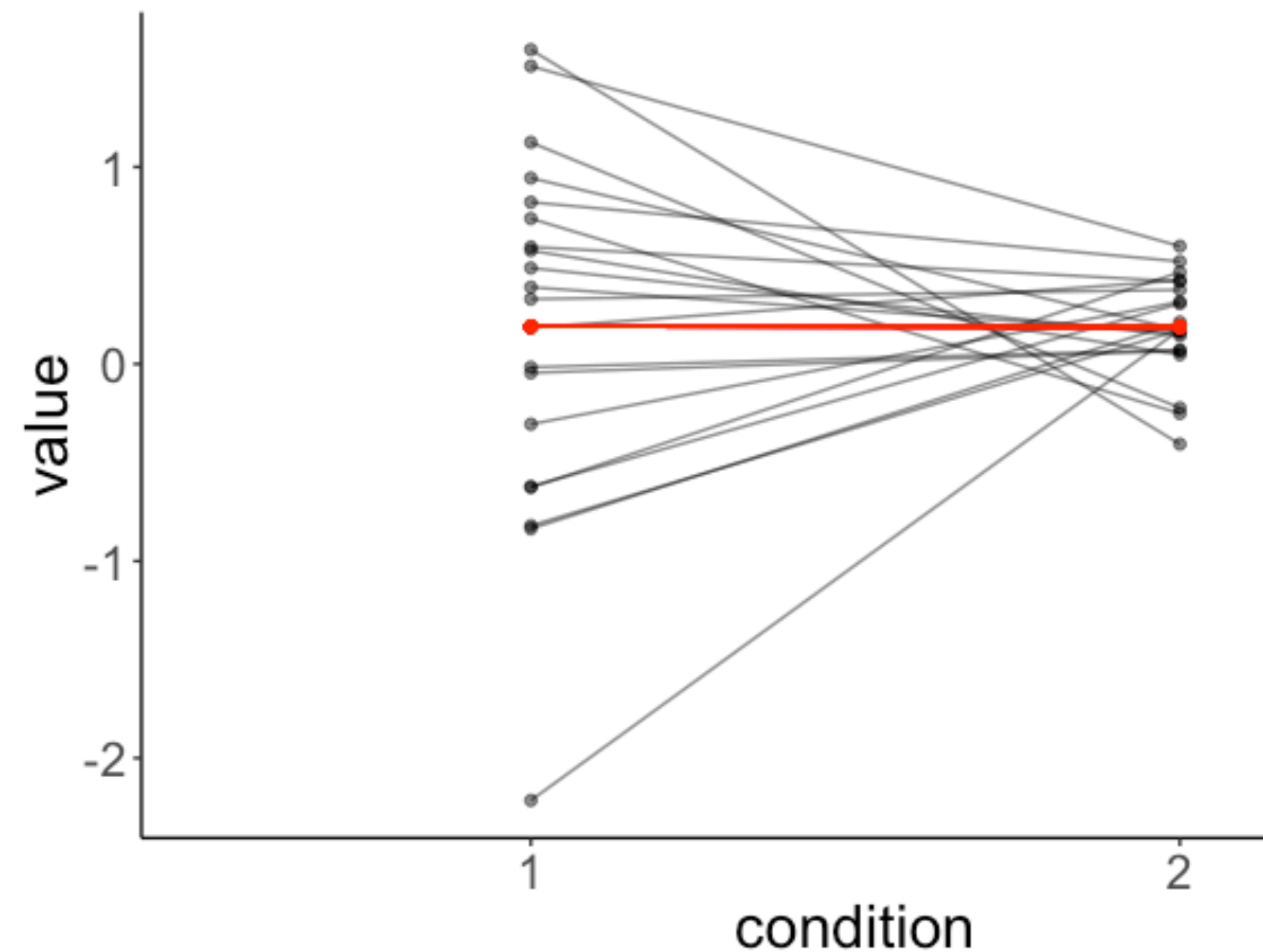
Correlation of Fixed Effects:
          (Intr)
condition1 -0.049
```

```
1 # fit model
2 fit.test = lmer(formula = value ~ 1 + condition + (1 | participant),
3                 data = df.test)
4
5 # simulate data
6 fit.test %>%
7   simulate()
```

simulated data



Non-equal variance



singular fit
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
Data: df.test

REML criterion at convergence: 83.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.5808	-0.3184	0.0130	0.4551	2.0913

Random effects:

Groups	Name	Variance	Std.Dev.
participant	(Intercept)	0.0000	0.0000
	Residual	0.4512	0.6717

Number of obs: 40, groups: participant, 20

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	0.190524	0.150197	1.268
condition2	-0.001941	0.212411	-0.009

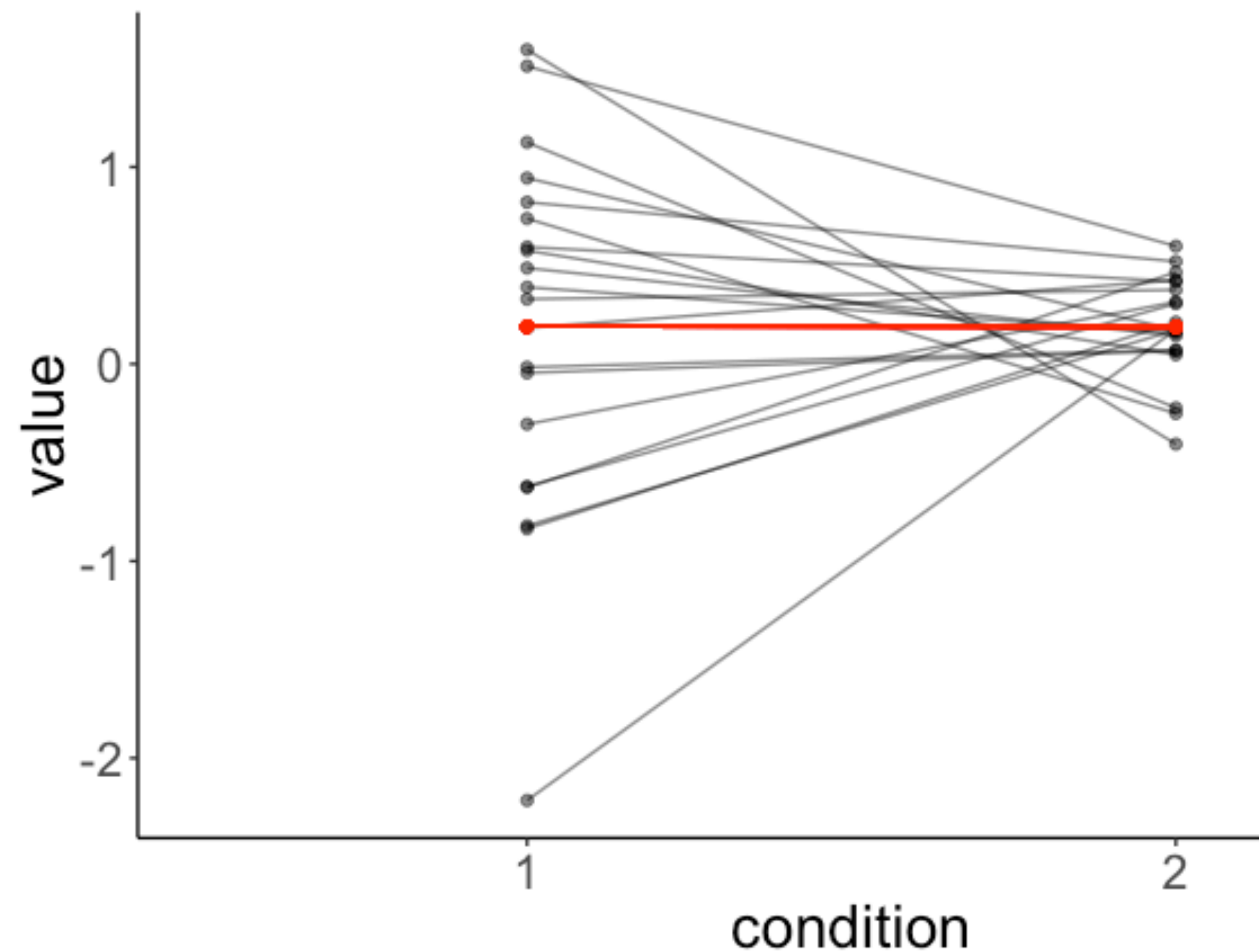
Correlation of Fixed Effects:

	(Intr)
condition2	-0.707

convergence code: 0
singular fit

clearly there are interindividual differences though?!

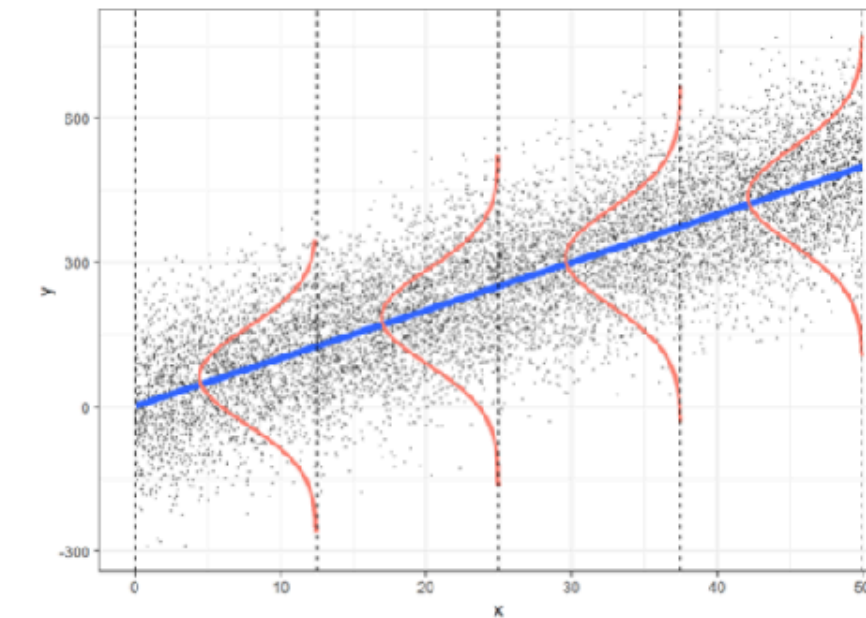
Non-equal variance



Model assumptions of simple regression

- independent observations
- Y is continuous
- errors are normally distributed
- errors have constant variance
- error terms are uncorrelated

**assumption
violated**



**the "model" would
just reproduce the
data**

```
1 # fit model
2 lmer(formula = value ~ 1 + condition + (1 + condition | participant),
3       data = df.test)
```

random intercept

random slope

won't work

Error: number of observations (=40) <= number of random effects (=40) for term (1 + condition | participant); the random-effects parameters and the residual variance (or scale parameter) are probably unidentifiable

Some more examples

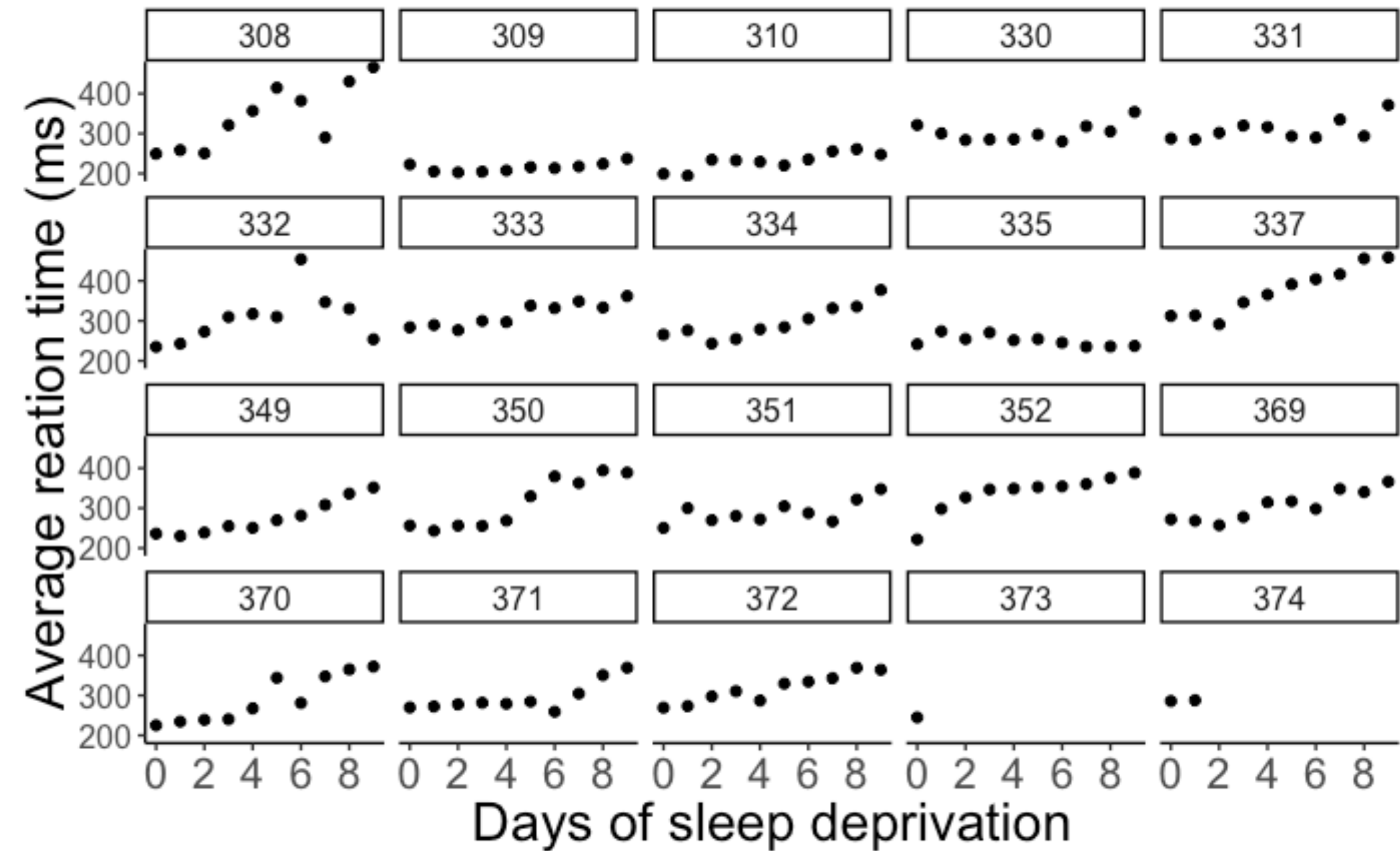
1. Sleep



Sleep data

How does sleep deprivation affect reaction time?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72



20 participants

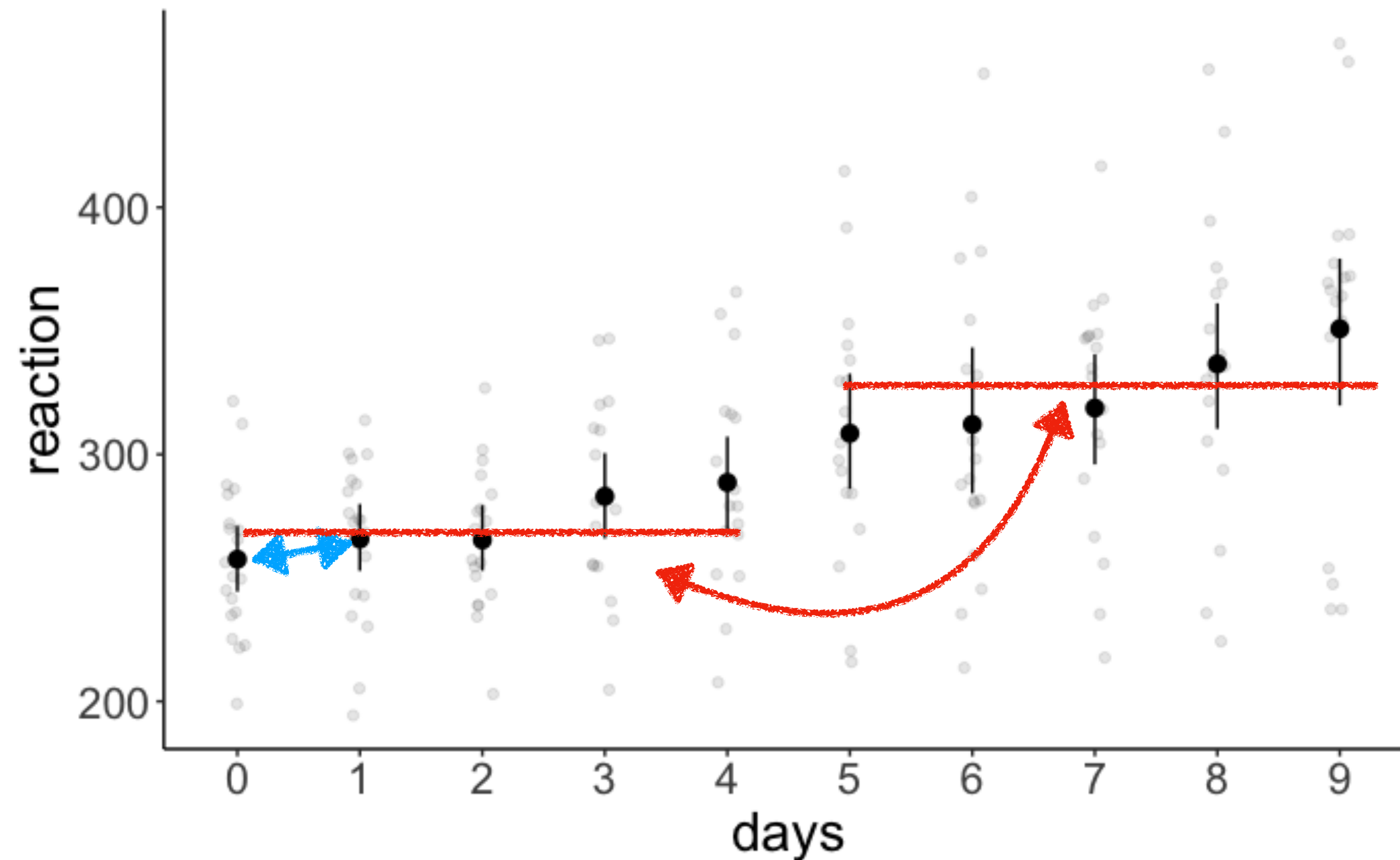
2 with incomplete information

Testing specific hypotheses with linear contrasts

1. Is there a significant difference between day 0 and day 1?

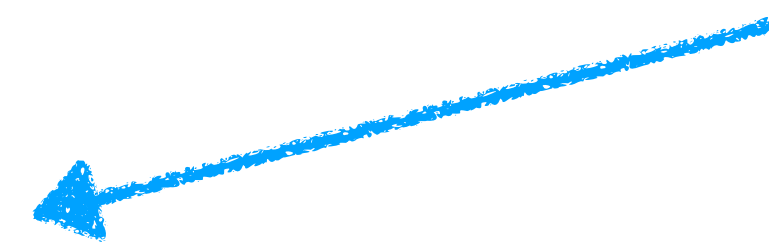
2. Is there a significant difference between the days 0-4 and days 5-9?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72



Sleep data

fit the model



```
1 fit = lmer(formula = reaction ~ 1 + days + (1 | subject),  
2           data = df.sleep %>%  
3           mutate(days = as.factor(days)))
```

Sleep data

fit the model

```
1 fit = lmer(formula = reaction ~ 1 + days + (1 | subject),
2           data = df.sleep %>%
3             mutate(days = as.factor(days)))
4
5 contrast = list(first_vs_second = c(-1, 1, rep(0, 8)),
6               early_vs_late = c(rep(-1, 5)/5, rep(1, 5)/5))
7
8 fit %>%
9   emmeans(specs = "days",
10          contr = contrast) %>%
11   pluck("contrasts")
```

define the contrasts

test the contrasts

contrast	estimate	SE	df	t.ratio	p.value
first_vs_second	7.82	10.10	156	0.775	0.4398
early_vs_late	53.66	4.65	155	11.534	<.0001

Degrees-of-freedom method: kenward-roger

days	reaction
0	257.54
1	265.73

index	reaction
early	271.67
late	325.39

2. Weight loss



Weight loss data

id	diet	exercises	timepoint	score
1	no	no	t1	10.43
1	no	no	t2	13.21
1	no	no	t3	11.59
1	yes	no	t1	10.20
1	yes	no	t2	12.51
1	yes	no	t3	14.60
2	no	no	t1	11.59
2	no	no	t2	10.66
2	no	no	t3	13.21
2	yes	no	t1	12.98
2	yes	no	t2	12.98
2	yes	no	t3	14.60

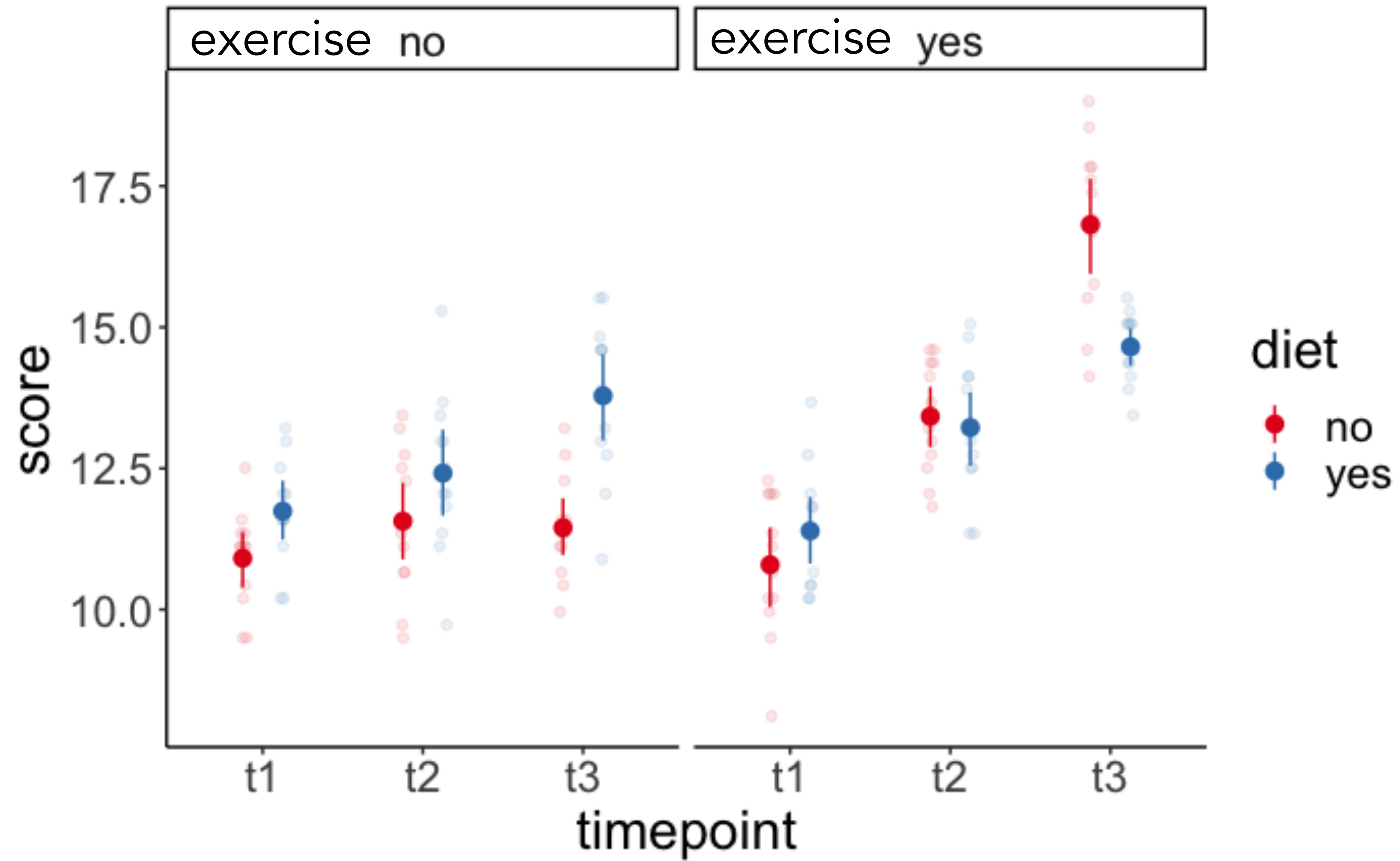
between participants: exercise yes/no

within participants: diet yes/no

within participants: time points

**one observation in each cell, so
we can use an ANOVA**

Weight loss data



Weight loss data

```
1 fit = aov_ez(id = "id",
2             dv = "score",
3             between = "exercises",
4             within = c("diet", "timepoint"),
5             data = df.weightloss)
```

id	diet	exercises	timepoint	score
1	no	no	t1	10.43
1	no	no	t2	13.21
1	no	no	t3	11.59
1	yes	no	t1	10.20
1	yes	no	t2	12.51
1	yes	no	t3	14.60
2	no	no	t1	11.59

Anova Table (Type 3 tests)									
Response: score									
	Effect		df	MSE		F	ges	p.value	
1	exercises		1, 22	1.84	38.77	***	.284	<.001	
2	diet		1, 22	0.65	7.91	*	.028	.010	
3	exercises:diet		1, 22	0.65	51.70	***	.157	<.001	
4	timepoint	1.74,	38.26	1.48	82.20	***	.541	<.001	
5	exercises:timepoint	1.74,	38.26	1.48	26.22	***	.274	<.001	
6	diet:timepoint	1.61,	35.44	1.92	0.78		.013	.439	
7	exercises:diet:timepoint	1.61,	35.44	1.92	9.97	***	.147	<.001	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 '.' 1									

main effects and interactions

Weight loss data

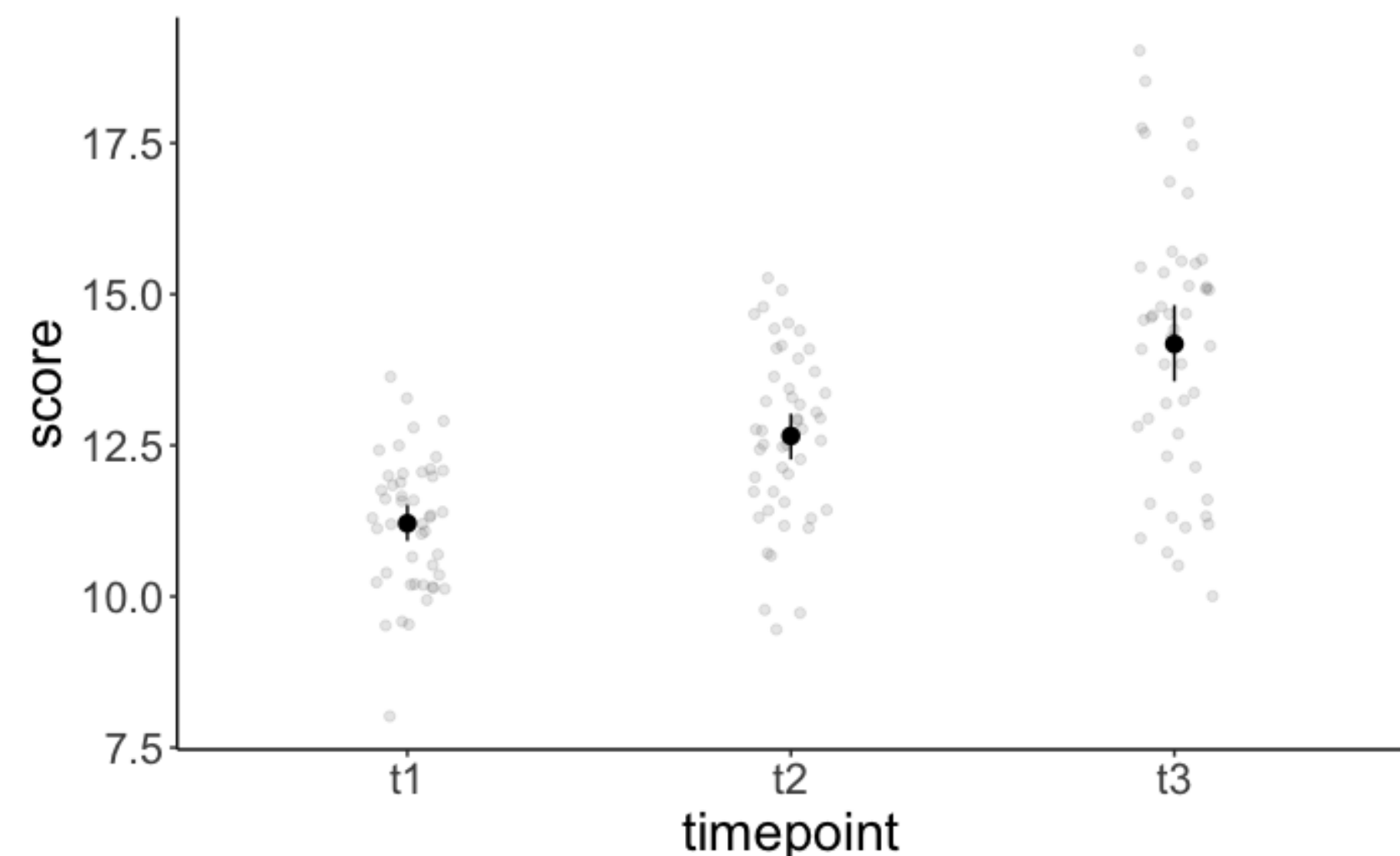
1. Is the score at the third time point different from the other two time points?
2. Is there a linear increase across time points?

```
1 fit = aov_ez(id = "id",  
2             dv = "score",  
3             between = "exercises",  
4             within = c("diet", "timepoint"),  
5             data = df.weightloss)  
6  
7 contrasts = list(first_two_vs_last = c(-0.5, -0.5, 1),  
8                 linear_increase = c(-1, 0, 1))  
9  
10 fit %>%  
11   emmeans(spec = "timepoint",  
12          contr = contrasts)
```

contrast	estimate	SE	df	t.ratio	p.value
first_two_vs_last	2.24	0.200	44	11.194	<.0001
linear_increase	2.97	0.231	44	12.820	<.0001

df.weightloss

id	diet	exercises	timepoint	score
1	no	no	t1	10.43
1	no	no	t2	13.21
1	no	no	t3	11.59
1	yes	no	t1	10.20
1	yes	no	t2	12.51
1	yes	no	t3	14.60
2	no	no	t1	11.59



3. politeness



Politeness

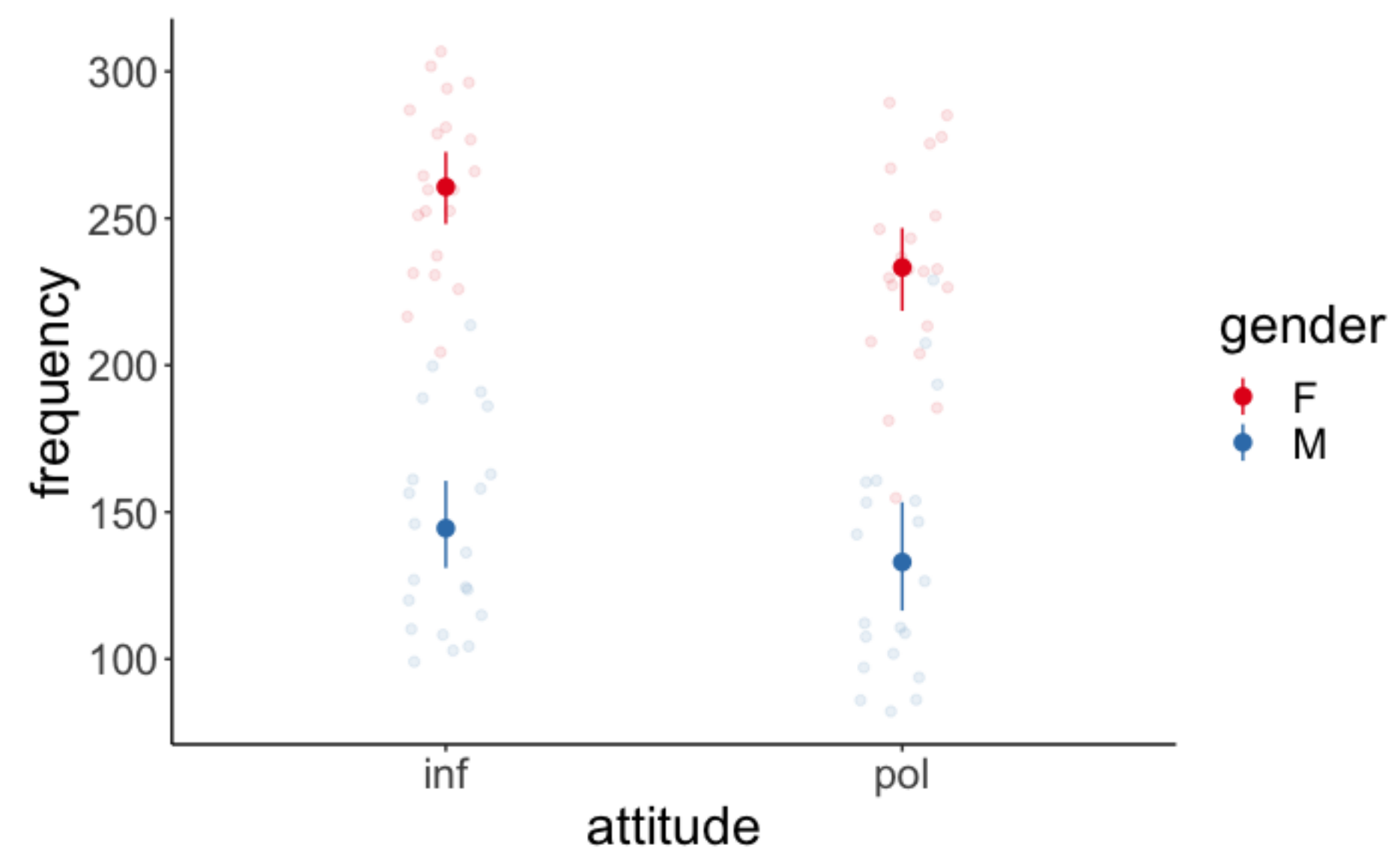
subject	gender	scenario	attitude	frequency
F1	F	1	pol	213.3
F1	F	1	inf	204.5
F1	F	2	pol	285.1
F1	F	2	inf	259.7
F1	F	3	pol	203.9
F1	F	3	inf	286.9
F1	F	4	pol	250.8
F1	F	4	inf	276.8
F1	F	5	pol	231.9
F1	F	5	inf	252.4
F1	F	6	pol	181.2
F1	F	6	inf	230.7
F1	F	7	inf	216.5
F1	F	7	pol	154.8
F3	F	1	pol	229.7

gender: female, male

scenario: different text prompt

attitude: informal vs. polite

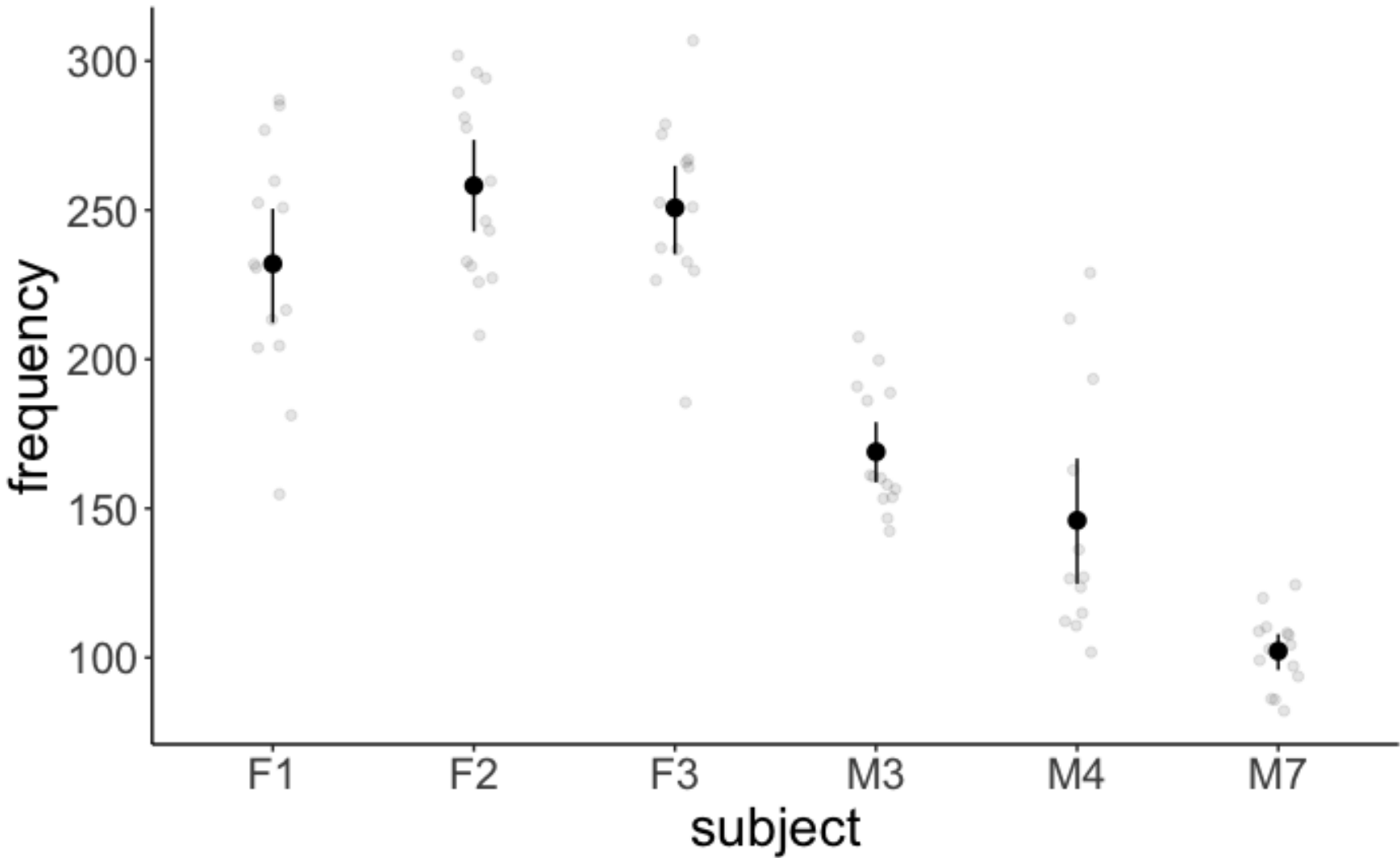
frequency: pitch of voice



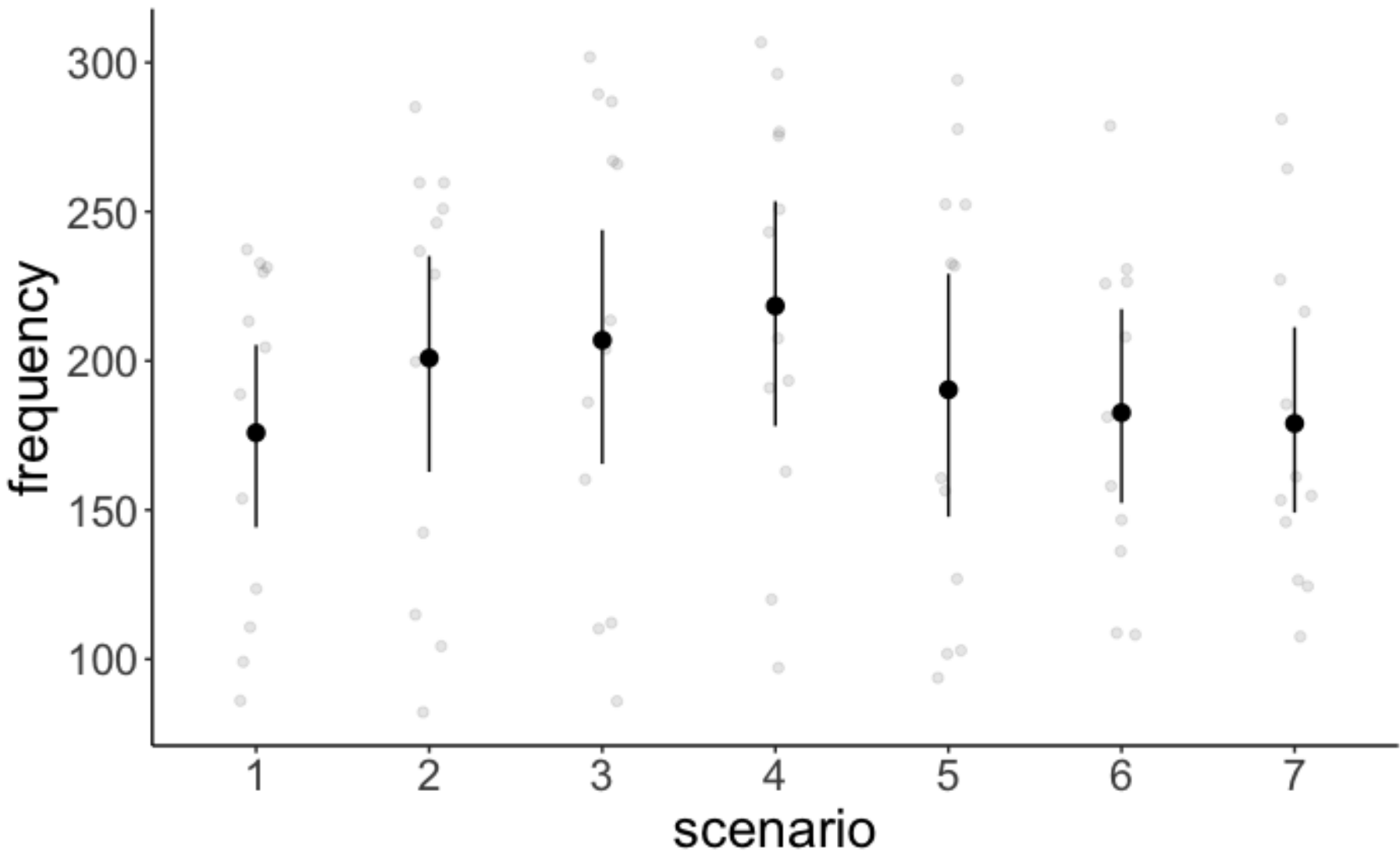
Politeness

subject	gender	scenario	attitude	frequency
F1	F	1	pol	213.3
F1	F	1	inf	204.5
F1	F	2	pol	285.1
F1	F	2	inf	259.7
F1	F	3	pol	203.9
F1	F	3	inf	286.9
F1	F	4	pol	250.8
F1	F	4	inf	276.8
F1	F	5	pol	231.9
F1	F	5	inf	252.4
F1	F	6	pol	181.2
F1	F	6	inf	230.7
F1	F	7	inf	216.5
F1	F	7	pol	154.8
F3	F	1	pol	229.7

variation across subjects



variation across scenarios

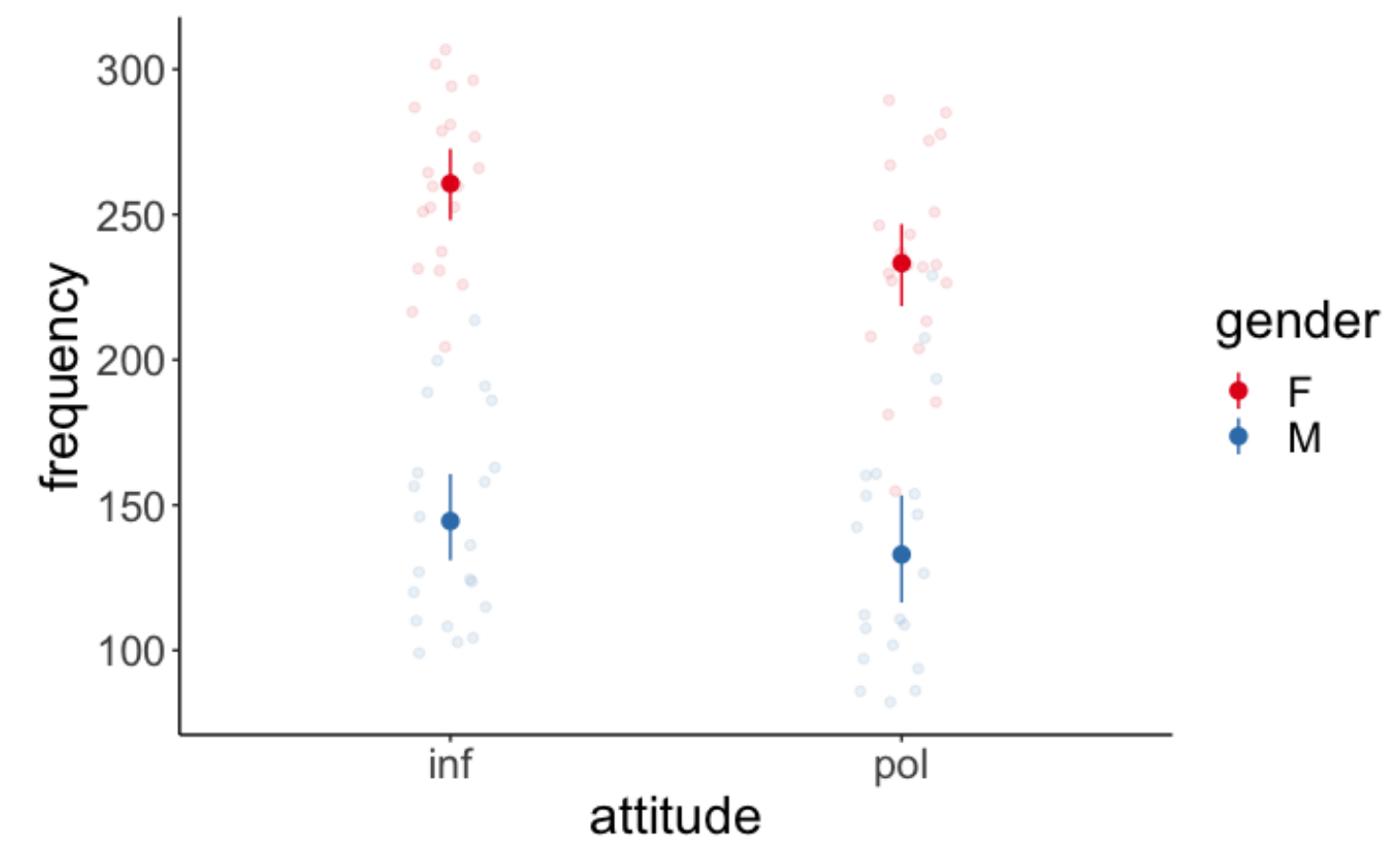


Politeness

Was there an effect of gender and attitude on pitch?

```
1 lmer(formula = frequency ~ 1 + attitude * gender + (1 + attitude | subject) + (1 | scenario),  
2     data = df.politeness) %>%  
3   joint_tests()
```

model	term	df1	df2	F.ratio	p.value
	attitude	1	3.99	12.411	0.0244
	gender	1	4.00	26.570	0.0067
	attitude:gender	1	3.99	1.959	0.2342



main effect of attitude, main effect of gender, no significant interaction effect

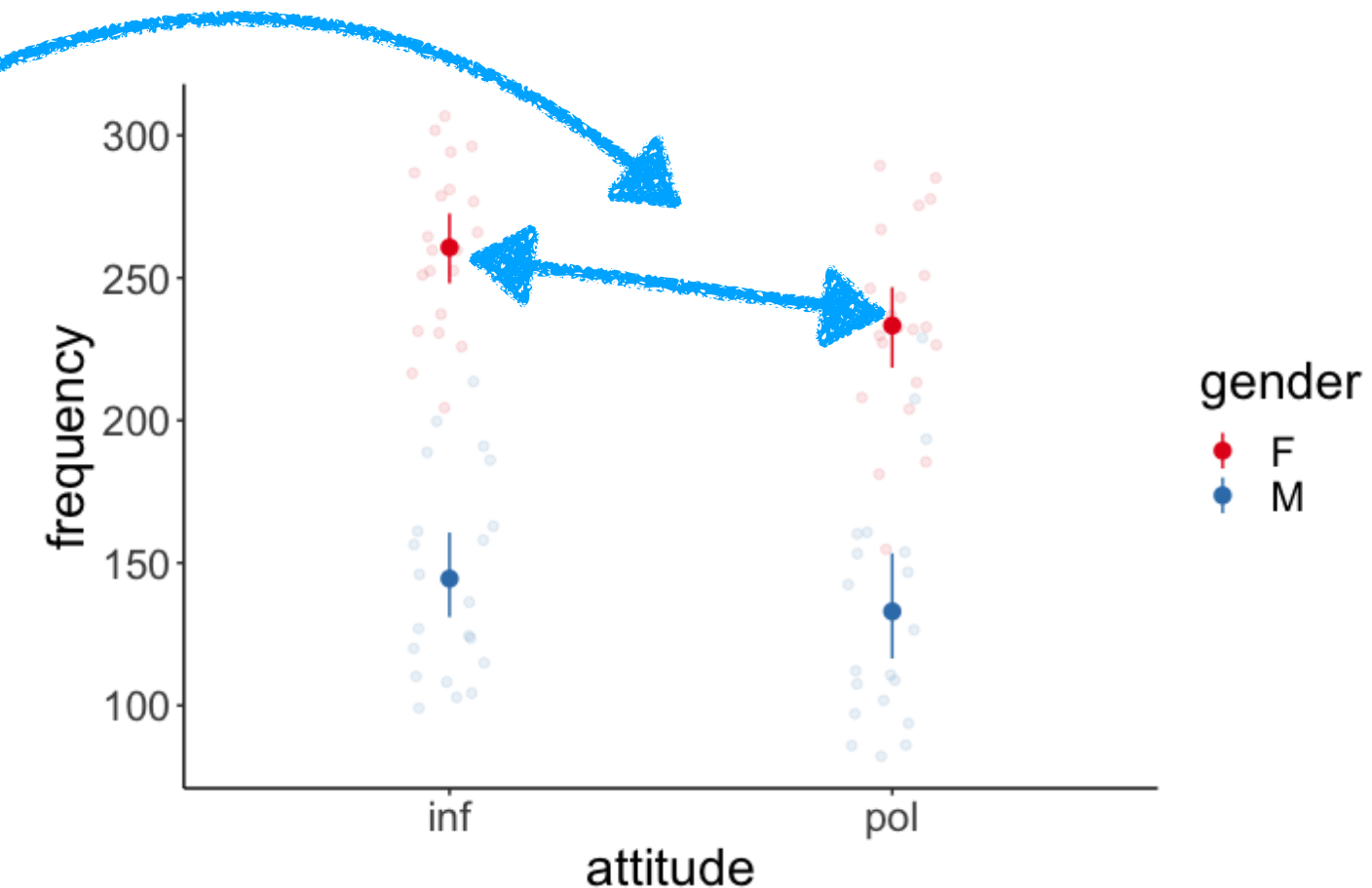
Politeness

Was there a difference between informal and polite speech for female participants?

```
1 fit = lmer(formula = frequency ~ 1 + attitude * gender + (1 | subject) + (1 | scenario),  
2           data = df.politeness)  
3  
4 fit %>%  
5   emmeans(specs = pairwise ~ attitude + gender,  
6           adjust = "none")
```

contrast	estimate	SE	df	t.ratio	p.value
inf F - pol F	27.4	7.81	3.89	3.508	0.0259
inf F - inf M	116.2	21.35	4.00	5.443	0.0055
inf F - pol M	128.0	21.78	4.57	5.879	0.0027
pol F - inf M	88.8	21.73	4.54	4.086	0.0116
pol F - pol M	100.6	22.16	4.00	4.541	0.0105
inf M - pol M	11.8	7.93	4.10	1.490	0.2088

Degrees-of-freedom method: kenward-roger



yes, there was significant difference in pitch for women between informal and formal speech

Politeness

Was there an effect of gender and attitude on pitch?

ANOVA

```
1 aov_ez(id = "subject",
2       dv = "frequency",
3       between = "gender",
4       within = "attitude",
5       data = df.politeness)
```

```
More than one observation per cell, aggregating the data using mean (i.e.,
fun_aggregate = mean)!Missing values for following ID(s):
M4
Removing those cases from the analysis.Anova Table (Type 3 tests)

Response: frequency
      Effect    df    MSE      F ges p.value
1      gender 1, 3 1729.42   17.22 * .851   .025
2      attitude 1, 3   3.65 309.71 *** .179  <.001
3 gender:attitude 1, 3   3.65  21.30 * .015   .019
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ignores variation between scenarios,
and just takes the mean

interaction effect

LMER

```
1 lmer(formula = frequency ~ 1 + attitude * gender +
2       (1 + attitude | subject) + (1 | scenario),
3       data = df.politeness) %>%
  joint_tests()
```

model term	df1	df2	F.ratio	p.value
attitude	1	3.99	12.411	0.0244
gender	1	4.00	26.570	0.0067
attitude:gender	1	3.99	1.959	0.2342

no interaction effect

Different random effect structures



Contents

Data format

Power analysis, and simulating these models

Longitudinal two-level model ▾

Three-level models ▾

More on level 1 specification ▾

Hypothesis tests ▾

Book recommendations

Suggestions, errors or typos

Using R and lme/lmer to fit different two- and three-level longitudinal models

April 21, 2015

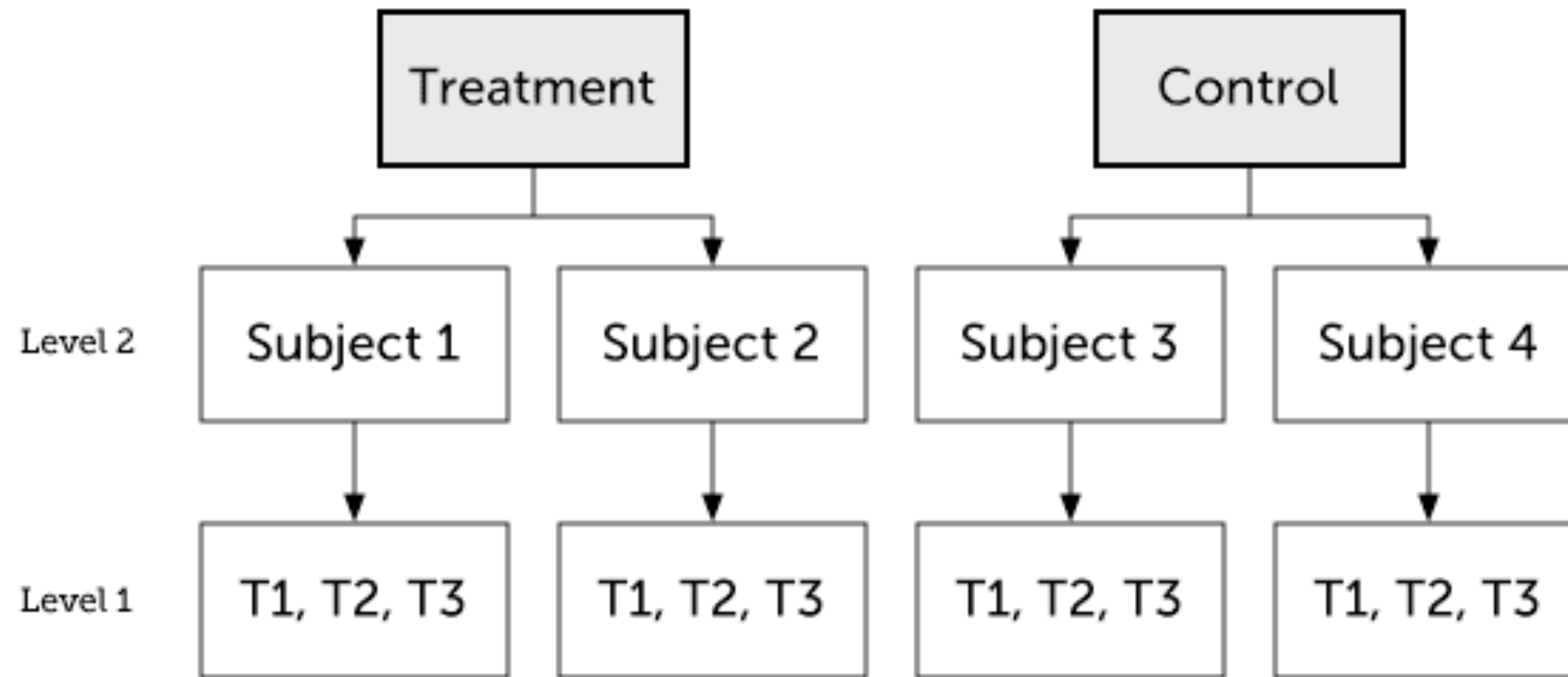


I often get asked how to fit different multilevel models (or individual growth models, hierarchical linear models or linear mixed-models, etc.) in R. In this guide I have compiled some of the more common and/or useful models (at least common in clinical psychology), and how to fit them using `nlme::lme()` and `lme4::lmer()`. I will cover the common two-level random intercept-slope model, and three-level models when subjects are clustered due to some higher level grouping (such as therapists), partially nested models where there are clustering in one group but not the other, and different level 1 residual covariances (such as AR(1)). The point of this post is to show how to fit these longitudinal models in R, not to cover the statistical theory behind them, or how to interpret them.

<https://rpsychologist.com/r-guide-longitudinal-lme-lmer#power-analysis-and-simulating-these-models>

Two-level model

Graphical representation



Simulate data

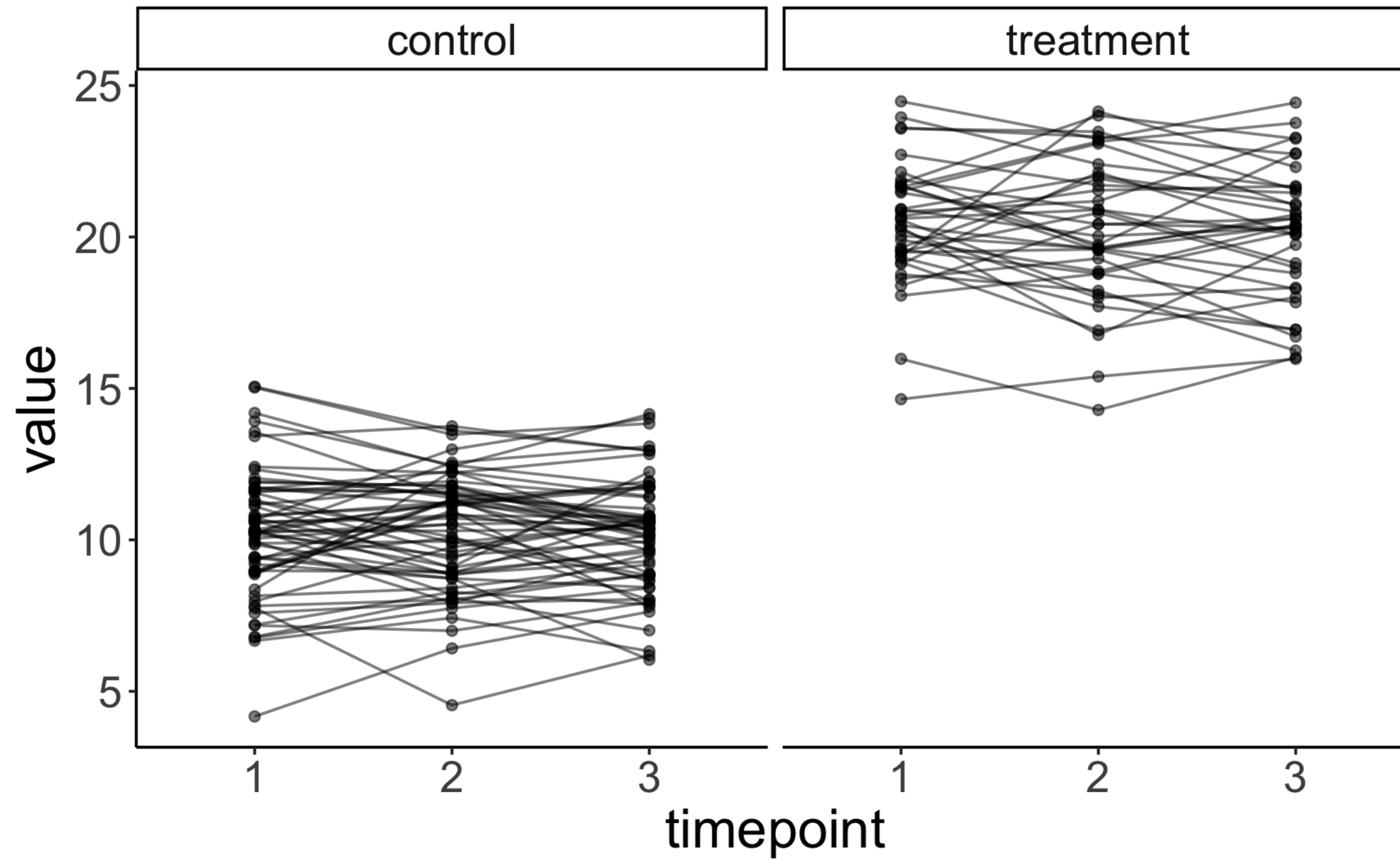
```

1 set.seed(1)
2
3 n_participants = 100
4 n_timepoints = 3
5 n_conditions = 2
6 p_condition = 0.5
7 b0 = 10
8 b1 = 10
9 sd_participant = 2
10 sd_residual = 1
11
12 df.data = tibble(participant = rep(1:n_participants, each = n_timepoints),
13                  timepoint = rep(1:n_timepoints, times = n_participants),
14                  intercept_participant = rep(rnorm(n_participants, sd = sd_participant),
15                                                each = n_timepoints)) %>%
16   group_by(participant) %>%
17   mutate(condition = rbinom(n = 1, size = 1, prob = p_condition)) %>%
18   ungroup() %>%
19   mutate(value = b0 + b1 * condition + intercept_participant +
20            rnorm(n_participants * n_timepoints, sd = sd_residual))

```

	participant	timepoint	intercept_participant	condition	value
1	1	1	-1.25290762	0	9.197279
2	1	2	-1.25290762	0	8.728533
3	1	3	-1.25290762	0	8.429024
4	2	1	0.36728665	0	9.437925
5	2	2	0.36728665	0	8.879826
6	2	3	0.36728665	0	9.292094
7	3	1	-1.67125722	1	19.328772
8	3	2	-1.67125722	1	17.707476
9	3	3	-1.67125722	1	16.944316
10	4	1	3.19056160	0	15.059852
11	4	2	3.19056160	0	13.615662
12	4	3	3.19056160	0	12.951915

Plot data



Fit the model

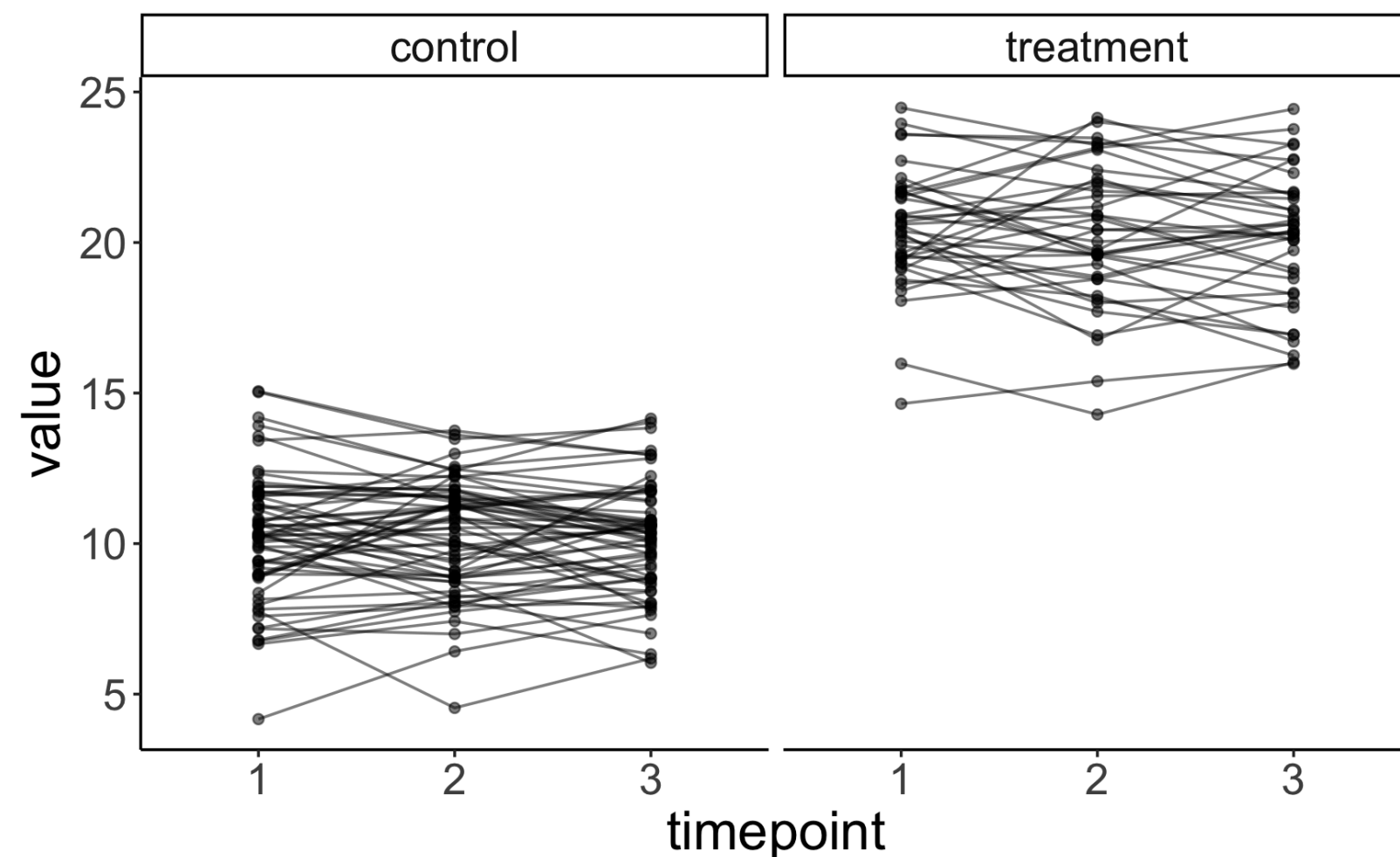
```
1 fit = lmer(formula = value ~ 1 + condition + (1 | participant),  
2           data = df.data)
```

```
Linear mixed model fit by REML ['lmerMod']  
Formula: value ~ 1 + condition + (1 | participant)  
Data: df.data  
  
REML criterion at convergence: 1102  
  
Scaled residuals:  
      Min       1Q   Median       3Q      Max  
-2.30522 -0.57146  0.03152  0.56826  2.28135  
  
Random effects:  
Groups      Name      Variance Std.Dev.  
participant (Intercept) 3.106    1.762  
Residual      1.087    1.043  
Number of obs: 300, groups: participant, 100  
  
Fixed effects:  
              Estimate Std. Error t value  
(Intercept)  10.2199    0.2365   43.21  
condition    10.0461    0.3837   26.18  
  
Correlation of Fixed Effects:  
              (Intr)  
condition -0.616
```

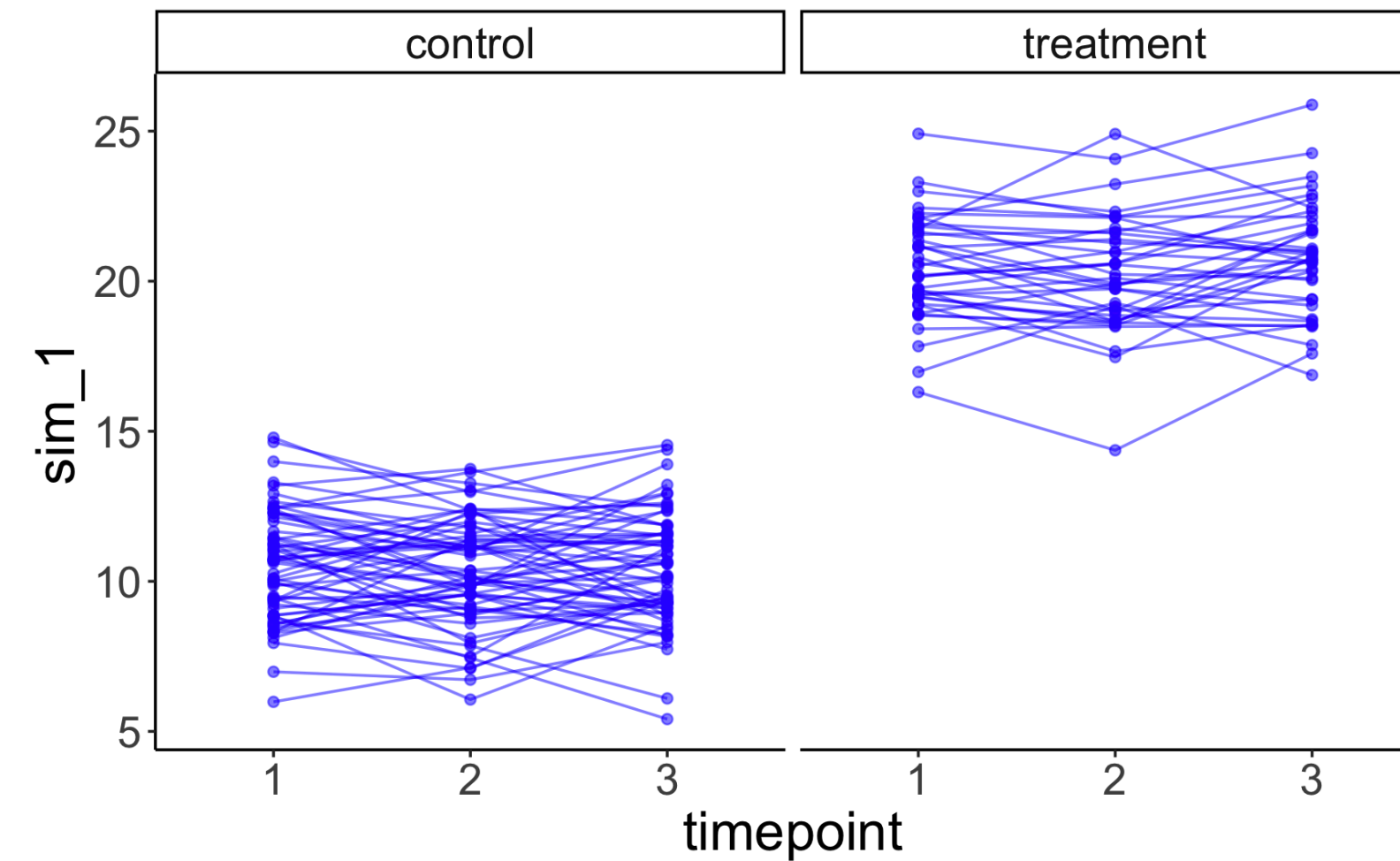
Simulate new data from the model

```
1 set.seed(1)
2
3 fit %>%
4   simulate() %>%
5   bind_cols(df.data) %>%
6   mutate(condition = factor(condition,
7                               levels = c(0, 1),
8                               labels = c("control", "treatment")),
9           timepoint = as.factor(timepoint)) %>%
10  ggplot(data = .,
11         mapping = aes(x = timepoint,
12                       y = sim_1,
13                       group = participant)) +
14    geom_point(alpha = 0.5,
15              color = "blue") +
16    geom_line(alpha = 0.5,
17             color = "blue") +
18    facet_grid(~ condition) +
19    labs(x = "timepoint")
```

original data

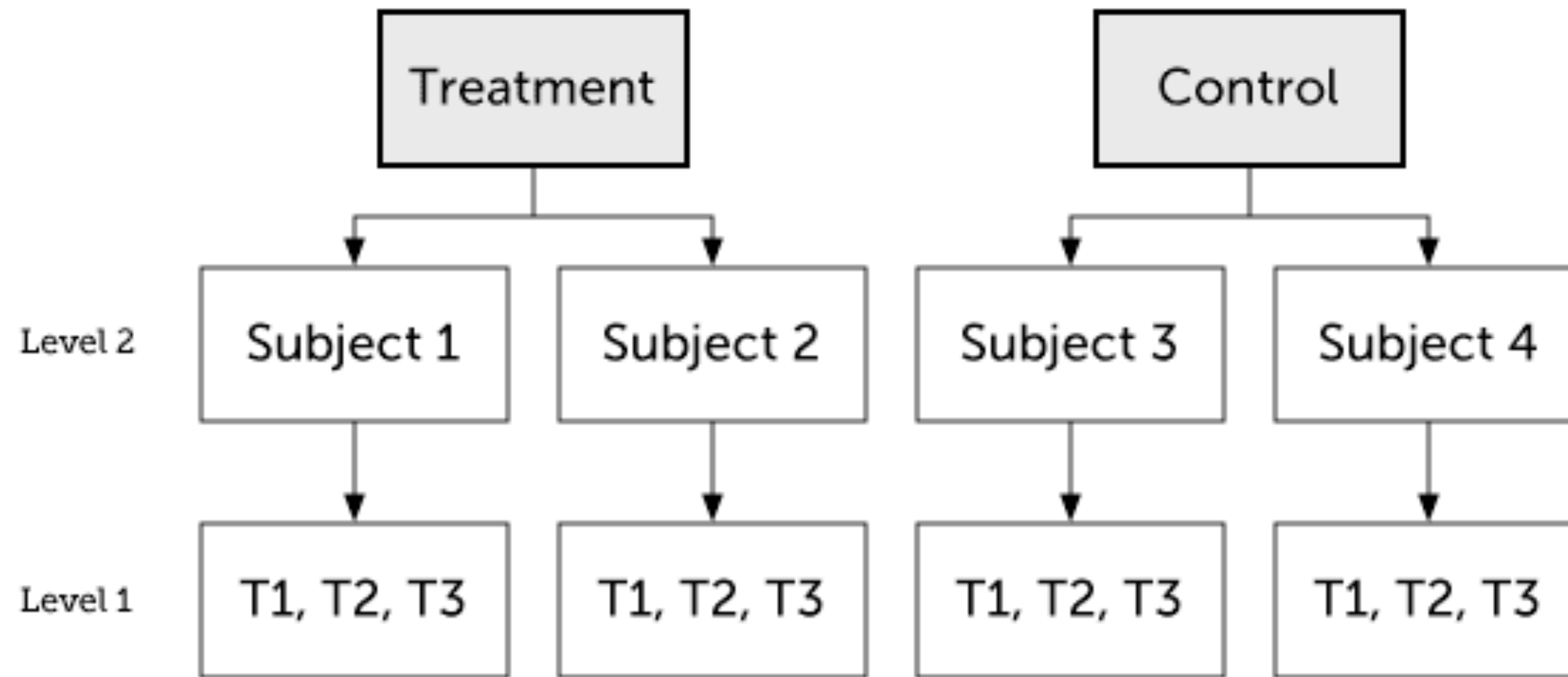


simulated data



Two-level growth model

Graphical representation



assume a linear effect of time

Simulate data

```

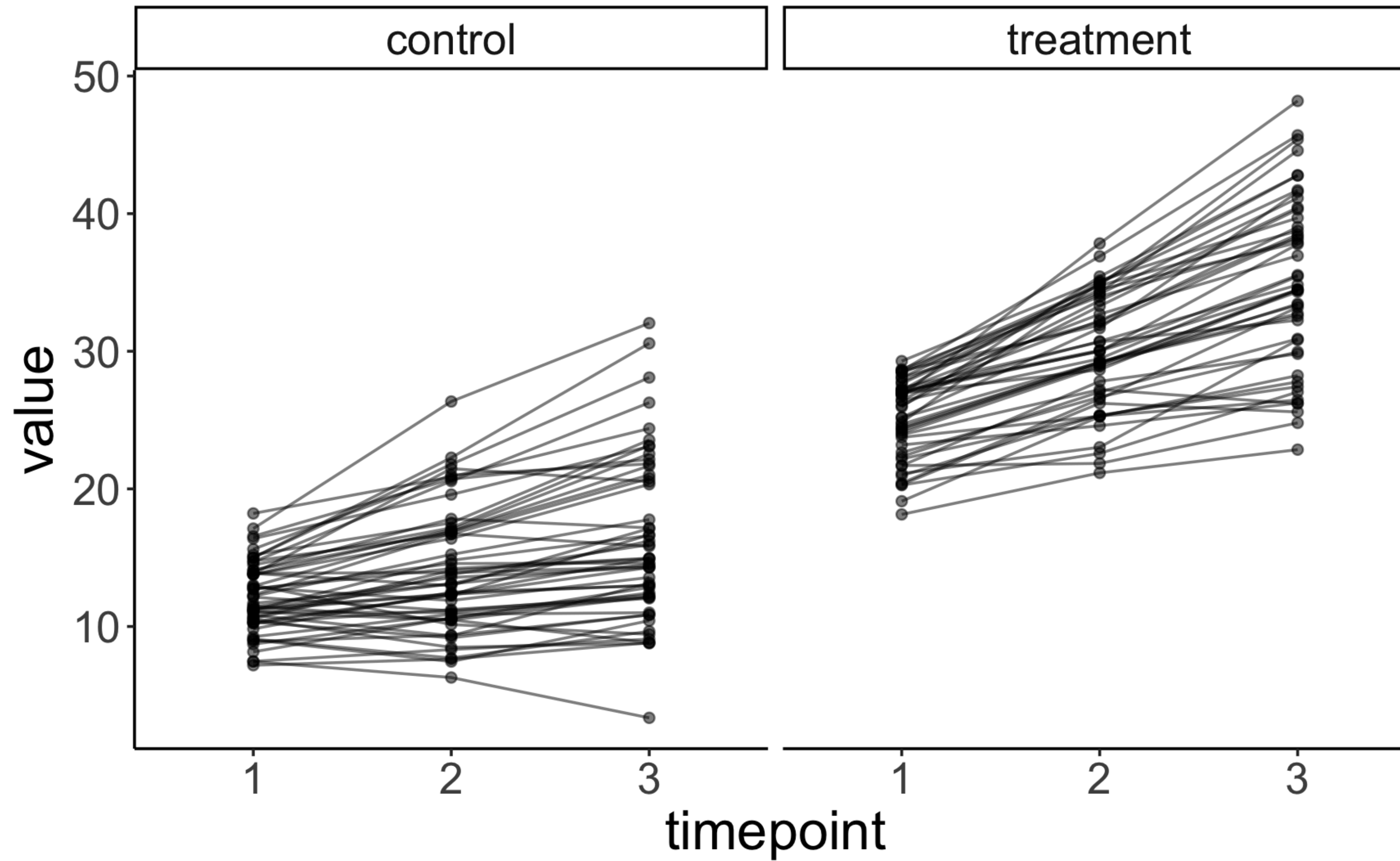
1 set.seed(1)
2
3 n_participants = 100
4 n_timepoints = 3
5 n_conditions = 2
6 p_condition = 0.5
7 b0 = 10 # intercept
8 b1 = 10 # condition
9 b2 = 2 # time
10 b3 = 3 # interaction
11 sd_intercept_participant = 2
12 sd_time_participant = 2
13 sd_residual = 1
14
15 df.data = tibble(participant = rep(1:n_participants, each = n_timepoints),
16                  timepoint = rep(1:n_timepoints, times = n_participants),
17                  intercept_participant = rep(rnorm(n_participants, sd = sd_intercept_participant),
18                                                each = n_timepoints),
19                  time_participant = rep(rnorm(n_participants, sd = sd_time_participant),
20                                          each = n_timepoints)) %>%
21   group_by(participant) %>%
22   mutate(condition = rbinom(n = 1, size = 1, prob = p_condition)) %>%
23   ungroup() %>%
24   mutate(value = b0 + intercept_participant +
25              b1 * condition +
26              (b2 + time_participant) * timepoint +
27              b3 * condition * timepoint +
28              rnorm(n_participants * n_timepoints, sd = sd_residual))

```

	participant	timepoint	intercept_participant	time_participant	condition	value
1	1	1	-1.252907621	-0.310183339	1	20.573131
2	1	2	-1.252907621	-0.310183339	1	22.844077
3	1	3	-1.252907621	-0.310183339	1	24.367254
4	2	1	0.367286648	0.021057937	0	12.119622
5	2	2	0.367286648	0.021057937	0	15.062837
6	2	3	0.367286648	0.021057937	0	17.513515
7	3	1	-1.671257225	-0.455460824	1	17.470186
8	3	2	-1.671257225	-0.455460824	1	22.446022
9	3	3	-1.671257225	-0.455460824	1	24.248006
10	4	1	3.190561604	0.079014386	1	24.844308
11	4	2	3.190561604	0.079014386	1	28.220589
12	4	3	3.190561604	0.079014386	1	28.880339
13	5	1	0.659015544	-0.327292322	1	22.047393
14	5	2	0.659015544	-0.327292322	1	25.189133
15	5	3	0.659015544	-0.327292322	1	28.051351

random slopes

Plot data

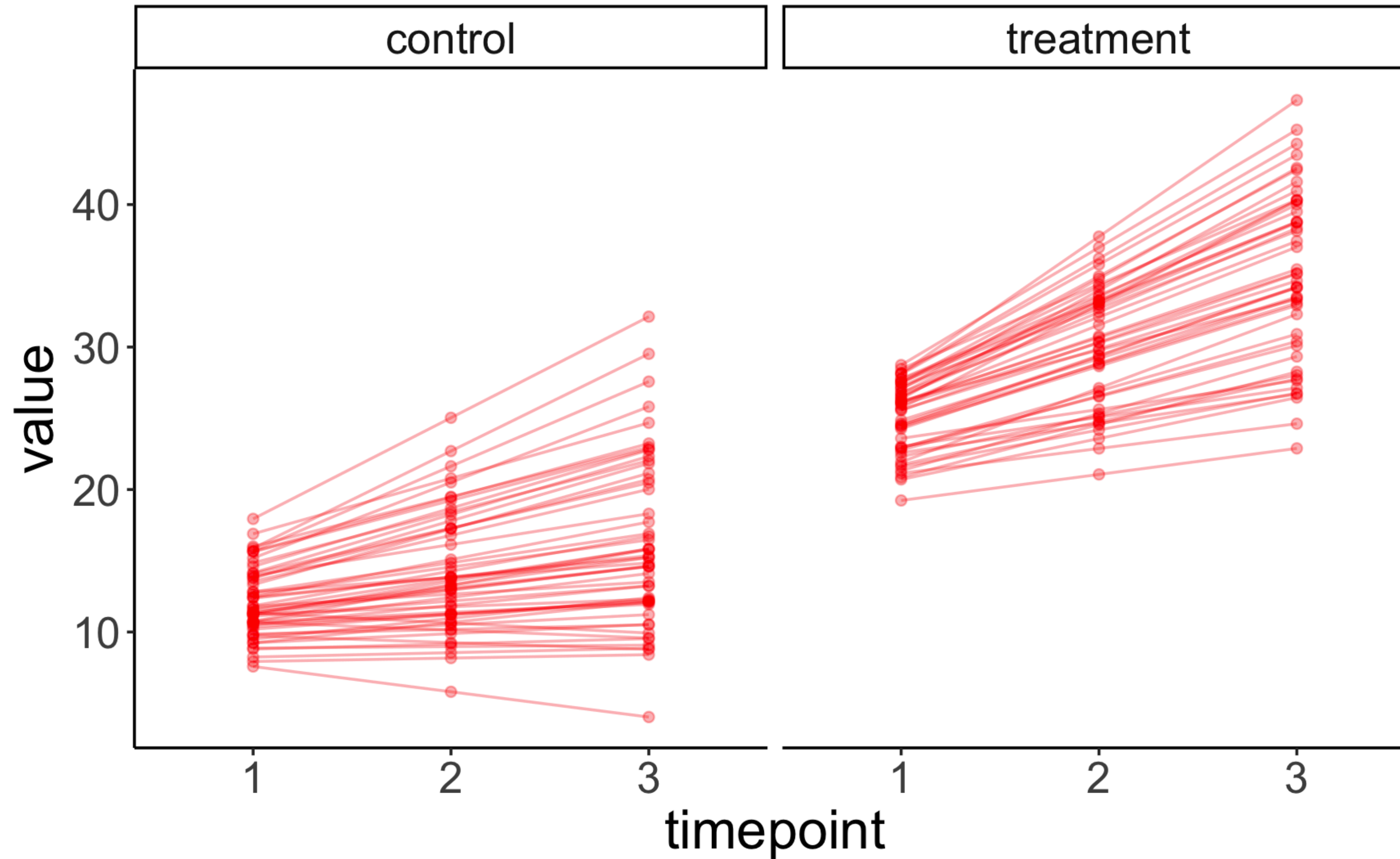


Fit the model

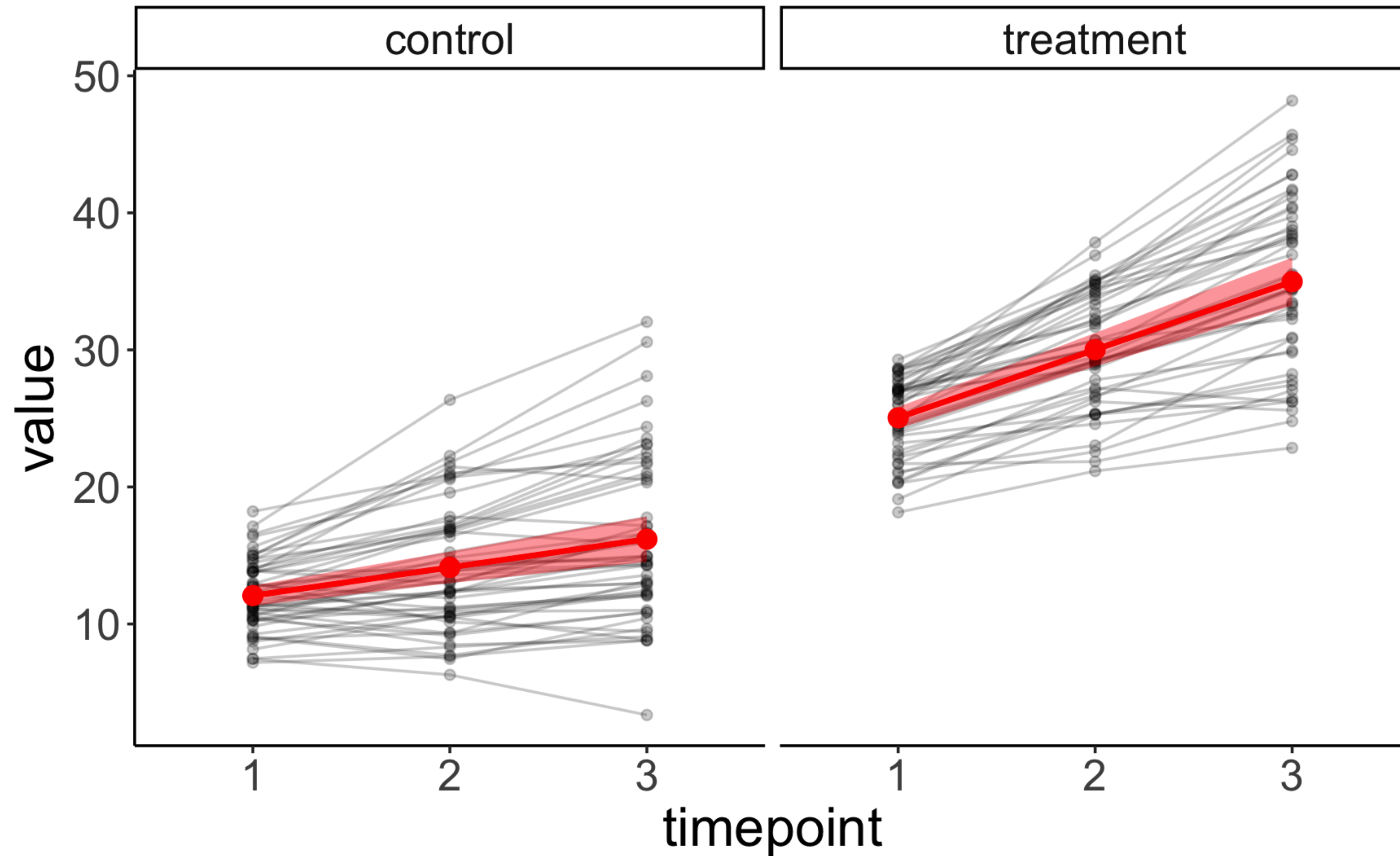
```
1 fit = lmer(formula = value ~ 1 + condition * timepoint + (1 + timepoint | participant),  
2           data = df.data)
```

```
Linear mixed model fit by REML ['lmerMod']  
Formula: value ~ 1 + condition * timepoint + (1 + timepoint | participant)  
Data: df.data  
  
REML criterion at convergence: 1360.3  
  
Scaled residuals:  
      Min       1Q   Median       3Q      Max  
-2.14633 -0.46360  0.03902  0.42302  2.82945  
  
Random effects:  
Groups      Name      Variance Std.Dev. Corr  
participant (Intercept) 3.190    1.786  
            timepoint   3.831    1.957   -0.06  
Residual          1.149    1.072  
Number of obs: 300, groups: participant, 100  
  
Fixed effects:  
              Estimate Std. Error t value  
(Intercept)    10.0101    0.3328   30.079  
condition       10.0684    0.4854   20.741  
timepoint       2.0595    0.2883    7.143  
condition:timepoint 2.9090    0.4205    6.917  
  
Correlation of Fixed Effects:  
              (Intr) condtn timpnt  
condition     -0.686  
timepoint     -0.266  0.182  
condtn:timpnt  0.182 -0.266 -0.686
```

Visualize model predictions (individual)



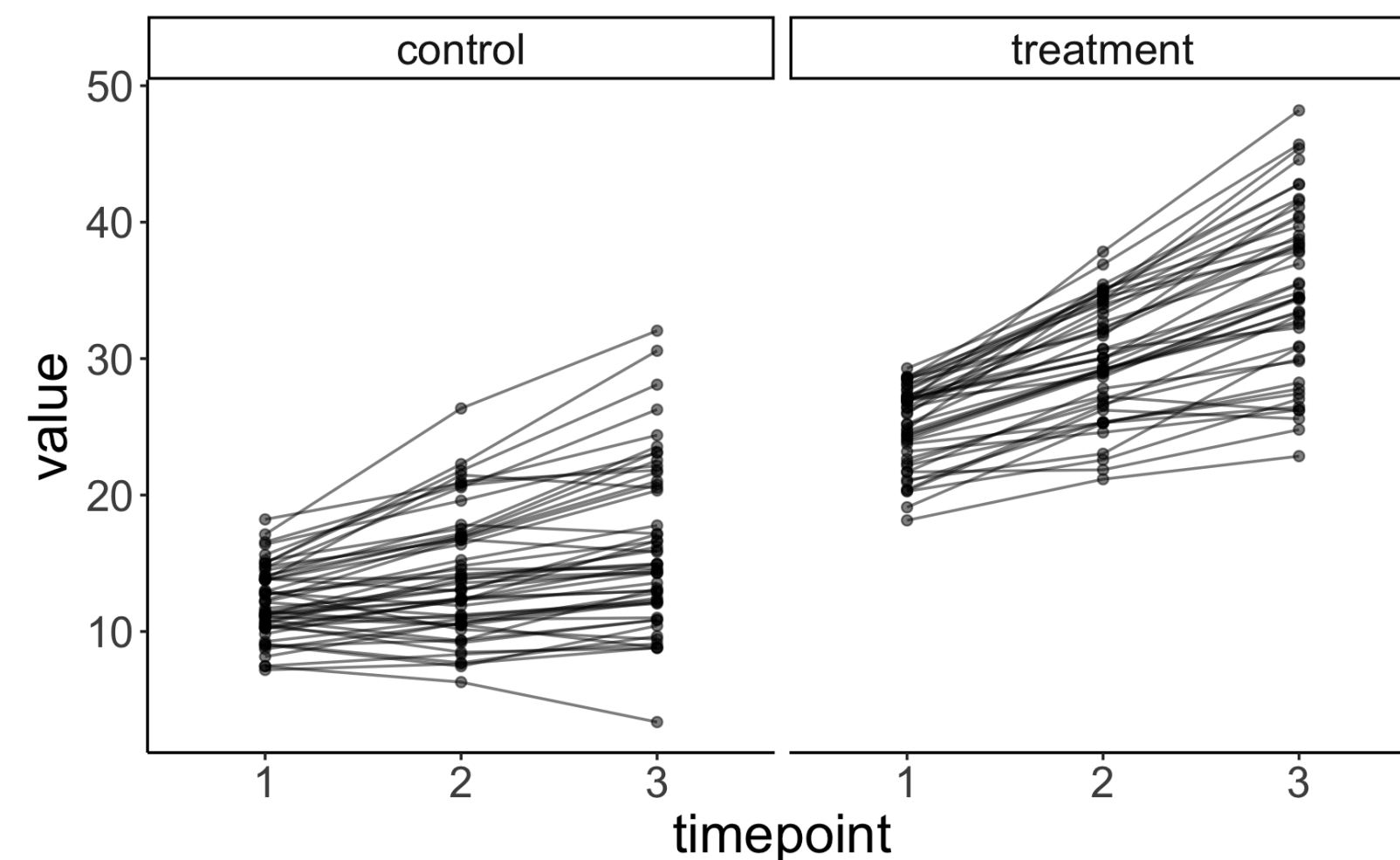
Visualize model predictions (overall)



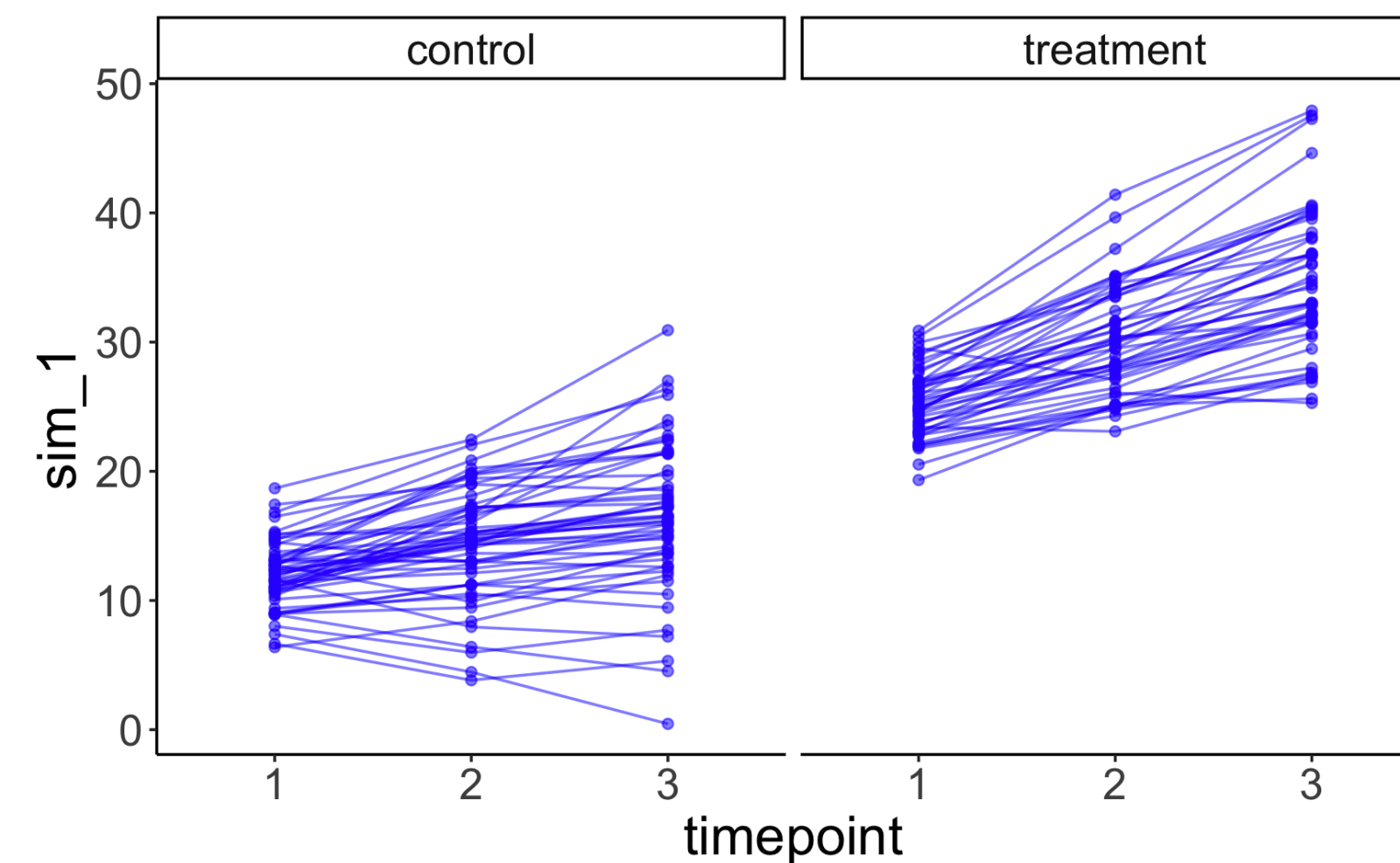
Simulate new data from the model

```
1 set.seed(1)
2
3 fit %>%
4   simulate() %>%
5   bind_cols(df.data) %>%
6   mutate(condition = factor(condition,
7                               levels = c(0, 1),
8                               labels = c("control", "treatment")),
9            timepoint = as.factor(timepoint)) %>%
10  ggplot(data = .,
11         mapping = aes(x = timepoint,
12                       y = sim_1,
13                       group = participant)) +
14    geom_point(alpha = 0.5,
15              color = "blue") +
16    geom_line(alpha = 0.5,
17             color = "blue") +
18    facet_grid(~ condition) +
19    labs(x = "timepoint")
```

original data

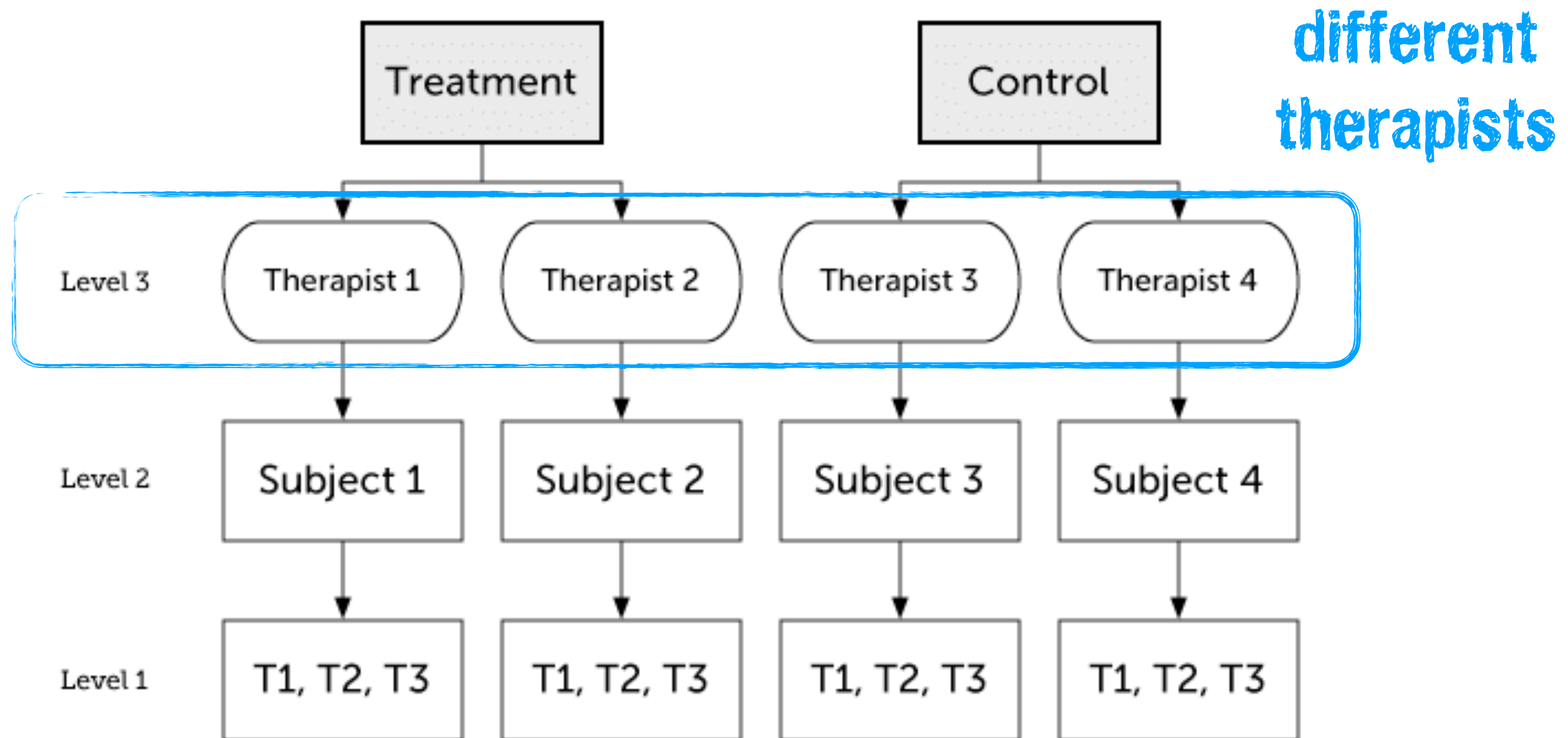


simulated data



Three-level model

Graphical representation



Fit the model

```
1 fit = lmer(formula = value ~ 1 + condition * timepoint +
2           (1 + timepoint | therapist) +
3           (1 + timepoint | therapist:participant),
4           data = df.data)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition * timepoint + (1 + timepoint | therapist) + (1 + timepoint | therapist:participant)
Data: df.data

REML criterion at convergence: 1237.9

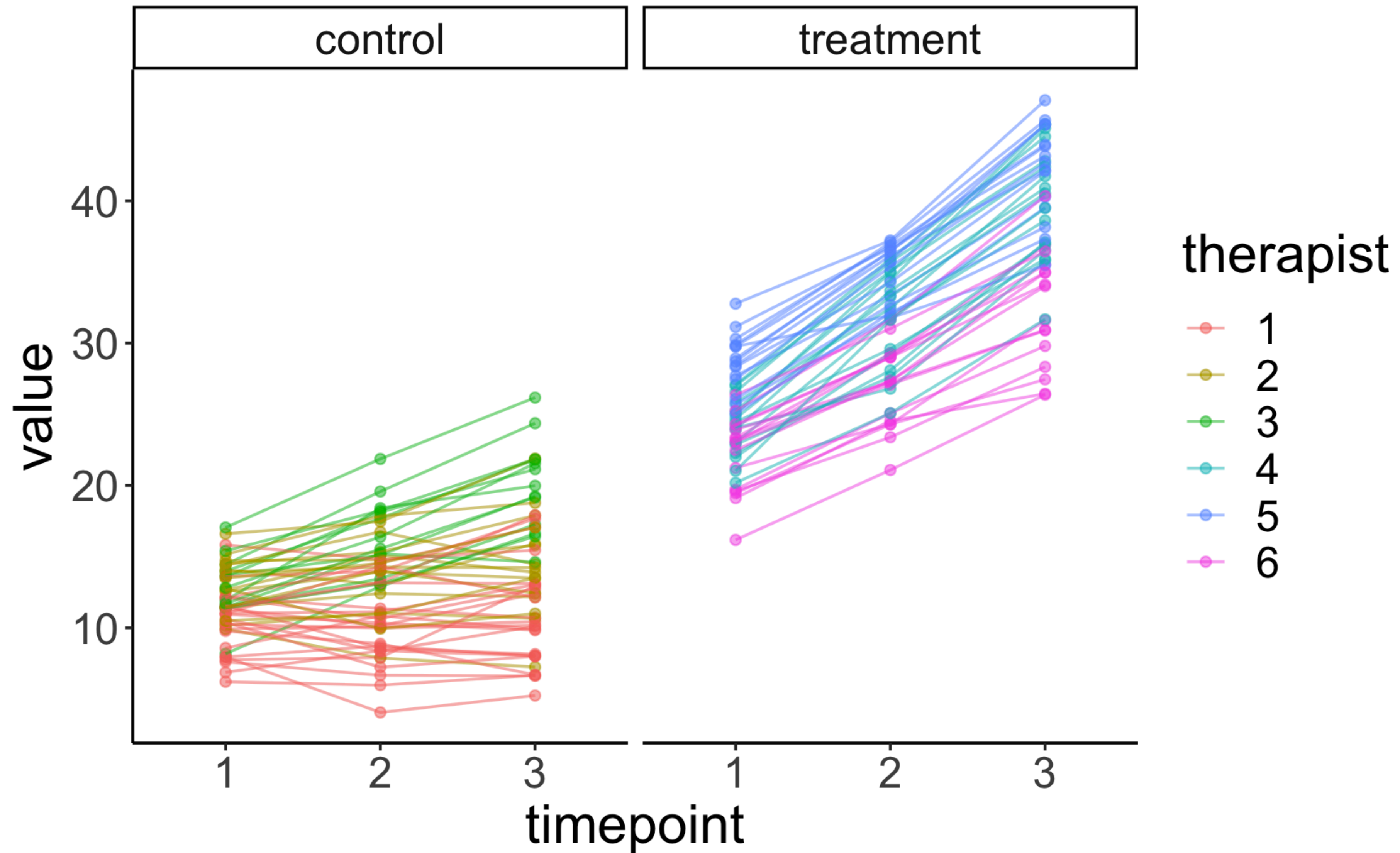
Scaled residuals:
    Min       1Q   Median       3Q      Max
-2.02925 -0.51105  0.01576  0.48074  2.12182

Random effects:
Groups                Name      Variance Std.Dev. Corr
therapist:participant (Intercept) 2.1361   1.4615
                    timepoint  0.8205   0.9058   0.33
therapist              (Intercept) 5.6373   2.3743
                    timepoint  2.4171   1.5547  -0.21
Residual                  1.0515   1.0254
Number of obs: 300, groups:  therapist:participant, 100; therapist, 6

Fixed effects:
              Estimate Std. Error t value
(Intercept)    10.5078     1.4040   7.484
condition        7.7672     1.9870   3.909
timepoint        1.5160     0.9125   1.661
condition:timepoint  5.0489     1.2912   3.910

Correlation of Fixed Effects:
              (Intr) condtn timpnt
condition    -0.707
timepoint    -0.208  0.147
cndtn:timpnt  0.147 -0.208 -0.707
```


Plot data



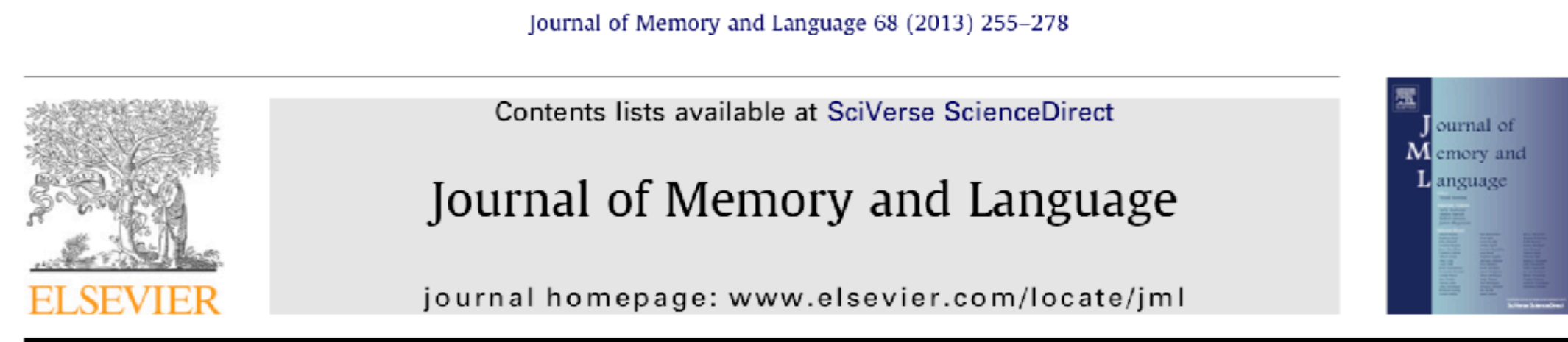
`lmer ( )` **standard operating procedures**

Standard Operating Procedures For Using Mixed-Effects Models	
Standard Operating Procedures For Using Mixed-Effects Models	
A Principled Workflow from the Decision, Development, and Psychopathology (D2P2) Lab	
document version 1.0.0 -- 28 June 2020	
[This document will be continuously updated and expanded; it may contain typos and other errors--both unintentional errors and errors based on incorrect or outdated knowledge--we will try to improve these things in future versions. Feel free to let us know if you spotted such things, how to further improve this document!]	
Authors (in alphabetical order except that the youngsters were so kind to put the oldest guy in the lab first; BF)	
Bernd Figner, Johannes Algermissen, Floor Burghoorn, Leslie Held, Afreen Khalid, Felix Klaassen, Farnaz Mosannenzadeh, Julian Quandt	
Content/Analysis Steps	
Content/Analysis Steps	1
1. Before data collection:	
Power/ design/ planning/ sample size	3
1.1. Power analysis	3
1.2. Sensitivity analysis	4
1.3. Sequential sampling with stopping rules	5
1.4. More readings	5
2. Preparing data	6
2.1. Categorical variables	6
2.2. Continuous variables	6
3. Running the model	7
3.1. Model specification and random effects	7
3.2. Addressing convergence warnings	7
3.2.1. Convergence warnings in R's lme4	7
3.2.2. Or we choose the Bayesian approach	9
3.2.3. MixedModels in Julia	9

http://decision-lab.org/wp-content/uploads/2020/07/SOP_Mixed_Models_D2P2_v1_0_0.pdf

What shall I include as random effects?

- mixed opinions on the topic
- go maximal!



Random effects structure for confirmatory hypothesis testing:
Keep it maximal



Dale J. Barr^{a,*}, Roger Levy^b, Christoph Scheepers^a, Harry J. Tily^c

^aInstitute of Neuroscience and Psychology, University of Glasgow, 58 Hillhead St., Glasgow G12 8QB, United Kingdom

^bDepartment of Linguistics, University of California at San Diego, La Jolla, CA 92093-0108, USA

^cDepartment of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

"Through theoretical arguments and Monte Carlo simulation, we show that LMEMs generalize best when they include the maximal random effects structure justified by the design. ...

Maximal LMEMs should be the 'gold standard' for confirmatory hypothesis testing in psycholinguistics and beyond."

What shall I include as random effects?

- general advice:
 - start maximal (as supported by the design)
 - random intercepts and slopes for participants
 - random intercepts for items
 - reduce complexity of the random effects structure step by step
 - remove the correlations between random effects first

Remove the correlation component from your model

```
1 # fit the model
2 fit.lmer = lmer(formula = reaction ~ 1 + days + (1 + days | subject),
3                 data = df.sleep)
4 # model summary
5 fit.lmer %>%
6   summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: reaction ~ 1 + days + (1 + days | subject)
Data: df.sleep

REML criterion at convergence: 1771.4

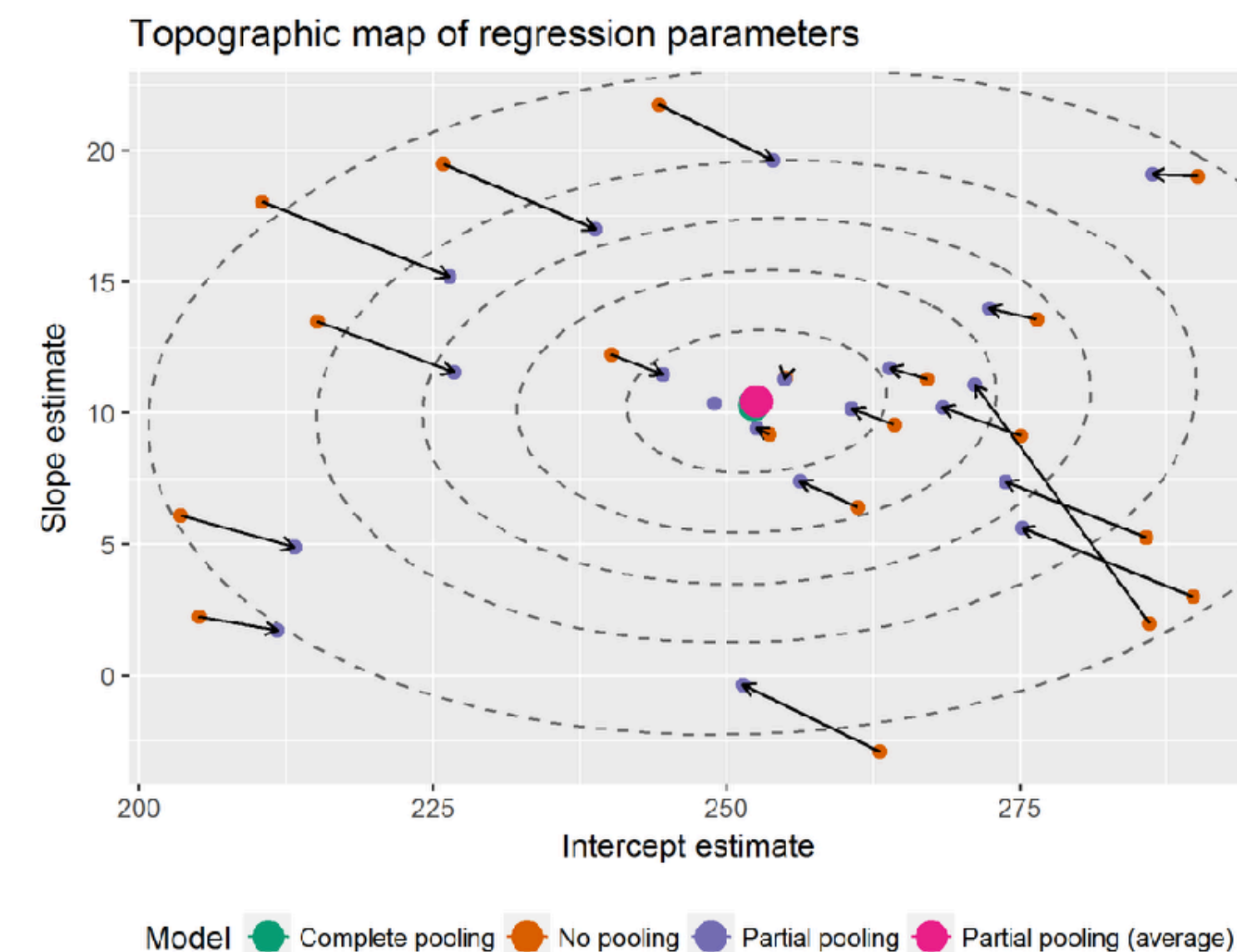
Scaled residuals:
    Min       1Q   Median       3Q      Max
-3.9707 -0.4703  0.0276  0.4594  5.2009

Random effects:
   Groups     Name      Variance Std.Dev. Corr
  subject (Intercept) 582.73    24.140
                days    35.03     5.919   0.07
  Residual                649.36    25.483
Number of obs: 183, groups: subject, 20

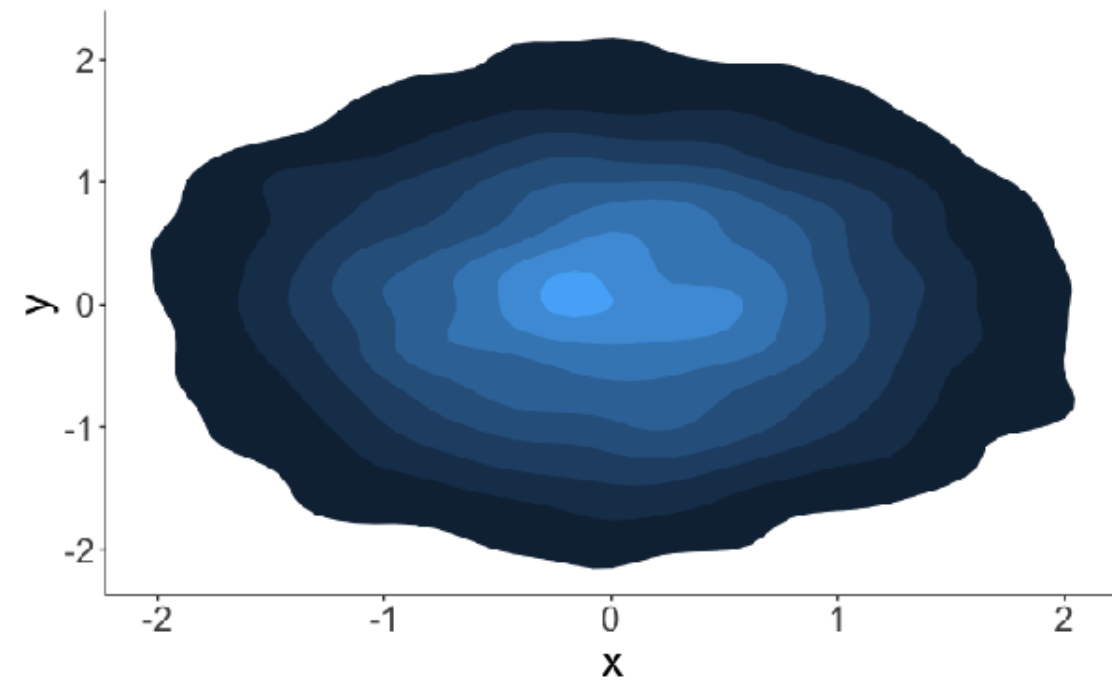
Fixed effects:
              Estimate Std. Error t value
(Intercept)  252.543     6.433   39.256
days         10.452     1.542    6.778

Correlation of Fixed Effects:
      (Intr)
days -0.137
```

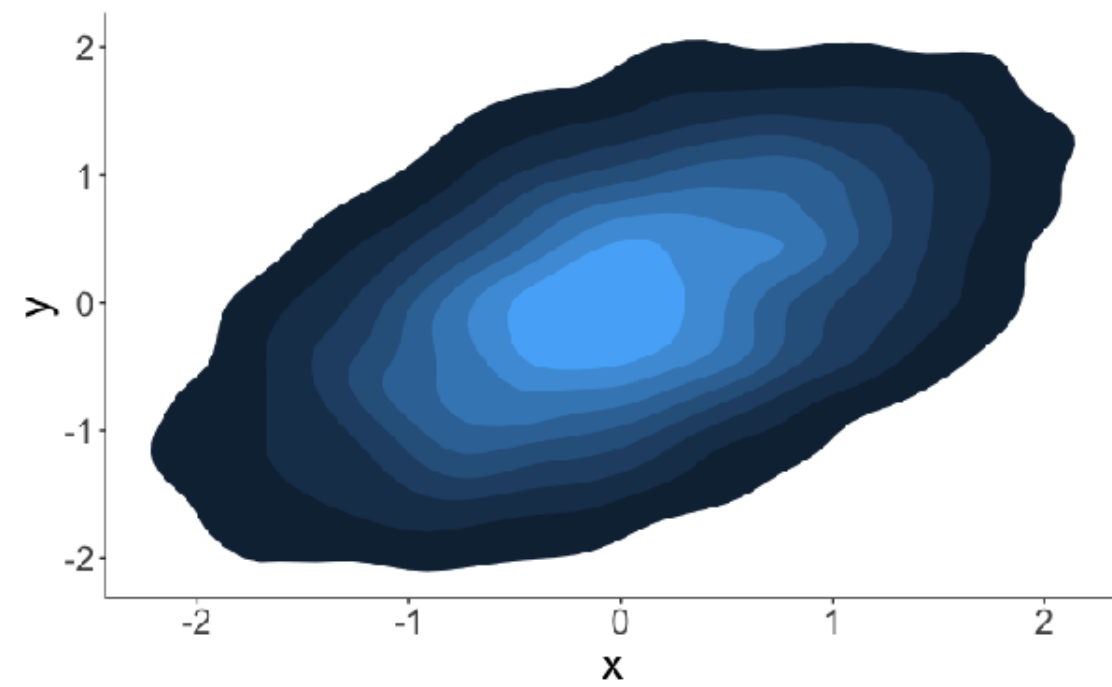
multivariate
Gaussian



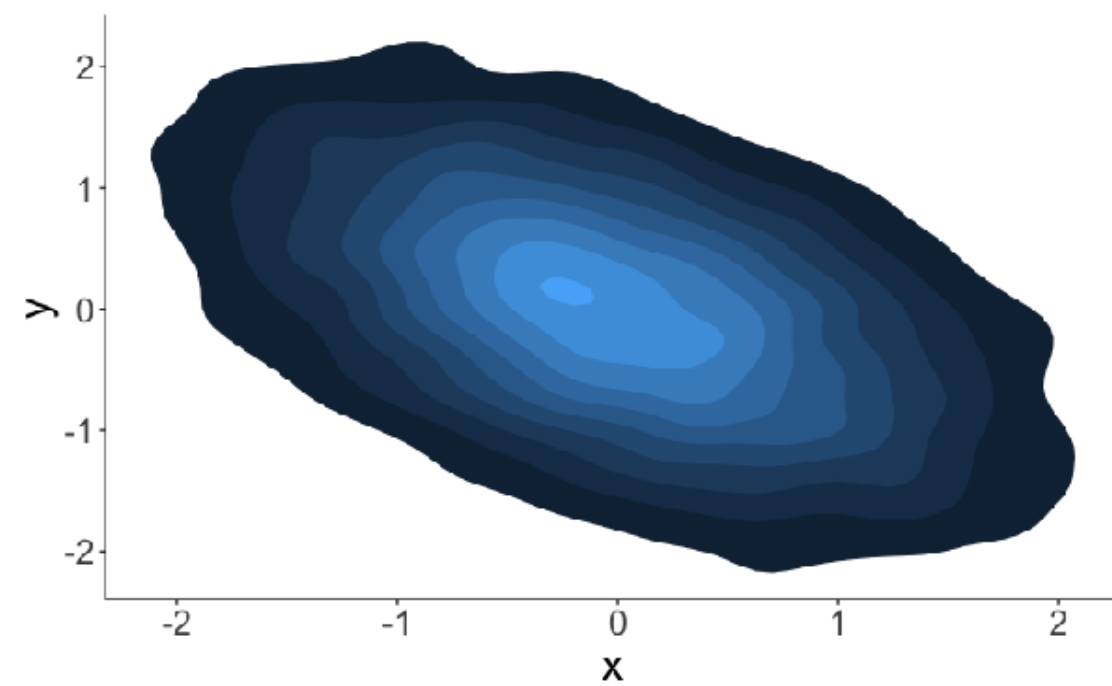
Remove the correlation component from your model



uncorrelated



positively correlated



negatively correlated

Remove the correlation component from your model

```
1 # fit the model
2 fit.lmer = lmer(formula = reaction ~ 1 + days + (0 + days | subject) + (1 | subject),
3                 data = df.sleep)
4 # model summary
5 fit.lmer %>%
6   summary()
```

↑
random
slopes

↑
random
intercepts

```
Linear mixed model fit by REML ['lmerMod']
Formula: reaction ~ 1 + days + (0 + days | subject) + (1 | subject)
Data: df.sleep

REML criterion at convergence: 1771.5

Scaled residuals:
    Min       1Q   Median       3Q      Max
-3.9805 -0.4673  0.0250  0.4589  5.2083

Random effects:
 Groups   Name      Variance Std.Dev.
subject  days      35.88     5.99
subject.1 (Intercept) 598.11    24.46
Residual                647.90    25.45
Number of obs: 183, groups:  subject, 20

Fixed effects:
              Estimate Std. Error t value
(Intercept)  252.550     6.491    38.907
days         10.439     1.556     6.708

Correlation of Fixed Effects:
      (Intr)
days -0.184
```

independent
Gaussians

Remove the correlation component from your model

```
1 # fit the model
2 fit.lmer = lmer(formula = reaction ~ 1 + days + (1 + days || subject),
3                 data = df.sleep)
4 # model summary
5 fit.lmer %>%
6   summary()
```

alternative syntax (doesn't
model correlation between
random effects)

```
Linear mixed model fit by REML ['lmerMod']
Formula: reaction ~ 1 + days + (0 + days | subject) + (1 | subject)
Data: df.sleep

REML criterion at convergence: 1771.5

Scaled residuals:
    Min       1Q   Median       3Q      Max
-3.9805 -0.4673  0.0250  0.4589  5.2083

Random effects:
 Groups   Name      Variance Std.Dev.
subject  days      35.88     5.99
subject.1 (Intercept) 598.11    24.46
Residual                647.90    25.45
Number of obs: 183, groups:  subject, 20

Fixed effects:
              Estimate Std. Error t value
(Intercept)  252.550     6.491    38.907
days         10.439     1.556     6.708

Correlation of Fixed Effects:
      (Intr)
days -0.184
```

independent
Gaussians

What if `lmer()` fails to converge?

1. We drop random effects in the following order: random correlations, random slopes of covariates (where significance is of no interest), random intercepts ("0+" instead "1+") (following [Barr et al., 2013](#)). We never remove the random slopes of the variables of interest (i.e., the ones for which we want to conduct significance tests).

Please note that removing random correlation terms can be tricky if random slopes are estimated for factors with 3 or more levels. In that case, it is probably easiest to use `afex::mixed()` with `expand_re = TRUE` (an alternative option is to create manually the relevant contrasts yourself and add them as predictors to your model, which allows you to suppress the random corrections using the double pipe symbol `| |`).

2. We try to run separate analyses: For example, one model to only test the fixed and random effect of A (with fixed effect of B present); then one model to only test the effect of B. If we really have to drop random slopes, we follow the next step:
3. We follow the PCA approach suggested by **rePsychLing** (see [Bates et al., 2015](#)) that is performing a PCA on the random effects and following the guidelines described in the paper.

- a. We use a likelihood ratio test to test whether the model fit becomes significantly worse. As we prefer a more conservative approach here (i.e., rather err on the side of keeping too many random effects; we prioritize avoiding inflated Type 2 errors for this kind of decision), we use larger alpha-level of .2 ([Matuschek et al., 2017](#)).
- b. Alternatively, we suggest an Information criterion approach to avoid using a *p* value for our inclusion/exclusion decision, but choose the best model based on *BIC* or *AIC*.



3.2.2. Or we choose a Bayesian approach

As an alternative to targeting convergence issues within **lme4**, we suggest fitting the same model with **brms** and comparing it to the **lme4** fit. We assume that both provide similar results when

Plan for today

- Quick recap
- Linear Mixed Model
 - Accommodating non-independence in data
 - Understanding lmer() syntax
 - A worked example
 - Simpson's Paradox
 - Reporting results
 - Understanding lmer() syntax
 - Reporting results
 - **Let's simulate some lmer()'s**
 - lmer() standard operating procedures
 - Some more examples

How was the pace of today's class?

much too slow a little too slow just right a little too fast much too fast

How happy were you with today's class overall?



What did you like about today's class? What could be improved next time?

Thank you!